

Provable Diffusion Posterior Sampling for Bayesian Inversion

Jinyuan Chang^{1,2}, Chenguang Duan³, Yuling Jiao⁴, Ruoxuan Li⁵, Jerry Zhijian Yang⁵, and Cheng Yuan⁴

¹Joint Laboratory of Data Science and Business Intelligence, Southwestern University of Finance and Economics, Chengdu, Sichuan 611130, China.

`changjinyuan@swufe.edu.cn`

²State Key Laboratory of Mathematical Sciences, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China

³Institut für Geometrie und Praktische Mathematik, RWTH Aachen University, Templergraben 55, Aachen 52056, Germany.

`duan@igpm.rwth-aachen.de`

⁴School of Artificial Intelligence, Wuhan University, Wuhan, Hubei 430072, China.

`{yulingjiaomath,yuancheng}@whu.edu.cn`

⁵School of Mathematics and Statistics, Wuhan University, Wuhan, Hubei 430072, China.

`{ruoxuanli.math,zjyang.math}@whu.edu.cn`

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Abstract. This paper proposes a novel diffusion-based posterior sampling method within a plug-and-play (PnP) framework. Our approach constructs a probability transport from an easy-to-sample terminal distribution to the target posterior, using a warm-start strategy to initialize the particles. To approximate the posterior score, we develop a Monte Carlo estimator in which particles are generated using Langevin dynamics, avoiding the heuristic approximations commonly used in prior work. The score governing the Langevin dynamics is learned from data, enabling the model to capture rich structural features of the underlying prior distribution. On the theoretical side, we provide non-asymptotic error bounds, showing that the method converges even for complex, multi-modal target posterior distributions. These bounds explicitly quantify the errors arising from posterior score estimation, the warm-start initialization, and the posterior sampling procedure. Our analysis further clarifies how the prior score-matching error and the condition number of the Bayesian inverse problem influence overall performance. Finally, we present numerical experiments demonstrating the effectiveness of the proposed method across a range of inverse problems.

Keywords: Posterior sampling, diffusion models, plug-and-play, inverse problems, convergence analysis

1 Introduction

Inverse problems play a crucial role across diverse fields, encompassing applications such as medical imaging reconstruction (Chung et al., 2022b), data assimilation (Reich and Cotter, 2015; Reich, 2019; Bao et al., 2024; Li et al., 2025b; Si and Chen, 2025; Ding et al., 2024b), physics sciences (Zheng et al., 2025a), digital twins (Thelen et al., 2022; Opper and Reich, 2025), and generative artificial intelligence (Chung et al., 2023; Uehara et al., 2025). The goal of a statistical inverse problem is to recover an unknown signal of interest $\mathbf{X}_0 \in \mathbb{R}^d$ from

indirect and noisy measurements $\mathbf{Y} \in \mathbb{R}^n$, which are linked through

$$(1.1) \quad \mathbf{Y} = \mathcal{F}(\mathbf{X}_0) + \mathbf{n}.$$

Here $\mathcal{F} : \mathbb{R}^d \rightarrow \mathbb{R}^n$ represents a known differentiable forward operator, and the random variable $\mathbf{n} \in \mathbb{R}^n$ denotes a measurement noise with a known density ρ but unknown realization. Consequently, the likelihood of observing $\mathbf{Y} = \mathbf{y}$ given $\mathbf{X}_0 = \mathbf{x}_0$ is $p_{\mathbf{Y}|\mathbf{X}_0}(\mathbf{y}|\mathbf{x}_0) := \rho(\mathbf{y} - \mathcal{F}(\mathbf{x}_0))$.

Inverse problems are typically ill-posed due to measurement noise, limited data availability, and intrinsic information loss introduced by the forward operator. As a result, a unique solution may not exist, or, if it exists, it may be sensitive to small perturbations in the measurements $\mathbf{y}$ (Hadamard, 1902). To ensure well-posedness, additional information about the unknown signal $\mathbf{X}_0$ must be incorporated to regularize the problem.

Within the Bayesian framework, prior knowledge about the unknown signal is encoded through a probability measure π_0 , referred to as the prior distribution. Applying Bayes' rule, the posterior distribution of $\mathbf{X}_0$ given $\mathbf{Y} = \mathbf{y}$ is expressed as

$$(1.2) \quad \underbrace{p_{\mathbf{X}_0|\mathbf{Y}}(\mathbf{x}_0|\mathbf{y})}_{\text{posterior}} \propto \underbrace{\exp(-\ell_{\mathbf{y}}(\mathbf{x}_0))}_{\text{likelihood}} \underbrace{\pi_0(\mathbf{x}_0)}_{\text{prior}},$$

where $\ell_{\mathbf{y}}(\cdot) := -\log p_{\mathbf{Y}|\mathbf{X}_0}(\mathbf{y}|\cdot)$ denotes the negative log-likelihood associated with the measurement $\mathbf{Y} = \mathbf{y}$. The posterior distribution (1.2) enforces fidelity to the data through the likelihood, while incorporating prior knowledge through the prior distribution. Extensive work has established conditions for stability and well-posedness of Bayesian inverse problems (Marzouk and Xiu, 2009; Stuart, 2010; Cotter et al., 2010; Dashti and Stuart, 2017; Latz, 2020).

To address the Bayesian inverse problem, two primary methodological paradigms have been developed over recent decades (Stuart, 2010; Sanz-Alonso et al., 2023; Bach et al., 2025; Oliviero-Durmus et al., 2025; Janati et al., 2025): optimization-based methods and sampling-based methods. Optimization-based approaches, also known as variational methods, seek a reconstruction of the unknown signal by identifying the most likely $\mathbf{x}_0$ under the posterior distribution (1.2). This leads to the maximum a posteriori (MAP) estimator

$$(1.3) \quad \mathbf{x}_0^* \in \arg \max_{\mathbf{x}_0 \in \mathbb{R}^d} p_{\mathbf{X}_0|\mathbf{Y}}(\mathbf{x}_0|\mathbf{y}) = \arg \min_{\mathbf{x}_0 \in \mathbb{R}^d} \ell_{\mathbf{y}}(\mathbf{x}_0) - \log \pi_0(\mathbf{x}_0).$$

Here, the data-fidelity term $\ell_{\mathbf{y}}$ enforces consistency with the observations, while the regularization term $-\log \pi_0$ incorporates prior information about the unknown signal. This line of work has close connection with Tikhonov regularization (Ito and Jin, 2014; Benning and Burger, 2018). Although optimization-based methods provide a single point estimate, they do not capture the full structure of the posterior distribution, and thus may fail to reflect uncertainty or represent multimodal posteriors. In contrast, sampling-based methods (Bach et al., 2025; Li et al., 2025a) aim to generate a collection of samples that approximate the posterior distribution (1.2):

$$(1.4) \quad \mathbf{X}_{0,1}^{\mathbf{y}}, \dots, \mathbf{X}_{0,n}^{\mathbf{y}} \sim p_{\mathbf{X}_0|\mathbf{Y}}(\cdot|\mathbf{y}).$$

Such samples can represent complex and multi-modal posterior distributions, enabling both point estimation and principled uncertainty quantification.

To effectively tackle Bayesian inverse problems via sampling, two central challenges must be addressed:

How can we construct a prior distribution π_0 that accurately reflects real-world data? How can we efficiently and accurately sample from the resulting potentially complex posterior distribution $p_{\mathbf{X}_0|\mathbf{Y}}(\cdot|\mathbf{Y})$?

Conventional priors are typically chosen as simple parametric distributions based on domain knowledge or intuition, such as Gaussian priors for smoothness and Laplacian priors for sparsity. Although computationally convenient, these priors present significant limitations. The choice of their parameters, such as the mean and covariance of a Gaussian prior, must be determined based on expert knowledge. Furthermore, these simple priors often fail to capture the rich underlying structure present in real-world data. In contrast, non-parametric data-driven approaches, including deep image prior (Ulyanov et al., 2018; Antoran et al., 2023), generative adversarial networks (GANs) (Goodfellow et al., 2014), variational autoencoders (VAEs) (Kingma and Welling, 2013), normalizing flows (NFs), score-matching methods (Vincent, 2011), and consistency models (Song et al., 2023c), have proven highly effective at learning prior distributions directly from data. Such data-driven priors have been applied to a wide range of methods for Bayesian inverse problems. For example, Bohra et al. (2022); Dasgupta et al. (2024); Patel et al. (2022) employ GAN-based priors, Cai et al. (2024) employs NF-based priors, Laumont et al. (2022); Sun et al. (2024); Ding et al. (2024b) adopt score-based priors, and Purohit et al. (2025) develops methods based on consistency-model (CM) priors. Despite their strong ability to approximate complex prior distributions, there remains a need for methods that can effectively and reliably sample from multi-modal posteriors.

Classical posterior sampling is dominated by Markov chain Monte Carlo (MCMC) methods (Gamerman and Lopes, 2006; Meyn et al., 2009; Durmus et al., 2022; Flock et al., 2025). Several recent works combine data-driven priors with MCMC variants like the Unadjusted Langevin Algorithm (ULA) (Laumont et al., 2022; Cai et al., 2024; Purohit et al., 2025; Ding et al., 2024b). However, the convergence guarantees for classical MCMC typically rely on the log-concavity of the target distribution (Bakry et al., 2014), causing them to struggle in multi-modal settings. Recent advances in diffusion models (Ho et al., 2020; Song et al., 2021) and flow-based models (Lipman et al., 2023; Albergo and Vanden-Eijnden, 2023; Albergo et al., 2025) have led to substantial progress in sampling from multi-modal distributions (Huang et al., 2024a,b; Ding et al., 2023; Huang et al., 2025b; Saremi et al., 2024; Grenioux et al., 2024; Chen et al., 2024a; He et al., 2024; Chemseddine et al., 2025; Wu and Xie, 2025; Guo et al., 2025a). These models are particularly well suited for multimodal sampling because they learn a smooth probability transport from a simple Gaussian base distribution to a complex target distribution. By constructing this continuous transformation path, they can naturally explore and recover separate modes of the distribution, provided these modes are represented in the learned score (or velocity) network. These approaches avoid the limitations of local exploration inherent to classical MCMC, enabling efficient sampling in highly multimodal landscapes. A growing body of theoretical work further establishes their convergence guarantees in non-log-concave settings (Lee et al., 2023; Chen et al., 2023a; Li et al., 2024b,a; Oko et al., 2023; Tang and Yang, 2024; Ding et al., 2024a; Beyler and Bach, 2025; Kremling et al., 2025).

Bringing these developments together, integrating diffusion models with data-driven priors offers a promising avenue toward a unified framework that couples expressive, data-driven prior models with powerful multi-modal posterior sampling capabilities (Oliviero-Durmus et al., 2025). However, diffusion-based posterior sampling remains challenging in practice due to the difficulty of accurately estimating the posterior score. While the posterior score

for Gaussian likelihoods, i.e., linear forward operators with Gaussian noise, can be obtained directly from the unconditional score (Guo et al., 2024), the posterior score under general likelihoods is not directly accessible. As a result, most existing methods rely on heuristic approximations (Chung et al., 2023; Yu et al., 2023; Rout et al., 2023; Chung et al., 2022a; Bansal et al., 2024; Song et al., 2023a,b; Bao et al., 2024; Guo et al., 2024; Li and Wang, 2025). These heuristic approximations introduce intrinsic bias into posterior estimation and lack theoretical guarantees. Further discussion appears in Section 2.2.

To avoid these heuristic approximations, we introduce a novel diffusion-based posterior sampling method that bridges the gap between the multi-modal sampling capabilities of diffusion models and the rigorous theoretical guarantees required for Bayesian inverse problems. To estimate the posterior score, we develop a Monte Carlo procedure in which particles are generated via Langevin dynamics, thereby avoiding the heuristic approximations commonly used in prior approaches. A key feature of this estimator is that the prior score guiding the Langevin dynamics is learned from the data, incorporating data-driven prior information into the posterior sampling process. We show that, for a certain class of multi-modal posterior distributions, this posterior score estimator converges provided the diffusion time is not too large (Lemma 4.6). This result implies that the time-reversal process must be initiated at a small terminal time, at which point the distribution may differ significantly from a standard Gaussian. Nonetheless, due to the smoothing effect of the forward noising process, this terminal distribution remains much easier to sample from than the target posterior (Lemma 4.12). To accomplish this, we propose a warm-start strategy that efficiently samples particles from the terminal distribution. Theoretically, we deliver a complete non-asymptotic error analysis of our method, providing rigorous guarantees for posterior sampling even when the target posterior is multi-modal. Empirically, our algorithm exhibits strong performance across a wide range of inverse problems.

1.1 Contributions. Our main contributions are summarized as follows:

- (i) We introduce a novel diffusion-based posterior sampling method in which the posterior score is estimated via a Monte Carlo procedure using particles generated by Langevin dynamics. The prior score driving the Langevin dynamics is learned from data, independent of the measurement model. This allows it to function as a plug-and-play (PnP) module that can be paired with a wide range of likelihood functions, enabling flexible application across diverse inverse problems without retraining. Because our score estimation strategy requires a small diffusion terminal time, we further design an efficient warm-start mechanism to sample from this terminal distribution, which serves as the initialization for the time-reversal process.
- (ii) We provide a comprehensive non-asymptotic convergence analysis for our method, establishing error bounds in the 2-Wasserstein distance. A key strength of our theory is that these guarantees hold even when the target posterior distribution is multi-modal. The analysis explicitly bounds the key error components: the posterior score estimation error, the warm-start error, and the posterior sampling process error. Our theoretical findings reveal the influence of the pre-trained prior score’s accuracy and the problem’s condition number. They also offer practical guidance for setting crucial hyperparameters, such as the Langevin dynamics simulation horizon and the diffusion terminal time. To the best of our knowledge, this is the first work to provide rigorous error bounds for a diffusion-based posterior score estimator.
- (iii) We conduct extensive numerical experiments on several challenging imaging inverse

problems, including Gaussian denoising, Gaussian deblurring, and nonlinear deblurring on the FFHQ dataset. Our method exhibits strong empirical performance and achieves competitive results. In addition, we demonstrate its effectiveness for uncertainty quantification.

The code to reproduce is available at <https://github.com/Ruoxuan0077/PDPS>.

1.2 Organization. The remainder of this paper is organized as follows. Section 2 introduces the diffusion-based framework for posterior sampling. We review related work in Section 3. Section 4 introduces our diffusion-based posterior sampling method. Section 5 presents its nonasymptotic error analysis. Section 6 provides extensive numerical results demonstrating the effectiveness of our approach. Finally, Section 7 concludes the paper by summarizing our key findings and future works. All theoretical proofs and further experimental details are deferred to the appendices.

2 Preliminaries

This section provides a brief introduction to diffusion models for posterior sampling. We begin by establishing the forward and time-reversal processes for posterior sampling in Section 2.1. Then we review some methods on estimating the posterior score in the previous work in Section 2.2.

2.1 Posterior diffusion models. Before proceeding, we introduce the forward and time-reversal processes for sampling from the prior distribution. Let π_0 be the prior density of the unknown signal $\mathbf{X}_0$ in (1.1). Consider the Ornstein-Uhlenbeck process as the forward process:

$$(2.1) \quad d\mathbf{X}_t = -\mathbf{X}_t dt + \sqrt{2} d\mathbf{B}_t, \quad \mathbf{X}_0 \sim \pi_0, \quad t \in (0, T),$$

where $T > 0$ is the terminal time, and $(\mathbf{B}_t)_{t \geq 0}$ is a d -dimensional Brownian motion. The stochastic differential equation (SDE) (2.1) can be interpreted as Langevin dynamics with a Gaussian stationary distribution $N(\mathbf{0}, \mathbf{I}_d)$. The transition distribution of this linear SDE admits an explicit form (Särkkä and Solin, 2019):

$$(2.2) \quad (\mathbf{X}_t | \mathbf{X}_0 = \mathbf{x}_0) \sim N(\mu_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}_d), \quad t \in (0, T),$$

where $\mu_t := \exp(-t)$ and $\sigma_t^2 := 1 - \exp(-2t)$. The marginal density π_t of the forward process $\mathbf{X}_t$ is given as:

$$(2.3) \quad \pi_t(\mathbf{x}) = \int p_{\mathbf{X}_t | \mathbf{X}_0}(\mathbf{x} | \mathbf{x}_0) \pi_0(\mathbf{x}_0) d\mathbf{x}_0 = \int \gamma_{d, \sigma_t^2}(\mathbf{x} - \mu_t \mathbf{x}_0) \pi_0(\mathbf{x}_0) d\mathbf{x}_0, \quad \mathbf{x} \in \mathbb{R}^d,$$

where $p_{\mathbf{X}_t | \mathbf{X}_0}(\cdot | \mathbf{x}_0)$ represents the conditional density of $\mathbf{X}_t$ given $\mathbf{X}_0 = \mathbf{x}_0$, and γ_d denotes the density of a d -dimensional standard Gaussian distribution.

Forward process for posterior sampling. We now consider the posterior forward process, which is governed by the same SDE as (2.1):

$$(2.4) \quad d\mathbf{X}_t^{\mathbf{y}} = -\mathbf{X}_t^{\mathbf{y}} dt + \sqrt{2} d\mathbf{B}_t, \quad \mathbf{X}_0^{\mathbf{y}} \sim q_0(\cdot | \mathbf{y}) := p_{\mathbf{X}_0 | \mathbf{Y}}(\cdot | \mathbf{y}), \quad t \in (0, T),$$

where $p_{\mathbf{X}_0 | \mathbf{Y}}(\cdot | \mathbf{y})$ denotes the conditional density of $\mathbf{X}_0$ given $\mathbf{Y} = \mathbf{y}$. Since this process shares the same transition distribution (2.2) as (2.1), the marginal density $q_t(\cdot | \mathbf{y})$ of the posterior forward process $\mathbf{X}_t^{\mathbf{y}}$ follows analogously to (2.3):

$$(2.5) \quad q_t(\mathbf{x} | \mathbf{y}) = \int \gamma_{d, \sigma_t^2}(\mathbf{x} - \mu_t \mathbf{x}_0) q_0(\mathbf{x}_0 | \mathbf{y}) d\mathbf{x}_0, \quad \mathbf{x} \in \mathbb{R}^d.$$

A key insight is that the posterior forward process (2.4) exhibits a fundamental connection with its prior counterpart (2.1). Specifically, the marginal density $q_t(\cdot|\mathbf{y})$ of the posterior forward process $\mathbf{X}_t^{\mathbf{y}}$ coincides with the conditional density of $\mathbf{X}_t$ given $\mathbf{Y} = \mathbf{y}$. To establish this equivalence, we verify that for each $\mathbf{x} \in \mathbb{R}^d$:

$$\begin{aligned}
 q_t(\mathbf{x}|\mathbf{y}) &= \int \gamma_{d,\sigma_t^2}(\mathbf{x} - \mu_t \mathbf{x}_0) \frac{p_{\mathbf{Y}|\mathbf{X}_0}(\mathbf{y}|\mathbf{x}_0) \pi_0(\mathbf{x}_0)}{p_{\mathbf{Y}}(\mathbf{y})} d\mathbf{x}_0 \\
 &= \int p_{\mathbf{X}_t|\mathbf{X}_0}(\mathbf{x}|\mathbf{x}_0) p_{\mathbf{Y}|\mathbf{X}_0}(\mathbf{y}|\mathbf{x}_0) \pi_0(\mathbf{x}_0) d\mathbf{x}_0 \frac{1}{p_{\mathbf{Y}}(\mathbf{y})} \\
 (2.6) \quad &= \frac{p_{\mathbf{Y}|\mathbf{X}_t}(\mathbf{y}|\mathbf{x}) \pi_t(\mathbf{x})}{p_{\mathbf{Y}}(\mathbf{y})} = p_{\mathbf{X}_t|\mathbf{Y}}(\mathbf{x}|\mathbf{y}),
 \end{aligned}$$

where $p_{\mathbf{Y}|\mathbf{X}_0}(\cdot|\mathbf{x}_0)$ denotes the conditional density of $\mathbf{Y}$ given $\mathbf{X}_0 = \mathbf{x}_0$, $p_{\mathbf{Y}|\mathbf{X}_t}(\cdot|\mathbf{x})$ denotes the conditional density of $\mathbf{Y}$ given $\mathbf{X}_t = \mathbf{x}$, and $p_{\mathbf{Y}}$ denotes the marginal density of $\mathbf{Y}$. Here, the first equality follows from (2.5) combined with Bayes' rule, while the second equality is due to the transition density (2.2).

Time-reversal process for posterior sampling. The time-reversal process (Anderson, 1982) of the posterior forward process (2.4) is given by:

$$(2.7) \quad d\bar{\mathbf{X}}_t^{\mathbf{y}} = (\bar{\mathbf{X}}_t^{\mathbf{y}} + 2\nabla \log q_{T-t}(\bar{\mathbf{X}}_t^{\mathbf{y}}|\mathbf{y})) dt + \sqrt{2} d\mathbf{B}_t, \quad \bar{\mathbf{X}}_0^{\mathbf{y}} \sim q_T(\cdot|\mathbf{y}), \quad t \in (0, T).$$

From Anderson (1982), the marginal density of $\bar{\mathbf{X}}_t^{\mathbf{y}}$ is $q_{T-t}(\cdot|\mathbf{y})$, which coincides with the marginal density of the forward process $\mathbf{X}_{T-t}^{\mathbf{y}}$. This fundamental result enables us to generate samples from the target posterior density $q_0(\cdot|\mathbf{y})$ by simulating the time-reversal process (2.7).

However, a crucial question arises:

How can we estimate the posterior score $\nabla \log q_t(\cdot|\mathbf{y})$ for $t \in (0, T)$?

In the next section, we review previous work on this question.

2.2 Limitations of the previous work. In this section, we review existing approaches to diffusion-based posterior sampling. We focus specifically on how they estimate the time-dependent posterior score $\nabla \log q_t(\cdot|\mathbf{y})$.

As shown in (2.6), the time-dependent posterior density $q_t(\cdot|\mathbf{y})$ can be expressed as the sum of the gradient of the time-dependent log-likelihood and the time-dependent prior score:

$$\begin{aligned}
 \nabla \log q_t(\mathbf{x}|\mathbf{y}) &= \nabla \log p_{\mathbf{Y}|\mathbf{X}_t}(\mathbf{y}|\mathbf{x}) + \nabla \log \pi_t(\mathbf{x}) \\
 &= \nabla \log \left(\int \exp(-\ell_{\mathbf{y}}(\mathbf{x}_0)) p_t(\mathbf{x}_0|\mathbf{x}) d\mathbf{x}_0 \right) + \nabla \log \pi_t(\mathbf{x}) \\
 (2.8) \quad &= \underbrace{\nabla \log \mathbb{E}[\exp(-\ell_{\mathbf{y}}(\mathbf{X}_0)) | \mathbf{X}_t = \mathbf{x}]}_{\text{time-dependent likelihood}} + \underbrace{\nabla \log \pi_t(\mathbf{x})}_{\text{time-dependent prior score}},
 \end{aligned}$$

where the denoising density $p_t(\cdot|\mathbf{x})$ is the conditional density of $\mathbf{X}_0$ given $\mathbf{X}_t = \mathbf{x}$, defined as

$$(2.9) \quad p_t(\mathbf{x}_0|\mathbf{x}) \propto \exp \left(-\log \pi_0(\mathbf{x}_0) - \frac{\|\mathbf{x} - \mu_t \mathbf{x}_0\|_2^2}{2\sigma_t^2} \right).$$

It is worth noting that the time-dependent prior score in (2.8) can be effectively estimated from samples of the prior distribution using denoising score matching (Vincent, 2011). The main challenge, therefore, lies in estimating the time-dependent likelihood term in (2.8).

Approximating the Time-Dependent Likelihood. Existing works on diffusion-based posterior sampling construct approximations to the time-dependent likelihood term based on an approximation of the denoising density $p_t(\cdot|\mathbf{x})$ in (2.9). Common approaches include:

- Dirac delta approximation (Chung et al., 2023; Yu et al., 2023; Rout et al., 2023; Chung et al., 2022a; Bansal et al., 2024). This line of work proposes a Dirac delta approximation for the denoising density:

$$p_t(\mathbf{x}_0|\mathbf{x}) \approx \delta_{\mathbb{E}[\mathbf{X}_0|\mathbf{X}_t=\mathbf{x}]}(\mathbf{x}_0),$$

where the conditional expectation $\mathbb{E}[\mathbf{X}_0|\mathbf{X}_t=\mathbf{x}]$ is a by-product of the time-dependent score, as shown by Tweedie’s formula (Robbins, 1992). This Dirac delta approximation implies the following approximation for the time-dependent likelihood:

$$\mathbb{E}[\exp(-\ell_{\mathbf{y}}(\mathbf{X}_0))|\mathbf{X}_t=\mathbf{x}] \approx \exp(-\ell_{\mathbf{y}}(\mathbb{E}[\mathbf{X}_0|\mathbf{X}_t=\mathbf{x}])).$$

This approximation can be viewed as interchanging the conditional expectation with the nonlinear function $\exp(-\ell_{\mathbf{y}}(\cdot))$, and thus suffers from a bias known as Jensen’s gap (Chung et al., 2023).

- Gaussian Approximation (Song et al., 2023a,b; Zhang et al., 2025). Another line of work proposes a Gaussian approximation for the denoising density $p_t(\cdot|\mathbf{x})$. According to (2.9), $p_t(\cdot|\mathbf{x})$ is dominated by the quadratic term for small t , meaning it behaves like a Gaussian distribution in this regime. However, for large t , its behavior is determined primarily by the prior density π_0 . If the prior π_0 is multi-modal, the denoising density $p_t(\cdot|\mathbf{x})$ will also tend to be multi-modal. In this scenario, the Gaussian approximation can lead to a large bias.

Estimating the Gradient of the Log-Likelihood. A second critical limitation is that these methods first approximate the time-dependent likelihood; however, the actual target is the gradient of its logarithm. Both of the aforementioned approaches use a direct plug-in gradient estimator. That is, they use the gradient of their likelihood estimator as an estimate of the true gradient.

The effectiveness of this plug-in approach is not generally guaranteed. An estimator that approximates a target function with low error in value can still have a gradient that is a poor approximation of the function’s true gradient.

Limitations. Taking these two issues together: (i) the potential large bias in the likelihood approximation; and (ii) the unreliability of the plug-in gradient estimator, we conclude that these existing methods are not sufficiently robust and accurate.

3 Related Work

3.1 Conditional diffusion models. Posterior sampling can be framed within the context of conditional generative learning. This section reviews some prominent conditional generative models, which is also known as diffusion guidance.

Conditional generative learning using paired data. This line of methods aims to estimate the conditional (or posterior) score $\nabla \log p_{\mathbf{X}_t|\mathbf{Y}}(\cdot|\mathbf{y})$ by leveraging paired samples drawn from the joint distribution $(\mathbf{X}_0, \mathbf{Y})$.

Classifier guidance (Dhariwal and Nichol, 2021) begins by training an unconditional score estimator $\hat{\mathbf{s}}_t(\mathbf{x}) \approx \nabla \log p_{\mathbf{X}_t}(\mathbf{x})$. Concurrently, it learns the time-dependent likelihood

$p_{\mathbf{Y}|\mathbf{X}_t}(\mathbf{y}|\mathbf{x})$ using a separate time-dependent classifier. This classifier is trained on time-dependent data pairs $\{(\mathbf{X}_t^i, \mathbf{Y}^i)\}_{i=1}^n$, which are generated by applying the forward corruption process to the $\mathbf{X}_0$ component of the original data pairs $\{(\mathbf{X}_0^i, \mathbf{Y}^i)\}_{i=1}^n$. Minimizing a cross-entropy loss yields the classifier $\hat{p}_{\mathbf{Y}|\mathbf{X}_t}(\mathbf{y}|\mathbf{x})$, an estimator for the time-dependent likelihood. An estimator for the conditional score is then derived via Bayes' rule:

$$\hat{\mathbf{s}}_t(\mathbf{x}|\mathbf{y}) := \nabla \log \hat{p}_{\mathbf{Y}|\mathbf{X}_t}(\mathbf{y}|\mathbf{x}) + \hat{\mathbf{s}}_t(\mathbf{x}) \approx \nabla \log p_{\mathbf{X}_t|\mathbf{Y}}(\mathbf{x}|\mathbf{y}).$$

However, classifier guidance has notable limitations. It is inherently restricted to discrete conditions (e.g., class labels) due to its reliance on a classifier, offering no direct method to estimate $p_{\mathbf{Y}|\mathbf{X}_t}(\mathbf{y}|\mathbf{x})$ when the condition $\mathbf{Y}$ is continuous. Furthermore, even if the likelihood estimator $\hat{p}_{\mathbf{Y}|\mathbf{X}_t}(\mathbf{y}|\mathbf{x})$ achieves high accuracy, the accuracy of its gradient of the log-likelihood $\nabla \log \hat{p}_{\mathbf{Y}|\mathbf{X}_t}(\mathbf{y}|\mathbf{x})$ is not guaranteed. This potential discrepancy can introduce significant errors into the final conditional score estimation.

In contrast, classifier-free guidance (Ho and Salimans, 2022) eliminates the need for an external classifier and allows for continuous conditions $\mathbf{Y}$. This approach typically obtains the conditional score via conditional score matching. A substantial body of literature has provided theoretical analysis for classifier-free guidance (Fu et al., 2024; Chang et al., 2024; Tang et al., 2024), and the method has been applied to inverse problems (Chung et al., 2024).

Despite their success in conditional generation, methods in this category are not well suited for posterior sampling in inverse problems for two primary reasons. First, inverse problems typically involve recovering a signal from a single measurement, making the paired data required to train a posterior score unavailable in practice. While one could synthetically generate paired data using the prior distribution and the measurement model (1.1), doing so is unnecessary and inefficient. Second, and more critically, these approaches require training a new, large-scale conditional score network for each specific likelihood. Any change to the measurement model, even a change in its parameters, necessitates training an entirely new network from scratch.

Zero-shot conditional generative learning. This line of work instead employs a training procedure that requires only samples from the prior distribution $p_{\mathbf{X}_0}$, with no need for paired data. The resulting prior score function acts as a plug-and-play prior that can be applied across a broad class of inverse problems with arbitrary likelihoods (Laumont et al., 2022; Chung et al., 2023, 2022a; Yu et al., 2023; Rout et al., 2023; Song et al., 2023b; Bansal et al., 2024; Guo et al., 2024; Wu et al., 2024b; Xu and Chi, 2024; Martin et al., 2025; Zhang et al., 2025; Sun et al., 2024; Li and Wang, 2025). Consequently, the training process is completely independent of the likelihood, eliminating the need to train a separate score network for each measurement model. The likelihood $p_{\mathbf{Y}|\mathbf{X}_0}$ is introduced only at inference time, a procedure referred to as inference-time alignment (Kim et al., 2025; Uehara et al., 2025). Our work belongs to this class of methods. Theoretical findings of diffusion guidance are investigated by Wu et al. (2024a); Chidambaram et al. (2024); Moufad et al. (2025); Jiao et al. (2025); Galashov et al. (2025).

3.2 Hybridizing diffusion models with classical sampling methods. To avoid heuristic approximations of posterior scores, recent work has integrated diffusion models with classical posterior sampling techniques. For instance, Xu and Chi (2024); Wu et al. (2024b); Coeurdoux et al. (2024); Zheng et al. (2025b); Chu et al. (2025) couple diffusion models with split Gibbs samplers, and Xu and Chi (2024); Wu et al. (2024b); Zheng et al. (2025b); Chu et al. (2025) further provide theoretical guarantees for this approach. Likewise, Wu et al. (2023); Cardoso et al. (2024); Kelvinius et al. (2025); Skreta et al. (2025); Lee et al. (2025);

Achituve et al. (2025); Chen et al. (2025); Kim et al. (2025) integrate diffusion models with sequential Monte Carlo methods, with Wu et al. (2023); Cardoso et al. (2024); Chen et al. (2025) offering rigorous theoretical analyses.

Our work differs fundamentally from these approaches in that they rely on a probability path connecting the initial distribution to the target posterior that is distinct from the diffusion model’s intrinsic time-reversal process.

4 Posterior Score Estimation and Warm-Start

We present a Monte Carlo technique for posterior score estimation in Section 4.1, and then propose a warm-start strategy for sampling from the terminal posterior density in Section 4.2. Finally, we integrate these components into a complete posterior sampling algorithm in Section 4.3.

4.1 Posterior score estimation. In this subsection, we propose a Monte Carlo-based approach for posterior score estimation. This method is inspired by the following lemma, which is an extension of Tweedie’s formula (Robbins, 1992).

Lemma 4.1 (Conditional Tweedie’s formula). *For each $t \in (0, T)$,*

$$(4.1) \quad \nabla \log q_t(\mathbf{x}|\mathbf{y}) = -\frac{1}{\sigma_t^2} \mathbf{x} + \frac{\mu_t}{\sigma_t^2} \mathbf{D}(t, \mathbf{x}, \mathbf{y}), \quad (\mathbf{x}, \mathbf{y}) \in \mathbb{R}^d \times \mathbb{R}^n,$$

where $\mathbf{D}(t, \mathbf{x}, \mathbf{y})$ is the posterior denoisor, defined as the conditional expectation:

$$(4.2) \quad \mathbf{D}(t, \mathbf{x}, \mathbf{y}) := \mathbb{E}[\mathbf{X}_0 | \mathbf{X}_t = \mathbf{x}, \mathbf{Y} = \mathbf{y}] = \int \mathbf{x}_0 p_t(\mathbf{x}_0 | \mathbf{x}, \mathbf{y}) d\mathbf{x}_0.$$

Here, the posterior denoising density $p_t(\mathbf{x}_0 | \mathbf{x}, \mathbf{y})$ is defined as:

$$(4.3) \quad p_t(\mathbf{x}_0 | \mathbf{x}, \mathbf{y}) := p_{\mathbf{X}_0 | \mathbf{X}_t, \mathbf{Y}}(\mathbf{x}_0 | \mathbf{x}, \mathbf{y}) \propto \pi_0(\mathbf{x}_0) \exp \left(-\frac{\|\mathbf{x} - \mu_t \mathbf{x}_0\|_2^2}{2\sigma_t^2} - \ell_{\mathbf{y}}(\mathbf{x}_0) \right).$$

The proof of Lemma 4.1 is deferred to Appendix D.1. This lemma reveals the relationship between the posterior score $\nabla \log q_t(\cdot | \mathbf{y})$ and the expectation of the posterior denoising distribution $p_t(\cdot | \mathbf{x}, \mathbf{y})$.

Sampling from the posterior denoising density. Due to Lemma 4.1, to estimate the posterior score function $\nabla \log q_t(\cdot | \mathbf{y})$, it suffices to generate samples that approximately follows the the posterior denoising density $p_t(\cdot | \mathbf{x}, \mathbf{y})$.

Sampling from the density with the form in (4.3) is known as the restricted Gaussian oracle (RGO) (Lee et al., 2021), whose potential function is a summation of a quadratic and the potential of the posterior density $q_0(\mathbf{x}_0 | \mathbf{y}) \propto \pi_0(\mathbf{x}_0) \exp(-\ell_{\mathbf{y}}(\mathbf{x}_0))$. Consequently, the RGO can be interpreted as a regularized version of sampling from the original posterior density $q_0(\cdot | \mathbf{y})$ and exhibits enhanced log-concavity (Lee et al., 2021). The RGO has attracted attention in diffusion-based sampling for estimating the time-dependent score function (Huang et al., 2024a,b; He et al., 2024; Grenioux et al., 2024). In He et al. (2024), the authors implement RGO using rejection sampling. However, this rejection sampling approach requires access to the minimum value of the posterior potential function and suffers from inefficiency in high-dimensional problems due to low acceptance rates.

In our work, we implement the RGO (4.3) using the Langevin dynamics, following the approach adopted in Huang et al. (2024a,b); Grenioux et al. (2024). Specifically, we consider the Langevin dynamics with stationary density $p_t(\cdot | \mathbf{x}, \mathbf{y})$ (4.3):

$$(4.4) \quad d\mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t} = \left(\nabla \log \pi_0(\mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t}) + \frac{\mu_t}{\sigma_t^2} (\mathbf{x} - \mu_t \mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t}) - \nabla \ell_{\mathbf{y}}(\mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t}) \right) ds + \sqrt{2} d\mathbf{B}_s, \quad s \in (0, S),$$

where $S > 0$ is the simulation horizon, and $(\mathbf{B}_s)_{s \geq 0}$ is a d -dimensional Brownian motion. Denote by $p_t^s(\cdot|\mathbf{x}, \mathbf{y})$ the marginal density of $\mathbf{X}_{0,s}^{\mathbf{x}, \mathbf{y}, t}$. The Langevin dynamics with such score has also been investigated in the literature of annealed Langevin dynamics (Guo et al., 2025b), and parallel tempering (Dong and Tong, 2022).

In practical applications, the exact prior score $\nabla \log \pi_0$ is typically intractable. To address this, one can approximate the prior score using samples from the prior density π_0 with standard techniques like the implicit score matching (Hyvärinen, 2005), sliced score matching (Song et al., 2020), and denoising score matching (Vincent, 2011). Let $\hat{\mathbf{s}}_{\text{prior}} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ denote an approximation to the prior score. We then consider the Langevin dynamics with this estimated prior score:

$$(4.5) \quad d\hat{\mathbf{X}}_{0,s}^{\mathbf{x}, \mathbf{y}, t} = \left(\hat{\mathbf{s}}_{\text{prior}}(\hat{\mathbf{X}}_{0,s}^{\mathbf{x}, \mathbf{y}, t}) + \frac{\mu_t}{\sigma_t^2} (\mathbf{x} - \mu_t \hat{\mathbf{X}}_{0,s}^{\mathbf{x}, \mathbf{y}, t}) - \nabla \ell_{\mathbf{y}}(\hat{\mathbf{X}}_{0,s}^{\mathbf{x}, \mathbf{y}, t}) \right) ds + \sqrt{2} d\mathbf{B}_s, \quad s \in (0, S).$$

Denote by $\hat{p}_t^s(\cdot|\mathbf{x}, \mathbf{y})$ the marginal density of $\hat{\mathbf{X}}_{0,s}^{\mathbf{x}, \mathbf{y}, t}$, which serves as an approximation to $p_t(\cdot|\mathbf{x}, \mathbf{y})$ under certain conditions. A rigorous error analysis will be provided in Section 5.5.

Monte Carlo approximation to the posterior score. With the aid of particles sampled by the Langevin dynamics with the estimated score (4.5), the posterior denoiser $\mathbf{D}(t, \mathbf{x}, \mathbf{y})$ in (4.2) can be estimated by Monte Carlo method:

$$(4.6) \quad \hat{\mathbf{D}}_m^S(t, \mathbf{x}, \mathbf{y}) := \frac{1}{m} \sum_{i=1}^m \hat{\mathbf{X}}_{0,S,i}^{\mathbf{x}, \mathbf{y}, t}, \quad \hat{\mathbf{X}}_{0,S,1}^{\mathbf{x}, \mathbf{y}, t}, \dots, \hat{\mathbf{X}}_{0,S,m}^{\mathbf{x}, \mathbf{y}, t} \sim \text{i.i.d.} \hat{p}_t^S(\cdot|\mathbf{x}, \mathbf{y}).$$

Then using Lemma 4.1 yields the following estimator of the posterior score function:

$$(4.7) \quad \hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y}) := -\frac{1}{\sigma_t^2} \mathbf{x} + \frac{\mu_t}{\sigma_t^2} \hat{\mathbf{D}}_m^S(t, \mathbf{x}, \mathbf{y}) \approx \nabla \log q_t(\mathbf{x}|\mathbf{y}).$$

The complete procedure of the posterior score estimation is summarized as Algorithm 1.

Algorithm 1: Posterior score estimation via Monte Carlo

Input: Measurement $\mathbf{y}$, prior score estimator $\hat{\mathbf{s}}_{\text{prior}}$, time index t of the diffusion model, evaluation point $\mathbf{x}$, simulation horizon S , initial density $\hat{p}_t^0(\cdot|\mathbf{y})$ for Langevin dynamics.

Output: Posterior score estimator $\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$.

- 1 **for** $i = 1, \dots, m$ **do**
 - 2 Initialize particle by $\hat{\mathbf{X}}_{0,0,i}^{\mathbf{x}, \mathbf{y}, t} \sim \hat{p}_t^0(\cdot|\mathbf{x}, \mathbf{y})$.
 - 3 Evolve particle to time S by simulating (4.5) to obtain $\hat{\mathbf{X}}_{0,S,i}^{\mathbf{x}, \mathbf{y}, t}$.
 - 4 **end**
 - 5 Compute the posterior denoiser estimate $\hat{\mathbf{D}}_m^S(t, \mathbf{x}, \mathbf{y})$ using $\{\hat{\mathbf{X}}_{0,S,i}^{\mathbf{x}, \mathbf{y}, t}\}_{i=1}^m$ via (4.6).
 - 6 Compute the posterior score estimate $\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$ using $\hat{\mathbf{D}}_m^S(t, \mathbf{x}, \mathbf{y})$ via (4.7).
 - 7 **return** $\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$
-

Log-concavity of the posterior denoising density. Recalling $\sigma_t^2 = 1 - \exp(-2t)$, we observe that σ_t^2 is sufficiently close to zero for small $t > 0$. Under this regime, the logarithmic posterior denoising density $\log p_t(\cdot|\mathbf{x}, \mathbf{y})$ in (4.3) is dominated by its quadratic component, resulting in behavior that closely resembles a unimodal Gaussian distribution. Therefore, the posterior denoising density $p_t(\cdot|\mathbf{x}, \mathbf{y})$ is easy to sample when $t > 0$ is small. Before formalizing this observation rigorously, we introduce the following assumption.

Assumption 1 (Semi-log-concavity of the posterior). For a fixed measurement $\mathbf{y} \in \mathbb{R}^n$, $\log q_0(\cdot|\mathbf{y}) \in C^2(\mathbb{R}^d)$, and there exist a constant $\alpha > 0$, such that $-\nabla^2 \log q_0(\mathbf{x}_0|\mathbf{y}) + \alpha \mathbf{I}_d \succeq \mathbf{0}$ for each $\mathbf{x}_0 \in \mathbb{R}^d$.

The semi-log-concavity condition in Assumption 1, while imposing some regularity conditions on the log-density, still allows complex multi-modal structures in the posterior distribution.

According to the Bayes' rule, Assumption 1 is equivalent to:

$$-\nabla^2 \log \pi_0(\mathbf{x}_0) + \nabla^2 \ell_{\mathbf{y}}(\mathbf{x}_0) + \alpha \mathbf{I}_d \succeq \mathbf{0}, \quad \mathbf{x}_0 \in \mathbb{R}^d.$$

This condition holds, for example, if the Hessian of the negative log-likelihood $\nabla^2 \ell_{\mathbf{y}}$ is uniformly bounded below, and the Hessian of the negative log-prior $-\nabla^2 \log \pi_0$ is uniformly bounded below. We provide two concrete examples focusing on the likelihood component.

Example 4.2 (Linear forward operator with Gaussian noise). Consider a linear forward model with Gaussian noise: $\mathbf{Y} = \mathbf{A}\mathbf{X}_0 + \mathbf{n}$, where $\mathbf{n} \sim N(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{A} \in \mathbb{R}^{n \times d}$. The Hessian of the negative log-likelihood is constant: $\nabla^2 \ell_{\mathbf{y}} = \mathbf{A}^\top \mathbf{A}$. As this matrix is positive semi-definite, it is bounded below. Specifically, $\nabla^2 \ell_{\mathbf{y}}(\mathbf{x}_0) - \lambda_{\min}(\mathbf{A}^\top \mathbf{A}) \mathbf{I}_d \succeq \mathbf{0}$, where $\lambda_{\min}(\cdot)$ denotes the minimum eigenvalue.

The assumption can also hold for more general nonlinear forward operators with Gaussian noise.

Example 4.3 (General forward operator with Gaussian noise). Consider a general forward model with Gaussian noise: $\mathbf{Y} = \mathcal{F}(\mathbf{X}_0) + \mathbf{n}$ with $\mathbf{n} \sim N(\mathbf{0}, \mathbf{I}_n)$. The Hessian of the negative log-likelihood is given as $\nabla^2 \ell_{\mathbf{y}}(\mathbf{x}_0) = \nabla \mathcal{F}(\mathbf{x}_0)^\top \nabla \mathcal{F}(\mathbf{x}_0) - (\mathbf{y} - \mathcal{F}(\mathbf{x}_0)) \nabla^2 \mathcal{F}(\mathbf{x}_0)$, where $\nabla^2 \mathcal{F}$ and $\nabla \mathcal{F}$ is the Hessian and Jacobian of $\mathcal{F}$, respectively. If $\mathcal{F}$, its Jacobian $\nabla \mathcal{F}$, and its Hessians $\nabla^2 \mathcal{F}$ are uniformly bounded in norm, then $\nabla^2 \ell_{\mathbf{y}}$ is also uniformly bounded both above and below. This condition is met by many highly nonlinear operators. For example, a deep neural network with tanh or sigmoid activation functions has uniformly bounded function values and derivatives of all orders. Thus, its associated negative log-likelihood Hessian is bounded.

We next consider the concrete examples of prior distributions. The first example is the Gaussian mixture. Gaussian mixtures are foundational in statistics, physics, and machine learning, largely due to their capability as universal approximators, i.e., any continuous density can be arbitrarily well-approximated by a Gaussian mixture.

Example 4.4 (Gaussian mixture). A mixture of Gaussian distributions $\{N(\mathbf{m}_k, \sigma_k^2 \mathbf{I}_d)\}_{k=1}^K$ is defined as:

$$(4.8) \quad \pi_0(\mathbf{x}_0) = \sum_{k=1}^K w_k \gamma_{d, \sigma_k^2}(\mathbf{x}_0 - \mathbf{m}_k),$$

where γ_d is the density of a d -dimensional standard Gaussian distribution, and the weights $w_k > 0$ satisfying $\sum_{k=1}^K w_k = 1$. It is well-known that Gaussian mixture can be multi-modal and thus are generally not log-concave. On the other hand, the eigenvalues of the Hessian of the log-density $\nabla^2 \log \pi_0$ are uniformly upper and lower bounded. Consequently, π_0 satisfies the semi-log-concavity condition in Assumption 1. A detailed derivation can be found in Appendix C.1.

To provide further intuition, we consider another concrete example considered in several related studies (Chen et al., 2021; Saremi et al., 2024; Grenioux et al., 2024; Ding et al., 2024a; Beyler and Bach, 2025). This scenario is particularly relevant in computer vision and image processing applications, where pixel values are inherently bounded and images are typically perturbed by small Gaussian noise.

Example 4.5 (Gaussian convolution). Let ν be a probability distribution on $\mathbb{R}^d$ with $\text{supp}(\nu) \subseteq [0, 1]^d$. Suppose that the prior density π_0 is a Gaussian convolution of ν , that is, there exists a constant $\sigma > 0$ such that

$$(4.9) \quad \pi_0(\mathbf{x}_0) = \int \gamma_{d, \sigma^2}(\mathbf{x}_0 - \mathbf{z}) \, d\nu(\mathbf{z}).$$

As demonstrated by Chen et al. (2021, Corollary 1), this prior density has sub-Gaussian tails. This setting (4.9) includes non-log-concave distributions, as ν can be an arbitrary multi-modal distribution and $\sigma > 0$ can be sufficiently small. As shown by Ding et al. (2024a, Proposition 3.5), the eigenvalues of the Hessian of the log-density $\nabla^2 \log \pi_0$, are uniformly bounded above and below. Hence, π_0 satisfies the semi-log-concavity condition in Assumption 1.

The following lemma establishes that the posterior denoising density $p_t(\cdot | \mathbf{x}, \mathbf{y})$ (4.3) exhibits log-concavity when $t > 0$ is sufficiently small. This log-concavity property is crucial because it ensures the convergence of Langevin dynamics (4.4), as shown by Bakry et al. (2014); Dalalyan (2017); Vempala and Wibisono (2019); Chewi et al. (2024).

Lemma 4.6 (Log-concavity of RGO). *Suppose Assumption 1 holds. For each $t \in (0, T)$,*

$$\left(\frac{\mu_t^2}{\sigma_t^2} - \alpha \right) \mathbf{I}_d \preceq -\nabla_{\mathbf{x}_0}^2 \log p_t(\mathbf{x}_0 | \mathbf{x}, \mathbf{y}), \quad (\mathbf{x}_0, \mathbf{x}, \mathbf{y}) \in \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^n.$$

Further, the posterior denoising density is log-concave provided that $t < \bar{t} := \frac{1}{2} \log(1 + \alpha^{-1})$.

The proof of Lemma 4.6 is deferred to Appendix D.2. Lemma 4.6 demonstrates that posterior denoising density $p_t(\cdot | \mathbf{x}, \mathbf{y})$ serves as a regularized counterpart to the original posterior density $q_0(\cdot | \mathbf{y})$, whereby the former can achieve log-concavity even when the latter is multi-modal.

Remark 4.7 (Choice of the terminal time of the diffusion model, I). Lemma 4.6 demonstrates that the posterior denoising density $p_t(\cdot | \mathbf{x}, \mathbf{y})$ is log-concave for $t \in (0, \bar{t})$. This log-concavity, in turn, guarantees the convergence of the Langevin dynamics (4.4) for the same time interval. The theoretical guarantees for the posterior score estimation (4.7) are provided in Lemma 5.5. Consequently, to ensure the validity of our posterior score estimation throughout the time-reversal process, the terminal time T must satisfy $T < \bar{t}$.

4.2 Warm-start strategy. This subsection presents a method for sampling from the terminal posterior density $q_T(\cdot | \mathbf{y})$, which serves as a warm-start procedure for the posterior time-reversal process (2.7).

Observe that the marginal density $q_T(\cdot | \mathbf{y})$ (2.5) approaches the standard Gaussian distribution in the limit of large T . Exploiting this asymptotic behavior, the prevailing methods in diffusion-based generative modeling (Ho et al., 2020; Song et al., 2021) and sampling (Huang et al., 2024a,b; He et al., 2024) involve choosing sufficiently large terminal time T and directly substituting the standard Gaussian density as a surrogate for the terminal posterior density $q_T(\cdot | \mathbf{y})$. This Gaussian approximation strategy, while computationally convenient, becomes

problematic in our work since the terminal time T should not be too large, as highlighted in Remark 4.7.

To address this limitation, we propose a Langevin dynamics to sample directly from the terminal posterior distribution without relying on Gaussian approximations. Specifically, we consider the Langevin dynamics with invariant density $q_T(\cdot|\mathbf{y})$:

$$(4.10) \quad d\mathbf{X}_{T,u}^{\mathbf{y}} = \nabla \log q_T(\mathbf{X}_{T,u}^{\mathbf{y}}|\mathbf{y}) du + \sqrt{2} d\mathbf{B}_u, \quad u \in (0, U),$$

where $U > 0$ is the simulation horizon, and $(\mathbf{B}_u)_{u \geq 0}$ is a d -dimensional Brownian motion. Denote the marginal density of $\mathbf{X}_{T,u}^{\mathbf{y}}$ by $q_T^u(\cdot|\mathbf{y})$. Since the terminal posterior score function $\nabla \log q_T(\cdot|\mathbf{y})$ cannot be computed analytically, we approximate it using the estimator $\hat{\mathbf{s}}_m^S(T, \cdot, \mathbf{y})$ developed in (4.7). Substituting this approximation into equation (4.10) results in the following approximate Langevin dynamics:

$$(4.11) \quad d\hat{\mathbf{X}}_{T,u}^{\mathbf{y}} = \hat{\mathbf{s}}_m^S(T, \hat{\mathbf{X}}_{T,u}^{\mathbf{y}}, \mathbf{y}) du + \sqrt{2} d\mathbf{B}_u, \quad u \in (0, U).$$

We denote the marginal density of $\hat{\mathbf{X}}_{T,u}^{\mathbf{y}}$ by $\hat{q}_T^u(\cdot|\mathbf{y})$. Algorithm 2 provides a complete description of the sampling procedure for the terminal posterior distribution.

Algorithm 2: Warm-start for diffusion posterior sampling

Input: Measurement $\mathbf{y}$, simulation horizon U , terminal density $\hat{q}_T^0(\cdot|\mathbf{y})$ for Langevin dynamics.

Output: Sample $\hat{\mathbf{X}}_{T,U}^{\mathbf{y}}$ approximately distributed according to $q_T(\cdot|\mathbf{y})$.

- 1 Initialize particle by $\hat{\mathbf{X}}_{T,0}^{\mathbf{y}} \sim \hat{q}_T^0(\cdot|\mathbf{y})$.
 - 2 Simulate the approximate Langevin dynamics from time 0 to U using the estimated score function $\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$ computed via Algorithm 1 to obtain $\hat{\mathbf{X}}_{T,U}^{\mathbf{y}}$.
 - 3 **return** $\hat{\mathbf{X}}_{T,U}^{\mathbf{y}}$
-

Log-Sobolev inequality. The preceding warm-start strategy is based on the Langevin dynamics (4.10). According to Bakry et al. (2014), the convergence of (4.10) can be guaranteed, provided that $q_T(\cdot|\mathbf{y})$ satisfies the log-concavity inequality (LSI).

As established in Bakry et al. (2014), the marginal density $q_T(\cdot|\mathbf{y})$ converges exponentially to the standard Gaussian distribution $N(\mathbf{0}, \mathbf{I}_d)$, which satisfies the log-Sobolev inequality. This convergence behavior suggests that there exists a threshold $\underline{t} > 0$ such that $q_T(\cdot|\mathbf{y})$ satisfies the log-Sobolev inequality for all $T > \underline{t}$.

To establish this result rigorously, we first introduce the following assumption regarding the concentration of the prior density π_0 .

Assumption 2 (Sub-Gaussian tails of the prior). The log-prior density $\log \pi_0 \in C^2(\mathbb{R}^d)$, and there exist constants $V_{\text{SG}} > 0$ and $C_{\text{SG}} > 0$, such that

$$\int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) \pi_0(\mathbf{x}_0) d\mathbf{x}_0 \leq C_{\text{SG}}.$$

Assumption 2 proposes mild conditions on the prior distribution. These conditions do not necessitate log-concavity and are weaker than the log-Sobolev inequality (Villani, 2003, Theorem 9.9).

Example 4.8 (Gaussian mixture). Gaussian mixture (4.8) mentioned in Example 4.4 satisfies Assumption 2, provided that $V_{\text{SG}}^2 > 2\sigma_k^2$ for each $1 \leq k \leq K$. A detailed derivation can be found in Appendix C.1.

Example 4.9 (Gaussian convolution). Gaussian convolution (4.9) in Example 4.5 has sub-Gaussian tails provided $V_{\text{SG}}^2 > 2\sigma^2$. A detailed derivation can be found in Appendix C.2.

Assumption 2 implies the sub-Gaussian tails of the posterior, as stated by the following lemma.

Lemma 4.10 (Sub-Gaussian tails of the posterior). *Suppose Assumption 2 holds. Then the posterior density $q_0(\cdot|\mathbf{y})$ also has sub-Gaussian tails, that is,*

$$\int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) q_0(\mathbf{x}_0|\mathbf{y}) d\mathbf{x}_0 \leq \kappa_{\mathbf{y}} C_{\text{SG}},$$

where $\kappa_{\mathbf{y}}$ is the condition number defined as:

$$(4.12) \quad \kappa_{\mathbf{y}} := \frac{\sup_{\mathbf{x}_0} \exp(-\ell_{\mathbf{y}}(\mathbf{x}_0))}{\int \exp(-\ell_{\mathbf{y}}(\mathbf{x}_0)) \pi_0(\mathbf{x}_0) d\mathbf{x}_0} < \infty.$$

The proof of Lemma 4.10 is deferred to Appendix D.3. A distribution with Gaussian tails are not necessarily log-concave and may not satisfy the log-Sobolev inequality (Villani, 2003, Theorem 9.9). In general, we cannot guarantee the fast convergence of the Langevin dynamics with the invariant density $q_0(\cdot|\mathbf{y})$ under this assumption.

Remark 4.11 (Condition number). The condition number $\kappa_{\mathbf{y}}$ in (4.12) serves as a fundamental measure of problem difficulty in posterior sampling. As demonstrated by Purohit et al. (2025, Theorem 4.1), this quantity reveals how well-posed the Bayesian inverse problem is. A moderate condition number occurs when the likelihood function $\exp(-\ell_{\mathbf{y}}(\cdot))$ is concentrated in regions where the prior density π_0 is also high. In this scenario, the denominator of (4.12) stays bounded away from zero. This indicates a well-posed inverse problem where the observed data is consistent with prior beliefs. Conversely, a large condition number indicates an ill-posed problem. This happens when the likelihood function is concentrated in regions with low prior probability, causing the denominator of (4.12) to approach zero. This situation typically arises when the observed data strongly conflicts with the prior information. We will show the impact of condition number rigorously in Lemma 5.5.

The following lemma shows that when T is sufficiently large, the terminal posterior density $q_T(\cdot|\mathbf{y})$ satisfies a log-Sobolev inequality. Further, we can explicitly estimate a log-Sobolev inequality constant using Chen et al. (2021, Theorem 2).

Lemma 4.12 (Log-Sobolev inequality of the terminal posterior). *Suppose Assumption 2 holds. For each $T > \underline{t} := \frac{1}{2} \log(1 + 2V_{\text{SG}}^2)$,*

$$C_{\text{LSI}}(q_T(\cdot|\mathbf{y})) \leq 12\sigma_T^2 \exp\left(2\frac{\sigma_T^2 + 2\mu_T^2 V_{\text{SG}}^2}{\sigma_T^2 - 2\mu_T^2 V_{\text{SG}}^2} \log(\kappa_{\mathbf{y}}^2 C_{\text{SG}}^2)\right).$$

The proof of Lemma 4.12 is deferred to Appendix D.4. This lemma shows that sufficient Gaussian smoothing upgrades a distribution $q_0(\cdot|\mathbf{y})$ with sub-Gaussian tails to one that satisfies the log-Sobolev inequality $q_T(\cdot|\mathbf{y})$. This transformation is crucial because the former distribution cannot be easily sampled by Langevin dynamics, while the latter can.

Remark 4.13 (Choice of the terminal time of the diffusion model, II). The threshold condition $T > \bar{t}$ established in Lemma 4.12 is essential for the log-Sobolev inequality of the terminal posterior distribution $q_T(\cdot|\mathbf{y})$. According to Bakry et al. (2014), this guarantees the convergence of Langevin dynamics (4.10) for sampling from the terminal posterior density $q_T(\cdot|\mathbf{y})$.

Remark 4.14. Similar findings are reported by Grenioux et al. (2024); Beyler and Bach (2025). In particular, Grenioux et al. (2024, Theorem 1) show that, for sufficiently large diffusion times, the marginal density of the forward process becomes strongly log-concave and therefore satisfies a log-Sobolev inequality by the Bakry-Émery theorem (Bakry and Émery, 1985), under the assumption that the target distribution is a Gaussian convolution of a compactly supported distribution. Likewise, Beyler and Bach (2025, Lemma 3) establish a comparable result under the assumption that the target posterior density has compact support.

4.3 Posterior sampling. Utilizing the posterior score estimation approach and warm-start strategy introduced in Sections 4.1 and 4.2, respectively, we obtain an approximation to the posterior time-reversal process (2.7) given by:

$$(4.13) \quad d\hat{\mathbf{X}}_t^{\mathbf{y}} = (\hat{\mathbf{X}}_t^{\mathbf{y}} + 2\hat{\mathbf{s}}_m^S(T-t, \hat{\mathbf{X}}_t^{\mathbf{y}}, \mathbf{y})) dt + \sqrt{2} d\mathbf{B}_t, \quad \hat{\mathbf{X}}_0^{\mathbf{y}} \sim \hat{q}_T^U(\cdot|\mathbf{y}), \quad t \in (0, T - T_0),$$

where the posterior estimator $\hat{\mathbf{s}}_m^S(t, \cdot, \mathbf{y})$ is defined as (4.7), and the approximate terminal posterior density $\hat{q}_T^U(\cdot|\mathbf{y})$ is characterized by (4.11). Denote the marginal density of $\hat{\mathbf{X}}_t^{\mathbf{y}}$ by $\hat{q}_t(\cdot|\mathbf{y})$.

Remark 4.15 (Early-stopping). Here $T_0 \in (0, T)$ is the early-stopping time. As we will establish in Section 5.3, the error of $\hat{\mathbf{s}}_m^S(t, \cdot, \mathbf{y})$ diverges as t approaches zero, creating a singularity that must be avoided. We therefore terminate the process (4.13) at $T - T_0$ to maintain numerical stability. The optimal choice of the early-stopping time T_0 will be addressed in Section 5.5. The numerical instability near $t = 0$ has been observed and investigated in the literature, such as Kim et al. (2022); Pidstrigach (2022); Duong et al. (2025).

The complete procedure of our posterior sampling method is summarized in Algorithm 3.

Algorithm 3: Diffusion-based posterior sampling

Input: Measurement $\mathbf{y}$, diffusion terminal time T , early-stopping time T_0 .

Output: Posterior sample $\hat{\mathbf{X}}_{T-T_0}^{\mathbf{y}}$ approximately drawn from $q_0(\cdot|\mathbf{y})$.

- 1 **Warm-start:** Set the initial particle $\hat{\mathbf{X}}_0^{\mathbf{y}} = \hat{\mathbf{X}}_{T,U}^{\mathbf{y}}$, where the warm-start sample $\hat{\mathbf{X}}_{T,U}^{\mathbf{y}}$ is obtained using Algorithm 2.
 - 2 **Reverse diffusion:** Simulate the approximate time-reversal stochastic process defined in (4.13) over the time interval $[0, T - T_0]$, employing the estimated score function $\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$ computed via Algorithm 1, to generate $\hat{\mathbf{X}}_{T-T_0}^{\mathbf{y}}$.
 - 3 **return** $\hat{\mathbf{X}}_{T-T_0}^{\mathbf{y}}$
-

Duality of the convergence. As discussed in Remark 4.7, the terminal time T must satisfy $T < \bar{t}$ to ensure convergence of the posterior score estimator $\hat{\mathbf{s}}_m^S(t, \cdot, \mathbf{y})$ for all $t \in (0, T)$. Conversely, as established in Remark 4.13, the terminal time T must also satisfy $T > \bar{t}$ to guarantee reliable convergence when sampling from the terminal posterior density $q_T(\cdot|\mathbf{y})$.

This dual constraint naturally gives rise to a fundamental question:

Does there exist a terminal time T such that both the posterior score estimation and the terminal posterior sampling procedures are theoretically guaranteed to converge?

The following theorem provides an affirmative answer to this question.

Theorem 4.16. *Suppose Assumptions 1 and 2 hold. If $2\alpha V_{\text{SG}}^2 \leq 1$, then there exists a terminal time $T > 0$ such that:*

- (i) *the posterior denoising density $p_t(\cdot|\mathbf{x}, \mathbf{y})$ is log-concave for all $t \in (0, T)$; and*
- (ii) *the terminal posterior density $q_T(\cdot|\mathbf{y})$ satisfies a log-Sobolev inequality.*

The proof of Theorem 4.16 is deferred to Appendix D.5. Under mild assumptions on the target posterior distribution, this theorem establishes the existence of a terminal time T that simultaneously guarantees convergence of both the posterior score estimation and sampling from the target posterior distribution, as illustrated in Figure 1.

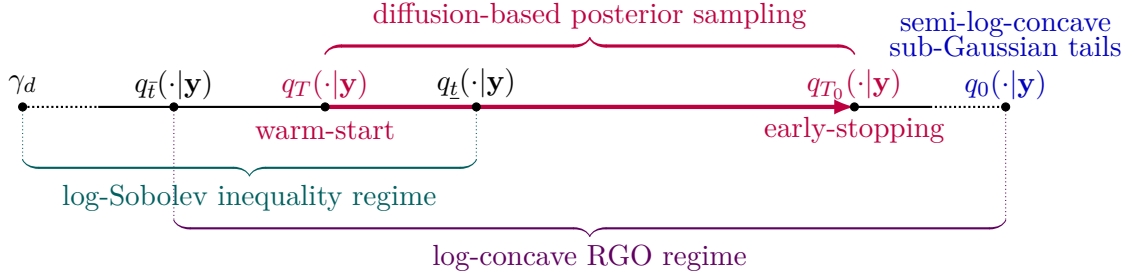


Figure 1: An illustrative schematic of the duality of convergence. Here, γ_d denotes the density of the d -dimensional standard Gaussian distribution. (i) The log-Sobolev inequality regime corresponds to sufficiently large time values $t \in (\bar{t}, +\infty)$, where the posterior density $q_t(\cdot|\mathbf{y})$ satisfies log-Sobolev inequality as established in Lemma 4.12. (ii) The log-concave RGO regime encompasses sufficiently small time values $t \in (0, \bar{t})$, where the posterior denoising density $p_t(\cdot|\mathbf{x}, \mathbf{y})$ is log-concave, thereby ensuring convergence of the score estimator $\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$ as shown in Lemma 4.6. (iii) Theorem 4.16 establishes the existence of an overlap between these regimes under mild assumptions, demonstrating that there exists a terminal time T such that T lies within the log-Sobolev inequality regime while the interval $(0, T)$ remains entirely contained in the log-concave RGO regime.

Discussions of related work. While prior studies have applied Monte Carlo score estimation to diffusion-based unconditional sampling (Huang et al., 2024a,b; He et al., 2024; Grenioux et al., 2024), to our knowledge, this work is the first to extend this approach to diffusion-based posterior sampling. Furthermore, our method differs fundamentally from these existing approaches in two key respects:

- (i) **Warm-start strategy.** Previous works (Huang et al., 2024a,b; He et al., 2024) approximate the terminal density of the time-reversal process using a standard Gaussian. Applying similar reasoning to the posterior context, one would approximate the terminal posterior density $q_T(\cdot|\mathbf{y})$ in (2.7) with a standard Gaussian. While this approximation is effective when the terminal time T is sufficiently large, such a requirement has a significant drawback: large T values can cause the denoising density $p_t(\cdot|\mathbf{x}, \mathbf{y})$ to lose log-concavity for $t \in (\bar{t}, T)$ (see Remark 4.7). This loss of log-concavity undermines convergence guarantees for sampling algorithms. In contrast, inspired by Grenioux et al. (2024), our work proposes a warm-start strategy (Section 4.2) to sample from the terminal posterior density $q_T(\cdot|\mathbf{y})$ without relying on this Gaussian approximation. Crucially, this approach achieves convergence only requiring a smaller T , which in

turn allows us to establish convergence guarantees for sampling from $p_t(\cdot|\mathbf{x}, \mathbf{y})$ for all $t \in (0, T)$.

- (ii) **Duality of the convergence.** The duality of log-concavity in stochastic localization has been established by Grenioux et al. (2024, Theorem 3) under a Gaussian convolution condition (Grenioux et al., 2024, Assumption A0). While this assumption is mild and permits multi-modal target distributions, it is impractical in the context of posterior sampling. Specifically, it is unclear what conditions on the prior and likelihood would be necessary to satisfy this Gaussian convolution assumption for the resulting posterior. In contrast, our Theorem 4.16 is established under assumptions on the prior and likelihood that are more transparent and practical. Our framework relies on semi-log-concavity (Assumption 1) and sub-Gaussian tails (Assumption 2), which are directly verifiable conditions. This accessibility makes our framework more readily applicable to practical posterior sampling scenarios.

5 Convergence Guarantee

In this section, we establish a rigorous convergence analysis for the diffusion-based posterior sampling method (4.13). Recall that our ultimate objective is to simulate the time-reversal process (2.7):

$$d\bar{\mathbf{X}}_t^{\mathbf{y}} = (\bar{\mathbf{X}}_t^{\mathbf{y}} + 2\nabla_{\mathbf{x}} \log q_{T-t}(\bar{\mathbf{X}}_t^{\mathbf{y}}|\mathbf{y})) dt + \sqrt{2} d\mathbf{B}_t, \quad \bar{\mathbf{X}}_0^{\mathbf{y}} \sim q_T(\cdot|\mathbf{y}), \quad t \in (0, T).$$

In practical implementations, we introduce three key modifications to make the algorithm computationally tractable. First, we terminate the reverse diffusion process early at time $T - T_0$ rather than running it for the full terminal time T , where $T_0 > 0$ represents the early-stopping time. Second, we replace the intractable true posterior score function $\nabla \log q_t(\cdot|\mathbf{y})$ with an estimator $\hat{\mathbf{s}}_m^S(t, \cdot, \mathbf{y})$ for $t \in (T_0, T)$ (4.7). Third, we approximate the terminal distribution $q_T(\cdot|\mathbf{y})$ using a warm-start strategy, yielding the surrogate terminal distribution $\hat{q}_T^U(\cdot|\mathbf{y})$ (4.11). These practical considerations lead to the implementable approximate process (4.13):

$$d\hat{\mathbf{X}}_t^{\mathbf{y}} = (\hat{\mathbf{X}}_t^{\mathbf{y}} + 2\hat{\mathbf{s}}_m^S(T - t, \hat{\mathbf{X}}_t^{\mathbf{y}}, \mathbf{y})) dt + \sqrt{2} d\mathbf{B}_t, \quad \hat{\mathbf{X}}_0^{\mathbf{y}} \sim \hat{q}_T^U(\cdot|\mathbf{y}), \quad t \in (0, T - T_0).$$

Let $\hat{q}_{T-T_0}(\cdot|\mathbf{y})$ denote the marginal probability density of $\hat{\mathbf{X}}_{T-T_0}^{\mathbf{y}}$ generated by this approximate process. Before the theoretical analysis, we need some additional operations to the particles generated by (4.13).

Truncation. For some truncation radius $R > 0$, which we will specify later, we define a truncation operator $\mathcal{T}_R$:

$$\mathcal{T}_R : \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad \mathbf{x} \mapsto \mathcal{T}_R \mathbf{x} := \mathbf{x} \mathbb{1}\{\|\mathbf{x}\|_2 \leq R\}.$$

This operator maps any particle $\mathbf{x}$ outside the ball of radius R centered at the origin, $BR(\mathbf{0})$, to zero. Consequently, if a distribution p is concentrated around the origin and has light tails, the resulting push-forward distribution $\mathcal{T}_R \# p$ provides a close approximation of p for a sufficiently large R . Define the truncated estimated density $\hat{q}_{T-T_0}^R(\cdot|\mathbf{y}) := \mathcal{T}_R \# \hat{q}_{T-T_0}(\cdot|\mathbf{y})$.

Scaling transformation. To bridge the gap arising from the early-stopping, we introduce a scaling transformation. We define the scaling operator $\mathcal{M}(\xi^{-1}) : \mathbf{x} \mapsto \xi^{-1} \mathbf{x}$ for $\xi \in \mathbb{R}$, with the induced pushforward measure $\mathcal{M}(\xi^{-1}) \# p(\mathbf{x}) = \xi^d p(\xi \mathbf{x})$ for any probability density p on $\mathbb{R}^d$. According to (2.2), we find $\mathbf{X}_{T_0}^{\mathbf{y}} \stackrel{d}{=} \mu_{T_0} \mathbf{X}_0^{\mathbf{y}} + \sigma_{T_0} \mathbf{W}$, where $\mathbf{W} \sim N(\mathbf{0}, \mathbf{I}_d)$ is independent of $\mathbf{X}_0^{\mathbf{y}}$. Therefore, $\mathbb{E}[\mathbf{X}_{T_0}^{\mathbf{y}}] = \mu_{T_0} \mathbb{E}[\mathbf{X}_0^{\mathbf{y}}]$. To eliminate this mean mismatch between the

early-stopped process and the target density, we consider the scaled and truncated density $\mathcal{M}(\mu_{T_0}^{-1})\sharp\hat{q}_{T-T_0}^R(\cdot|\mathbf{y})$ as an approximation to the target posterior density $q_0(\cdot|\mathbf{y})$, following the choice of Lee et al. (2023).

5.1 Additional assumptions. In addition to Assumptions 1 and 2, our analysis requires further assumptions.

Assumption 3 (Polynomial growth bounds of prior score). There exist constants $r \geq 1$ and $B > 1$ such that for each $\mathbf{x}_0 \in \mathbb{R}^d$,

$$\|\nabla \log \pi_0(\mathbf{x}_0)\|_2 \leq B(1 + \|\mathbf{x}_0\|_2^r).$$

Assumption 3 restricts the prior score, bounding their growth to be polynomial, which is a crucial condition for establishing the posterior score estimation error bound presented in Lemma 5.5.

The norm of the prior score $\|\nabla \log \pi_0(\mathbf{x}_0)\|_2$ quantifies the “steepness” of the log-prior landscape $\log \pi_0$ at the point $\mathbf{x}_0$. This assumption requires that the log-prior $\log \pi_0(\mathbf{x}_0)$ decays no faster than a polynomial of degree $r + 1$. This implies that the prior density π_0 itself cannot decay faster than a distribution like $\pi(\mathbf{x}_0) \propto \exp(-c\|\mathbf{x}_0\|_2^{r+1})$ for some constant $c > 0$.

Considered together, Assumptions 2 and 3 place complementary constraints on the prior’s tail behavior. While Assumption 2 ensures the prior’s tails are not too heavy, Assumption 3 ensures they are not too light. In the specific case where $r = 1$, Assumption 3 requires the prior tails decay no faster than a Gaussian. Meanwhile, Assumption 2 requires the tails to decay no slower than a Gaussian. Therefore, a distribution satisfying both assumptions must have tails that behave almost exactly like a Gaussian, i.e., decaying as $\exp(-c\|\mathbf{x}_0\|_2^2)$ for some constant c .

Example 5.1 (Gaussian mixture). Gaussian mixture (4.8) in Example 4.4 satisfies Assumption 3 with $r = 1$. See Appendix C.1 for details.

Example 5.2 (Gaussian convolution). Gaussian convolution (4.9) in Example 4.5 satisfies Assumption 3 with $r = 1$, as demonstrated by Ding et al. (2024a, Proposition 3.2).

Assumption 4. For a fixed measurement $\mathbf{y} \in \mathbb{R}^n$, there exists a constant $H_{\mathbf{y}} > 1$, such that $\|\nabla \log q_0(\mathbf{0}|\mathbf{y})\|_2 \leq H_{\mathbf{y}}$.

Assumption 4 requires the posterior score to be bounded at the origin. This is a direct consequence of assuming $\log q_0(\cdot|\mathbf{y}) \in C^2(\mathbb{R}^d)$ (see Assumptions 1 and 2), as a twice-continuously differentiable function will have a continuous gradient, which is necessarily bounded at the specific point $\mathbf{x}_0 = \mathbf{0}$. This assumption is essential in establishing the bounds of the terminal posterior score $\nabla \log q_T(\cdot|\mathbf{y})$ in Lemma 5.8.

Finally, we introduce two assumptions on the prior score estimator $\hat{\mathbf{s}}_{\text{prior}}$, which are used in the proof of Lemma 5.5.

Assumption 5 (Prior score matching error). Let $\hat{\mathbf{s}}_{\text{prior}}$ be an estimator of the prior score function $\nabla \log \pi_0$. There exists $\varepsilon_{\text{prior}} \in (0, 1)$, such that

$$\int \|\nabla \log \pi_0(\mathbf{x}_0) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2^2 \pi_0(\mathbf{x}_0) d\mathbf{x}_0 \leq \varepsilon_{\text{prior}}^2.$$

Assumption 6. For a fixed measurement $\mathbf{y} \in \mathbb{R}^n$, there exist constants $G > 1$, such that $\hat{\mathbf{s}}_{\text{prior}} - \nabla \ell_{\mathbf{y}}$ is G -Lipschitz.

Assumption 5 requires that the L^2 -error of the prior score estimator $\hat{\mathbf{s}}_{\text{prior}}$ be sufficiently small, where the error is measured with respect to the prior density π_0 . In our setting, numerous samples drawn from π_0 are available to estimate this score function. Methods such as implicit score matching (Hyvärinen, 2005), sliced score matching (Song et al., 2020), or denoising score matching (Vincent, 2011) can be used to obtain an estimator $\hat{\mathbf{s}}_{\text{prior}}$ with theoretical guarantees in this metric. While this assumption could be replaced by explicit score matching bounds (Tang and Yang, 2024; Oko et al., 2023; Fu et al., 2024; Ding et al., 2024a) obtained by some standard techniques in non-parametric regression using deep neural networks (Bauer and Kohler, 2019; Schmidt-Hieber, 2020; Kohler and Langer, 2021; Jiao et al., 2023), we maintain this formulation for the clarity of presentation.

Assumption 6 guarantees uniform Lipschitz continuity, which is also used in the previous work on error analysis for posterior sampling, such as Laumont et al. (2022, H1) and Cai et al. (2024, Assumptions 3.6 and 3.8). The Lipschitz continuity is essential for controlling the Wasserstein error of the Langevin dynamics in Lemma 5.5. In general, the convergence of diffusion or flow-based generative models in Wasserstein distance typically requires the Lipschitz assumption. See Kwon et al. (2022); Chen et al. (2023b); Huang et al. (2025a); Beyler and Bach (2025); Kremling et al. (2025) for example.

A sufficient condition for Assumption 6 is that both the prior score estimator $\hat{\mathbf{s}}_{\text{prior}}$ and the gradient of the negative log-likelihood $\nabla \ell_{\mathbf{y}}$ are individually uniformly Lipschitz continuous.

- (i) For the likelihood term, this condition is met in our previous examples. In both Example 4.2 (linear forward operator) and Example 4.3 (general forward operator), the Hessian $\nabla^2 \ell_{\mathbf{y}}$ is uniformly bounded. A bounded Hessian implies that the gradient $\nabla \ell_{\mathbf{y}}$ is Lipschitz continuous.
- (ii) Next, for the prior score estimator $\hat{\mathbf{s}}_{\text{prior}}$, it is often sufficient to require that the true prior score $\nabla \log \pi_0$ is uniformly Lipschitz. One can then employ an estimation method that finds an estimator $\hat{\mathbf{s}}_{\text{prior}}$ that is both close in the L^2 metric (to satisfy Assumption 5) and simultaneously preserves the Lipschitz property (to satisfy Assumption 6). As shown in Examples 4.4 and 4.5, the scores of a Gaussian mixture (4.8) and a Gaussian convolution (4.9) are indeed uniformly Lipschitz.

Remark 5.3. In practice, estimators are often parameterized by deep neural networks. Various techniques to enforce the Lipschitz constraint on such networks have been proposed, including weight clipping (Arjovsky et al., 2017), gradient penalty (Gulrajani et al., 2017), and spectral normalization (Miyato et al., 2018). From a theoretical standpoint, the approximation properties of deep neural networks with Lipschitz constraints have been studied by Chen et al. (2022); Huang et al. (2022); Jiao et al. (2023); Ding et al. (2025). Furthermore, Ding et al. (2025, Theorem IV.3) proposes a Sobolev-regularized framework that incorporates Lipschitz constraints on the estimator in the context of least-squares regression.

5.2 Error decomposition. The following lemma decomposes the 2-Wasserstein error of the estimated posterior density into the early-stopping error, the posterior score estimation error, and the warm-start error.

Lemma 5.4 (Error decomposition). *Suppose Assumption 2 hold.*

- (i) Let $T_0 \in (0, \frac{1}{2})$ and $R^2 = (4\mu_{T_0}^2 V_{\text{SG}}^2 + 16\sigma_{T_0}^2) \log(\kappa_{\mathbf{y}} \varepsilon_{T_0}^{-1})$. For $\varepsilon_{T_0} \in (0, 1)$, we have

$$\mathbb{W}_2^2(q_0(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1}) \# \hat{q}_{T-T_0}^R(\cdot|\mathbf{y})) \leq C \left\{ \frac{\sigma_{T_0}^2}{\mu_{T_0}^2} + \varepsilon_{T_0} \log \left(\frac{\kappa_{\mathbf{y}}}{\varepsilon_{T_0}} \right) \right\},$$

provided that

$$\|q_{T_0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0}(\cdot|\mathbf{y})\|_{\text{TV}}^2 = \varepsilon_{T_0}^2.$$

Here C is a constant only depending on d , V_{SG} , and C_{SG} .

(ii) For $0 < T_0 < T$, it holds that

$$\begin{aligned} & \|q_{T_0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0}(\cdot|\mathbf{y})\|_{\text{TV}}^2 \\ & \leq \underbrace{\int_{T_0}^T \mathbb{E}[\|\nabla \log q_t(\mathbf{X}_t^{\mathbf{y}}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(t, \mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|_2^2] dt}_{\text{posterior score estimation}} + \underbrace{2\|q_T(\cdot|\mathbf{y}) - \hat{q}_0^U(\cdot|\mathbf{y})\|_{\text{TV}}^2}_{\text{warm-start}}, \end{aligned}$$

where the expectation is taken with respect to $\mathbf{X}_t^{\mathbf{y}} \sim q_t(\cdot|\mathbf{y})$.

The proof of Lemma 5.4 can be found in Appendix E. Lemma 5.4 decomposes the posterior sampling error into three distinct components:

- (i) The **posterior score estimation error** quantifies the discrepancy between the true posterior score $\nabla \log q_t(\cdot|\mathbf{y})$ and its Monte Carlo estimator $\hat{\mathbf{s}}_m^S(t, \cdot, \mathbf{y})$ defined in (4.7). Section 5.3 provides a detailed convergence analysis, demonstrating that this error increases as the early-stopping time T_0 approaches zero.
- (ii) The **warm-start error** measures the difference between the true terminal posterior density $q_T(\cdot|\mathbf{y})$ and its estimator $\hat{q}_T^U(\cdot|\mathbf{y})$ obtained through the warm-start procedure (4.11). This error is analyzed in Section 5.4.
- (iii) The **early-stopping error** $\mathcal{O}(\sigma_{T_0}^2)$ captures the error introduced by terminating the reverse process at time $T - T_0$ rather than T . This error decreases as T_0 approaches zero, creating a fundamental trade-off with the posterior score estimation error. The optimal selection of T_0 to balance these competing effects is addressed in Section 5.5.

5.3 Error of posterior score estimation. In this subsection, we provide an error bound for the posterior score estimation (4.7).

Lemma 5.5 (Error bound of the posterior score estimation). *Suppose Assumptions 1, 2, 3, 5, and 6 hold. Let $0 < T_0 < T < \frac{1}{2} \log(1 + \alpha^{-1})$. For each time $t \in (T_0, T)$,*

$$\begin{aligned} & \mathbb{E}[\|\nabla \log q_t(\mathbf{X}_t^{\mathbf{y}}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(t, \mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|_2^2] \\ & \leq \underbrace{C \frac{\mu_{T_0}^2}{\sigma_{T_0}^4} \frac{\kappa_{\mathbf{y}}}{m}}_{\text{Monte Carlo}} + \underbrace{C \frac{\mu_{T_0}^2}{\sigma_{T_0}^4} \exp\left(-\frac{2(\mu_T^2 - \alpha\sigma_T^2)}{\sigma_T^2} S\right) \eta_{\mathbf{y}}^2}_{\text{convergence of Langevin dynamics}} \\ & \quad + \underbrace{C \frac{\mu_{T_0}^2}{\sigma_{T_0}^4} \left(\frac{\sigma_T^2 \eta_{\mathbf{y}}^2}{\mu_T^2 - \alpha\sigma_T^2} + S\right)^{\frac{1}{2}} \exp\left(2\left(G + \frac{\mu_{T_0}^2}{\sigma_{T_0}^2}\right) S\right) \kappa_{\mathbf{y}}^{\frac{1}{2}} \varepsilon_{\text{prior}}^{\frac{1}{2}}}_{\text{prior score estimation error}}, \end{aligned}$$

where S is the simulation horizon of (4.5), and C is a constant only depending on d , B , V_{SG} , and C_{SG} . The condition number $\kappa_{\mathbf{y}}$ is defined as (4.12), and the initial discrepancy $\eta_{\mathbf{y}}$ is defined as

$$\eta_{\mathbf{y}}^2 := \sup_{t \in (T_0, T)} \max \left\{ \mathbb{E}[\chi^2(\hat{p}_t^0(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}) \| p_t(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))], \mathbb{E}[\mathbf{W}_2^2(\hat{p}_t^0(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}), p_t(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))] \right\}.$$

Here the expectation is taken with respect to $\mathbf{X}_t^{\mathbf{y}} \sim q_t(\cdot|\mathbf{y})$ and $\hat{\mathbf{X}}_{0,S,1}^{\mathbf{x},\mathbf{y},t}, \dots, \hat{\mathbf{X}}_{0,S,m}^{\mathbf{x},\mathbf{y},t} \sim \hat{q}_t^S(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y})$.

The proof of Lemma 5.5 is provided in Appendix F. This result demonstrates that the posterior score estimation error decomposes into three fundamental components: the Monte Carlo approximation error (4.6), the Langevin dynamics convergence error (4.4), and the prior score estimation error (Assumption 5). Specifically, the Monte Carlo error decreases at a rate of $\mathcal{O}(m^{-1})$, where m denotes the number of particles. The Langevin dynamics error decreases exponentially with respect to the simulation horizon S . Conversely, the prior score error component grows with S . This creates an inherent trade-off in selecting S : a larger simulation horizon improves convergence but amplifies the impact of the prior score error. More importantly, the bound shows that the condition number $\kappa_{\mathbf{y}}$ also amplifies the impact of the prior score error, aligning with the explanation in Remark 4.11.

Lemma 5.5 provides actionable insights for hyperparameter selection in posterior score estimation. The optimal terminal time S must balance convergence gains of Langevin dynamics against prior score error amplification. The number of particles m can be chosen based on computational budget and desired accuracy, following predictable Monte Carlo scaling laws. These theoretical guidelines are formalized in the following result.

Corollary 5.6. *Under the same conditions as Lemma 5.5. Then for each $\varepsilon \in (0, 1)$, the following inequality holds*

$$\int_{T_0}^T \mathbb{E} [\|\nabla \log q_t(\mathbf{X}_t^{\mathbf{y}} | \mathbf{y}) - \hat{\mathbf{s}}_m^S(t, \mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|_2^2] dt \leq C \frac{\mu_{T_0}^2}{\sigma_{T_0}^4} \varepsilon^2,$$

provided that the terminal time S of Langevin dynamics (4.5), the prior score matching error $\varepsilon_{\text{prior}}$, and the number of Monte Carlo particles m are given, respectively, as:

$$S = \Theta\left(\frac{\sigma_T^2}{\mu_T^2 - \alpha\sigma_T^2} \log\left(\frac{T\eta_{\mathbf{y}}^2}{\varepsilon^2}\right)\right), \quad m = \Theta\left(\frac{T\kappa_{\mathbf{y}}}{\varepsilon^2}\right),$$

$$\varepsilon_{\text{prior}} = \Theta\left(\frac{\mu_T^2 - \alpha\sigma_T^2}{\sigma_T^2} \left(\eta_{\mathbf{y}}^2 + \log\left(\frac{T\eta_{\mathbf{y}}^2}{\varepsilon^2}\right)\right)^{-1} \exp\left(-4\frac{\sigma_T^2}{\sigma_{T_0}^2} \frac{\mu_{T_0}^2 + G\sigma_{T_0}^2}{\mu_T^2 - \alpha\sigma_T^2} \log\left(\frac{T\eta_{\mathbf{y}}^2}{\varepsilon^2}\right)\right) \frac{\varepsilon^4}{T^2\kappa_{\mathbf{y}}}\right).$$

Here the expectation is taken with respect to $\mathbf{X}_t^{\mathbf{y}} \sim q_t(\cdot | \mathbf{y})$ and $\hat{\mathbf{X}}_{0,S,1}^{\mathbf{x},\mathbf{y},t}, \dots, \hat{\mathbf{X}}_{0,S,m}^{\mathbf{x},\mathbf{y},t} \sim \hat{q}_t^S(\cdot | \mathbf{x}, \mathbf{y})$.

The proof of Corollary 5.6 appears in Appendix F.

Remark 5.7 (Condition number). Lemma 5.5 and Corollary 5.6 demonstrate that the condition number $\kappa_{\mathbf{y}}$ acts as an amplification factor, directly controlling how prior score estimation errors propagate to the posterior score estimation. This provides theoretical justification for the empirical observation that posterior sampling becomes increasingly challenging as the prior-likelihood mismatch grows, as discussed in Remark 4.11.

Discussions with related work. Prior research has examined error analysis for Monte Carlo-based score estimation. For example, He et al. (2024, Proposition 3.1) decomposes the estimation error into components from Langevin dynamics and Monte Carlo approximation. Lemma 5.5 extends this work in two significant ways:

- (i) In He et al. (2024), the authors assume that the error of Langevin dynamics (4.5) remains sufficiently small for all $t > 0$. This assumption proves difficult to satisfy in practice, since the inner Langevin dynamics may fail to converge when the denoising density $p_t(\cdot | \mathbf{x}, \mathbf{y})$ lacks log-concavity for large values of t . In contrast, we provide an error bound of Langevin dynamics (4.5) under some mild assumptions, and specify a time range for which the Langevin dynamics converges (4.5).

- (ii) The error bound established by He et al. (2024) is under the assumption that the score function of the target distribution is analytically available, which holds for standard unconditional sampling scenarios. However, the posterior sampling typically requires estimating the unknown prior score function from samples. Our work fills this gap by analyzing how the prior score estimation error $\varepsilon_{\text{prior}}^2$ affects the posterior sampling performance and introducing the condition number $\kappa_{\mathbf{y}}$ as a key characterization of this relationship.

5.4 Error of warm-start. This subsection establishes an error bound for the warm-start procedure (4.11).

Lemma 5.8 (Error bound of the warm-start). *Suppose Assumptions 1, 2, 3, 4, 5 and 6 hold. Assume $2\alpha V_{\text{SG}}^2 \leq 1$. Let the terminal time satisfy $T \in (\frac{1}{2} \log(1 + 2V_{\text{SG}}^2), \frac{1}{2} \log(1 + \alpha^{-1}))$. Then for $\varepsilon \in (0, 1)$, it holds that*

$$\|q_T(\cdot|\mathbf{y}) - \hat{q}_T^U(\cdot|\mathbf{y})\|_{\text{TV}}^2 \leq C\varepsilon,$$

provided that

$$U = \Theta\left(\exp\left(2\frac{\sigma_T^2 + 2\mu_T^2 V_{\text{SG}}^2}{\sigma_T^2 - 2\mu_T^2 V_{\text{SG}}^2} \log(\kappa_{\mathbf{y}}^2 C_{\text{SG}}^2)\right) \log\left(\frac{\zeta_{\mathbf{y}}^2}{\varepsilon}\right)\right),$$

$$\varepsilon_{\text{post}} = \Theta\left(\exp\left(-2\frac{\sigma_T^2 + 2\mu_T^2 V_{\text{SG}}^2}{\sigma_T^2 - 2\mu_T^2 V_{\text{SG}}^2} \log(\kappa_{\mathbf{y}}^2 C_{\text{SG}}^2)\right) \left(\zeta_{\mathbf{y}}^2 + \log\left(\frac{\zeta_{\mathbf{y}}^2}{\varepsilon}\right)\right)^{-1} (\mu_T^2 - \alpha\sigma_T^2)^3 \frac{\varepsilon^2}{\kappa_{\mathbf{y}}^{1/2}}\right),$$

where C is a constant only depending on d , B , $H_{\mathbf{y}}$, V_{SG} , and C_{SG} . The initial divergence $\zeta_{\mathbf{y}}^2$ and the posterior score matching error at the terminal time T are defined, respectively, as:

$$\zeta_{\mathbf{y}}^2 := \chi^2(\hat{q}_T^0(\cdot|\mathbf{y})\|q_T(\cdot|\mathbf{y})), \quad \varepsilon_{\text{post}}^2 := \mathbb{E}[\|\nabla \log q_T(\mathbf{X}_T^{\mathbf{y}}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T, \mathbf{X}_T^{\mathbf{y}}, \mathbf{y})\|_2^2].$$

where the expectation is taken with respect to $\mathbf{X}_t^{\mathbf{y}} \sim q_t(\cdot|\mathbf{y})$ and $\hat{\mathbf{X}}_{0,S,1}^{\mathbf{x},\mathbf{y},t}, \dots, \hat{\mathbf{X}}_{0,S,m}^{\mathbf{x},\mathbf{y},t} \sim \hat{q}_t^S(\cdot|\mathbf{x}, \mathbf{y})$. Furthermore, to achieve the required posterior score estimation error $\varepsilon_{\text{post}}^2$, the hyperparameters for the Langevin dynamics (4.5) are specified as:

$$S = \Theta\left(\frac{\sigma_T^2}{\mu_T^2 - \alpha\sigma_T^2} \log\left(\frac{T\eta_{\mathbf{y}}^2}{\varepsilon_{\text{post}}^2}\right)\right), \quad m = \Theta\left(\frac{T\kappa_{\mathbf{y}}}{\varepsilon_{\text{post}}^2}\right),$$

$$\varepsilon_{\text{prior}} = \Theta\left(\frac{\mu_T^2 - \alpha\sigma_T^2}{\sigma_T^2} (\eta_{\mathbf{y}}^2 + \log\left(\frac{T\eta_{\mathbf{y}}^2}{\varepsilon_{\text{post}}^2}\right))^{-1} \exp\left(-4\frac{\sigma_T^2}{\sigma_{T_0}^2} \frac{\mu_{T_0}^2 + G\sigma_{T_0}^2}{\mu_T^2 - \alpha\sigma_T^2} \log\left(\frac{T\eta_{\mathbf{y}}^2}{\varepsilon_{\text{post}}^2}\right)\right) \frac{\varepsilon_{\text{post}}^4}{T^2\kappa_{\mathbf{y}}}\right).$$

where S denotes the simulation horizon, $\varepsilon_{\text{prior}}^2$ represents the prior score matching error, and m is the number of Monte Carlo particles.

The proof of Lemma 5.8 is provided in Appendix G. The warm-start procedure in Section 4.2 employs a nested structure consisting of two Langevin dynamics components. The inner Langevin dynamics (4.5) aims to estimate the posterior score function $\nabla \log q_T(\cdot|\mathbf{y})$, while the outer Langevin dynamics (4.11) generates particles from the terminal posterior density $q_T(\cdot|\mathbf{y})$. Lemma 5.8 specifies the terminal time U and posterior score estimation error budget $\varepsilon_{\text{post}}$ for the outer Langevin dynamics (4.11). Further, to satisfy the constraint on the posterior score estimation error, the selection of hyperparameters for the inner dynamics are also given in Lemma 5.8.

5.5 Non-asymptotic convergence rate. Building upon Lemma 5.4, Corollary 5.5, and Lemma 5.8, we provide a non-asymptotic bound on the convergence of the proposed algorithm (4.13). This bound quantifies the approximation quality, measured in the expected squared Wasserstein-2 distance between the estimated distribution and the ground-truth posterior distribution.

Theorem 5.9 (Convergence of posterior sampling). *Suppose Assumptions 1, 2, 3, 4, 5 and 6 hold. Assume $2\alpha V_{\text{SG}}^2 \leq 1$. Let the terminal time satisfy $T \in (\frac{1}{2} \log(1 + 2V_{\text{SG}}^2), \frac{1}{2} \log(1 + \alpha^{-1}))$. For each time $t \in (T_0, T)$, Then for $\varepsilon \in (0, 1)$, the following inequality holds:*

$$\begin{aligned} & \mathbb{E}[\mathbb{W}_2^2(q_0(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\# \hat{q}_{T-T_0}^R(\cdot|\mathbf{y}))] \\ & \leq C \left\{ \frac{\sigma_{T_0}^2}{\mu_{T_0}^2} + \left(\frac{\mu_{T_0}}{\sigma_{T_0}^2} \varepsilon + \varepsilon^{\frac{1}{2}} \right) \log \left(\kappa_{\mathbf{y}} \left(\frac{\mu_{T_0}}{\sigma_{T_0}^2} \varepsilon + \varepsilon^{\frac{1}{2}} \right)^{-1} \right) \right\}, \end{aligned}$$

where the hyperparameter selections follow Corollary 5.5 and Lemma 5.8, and C is a constant only depending on $d, \kappa_{\mathbf{y}}, B, H_{\mathbf{y}}, V_{\text{SG}}$, and C_{SG} . Furthermore, by setting $T_0 = \sqrt{\varepsilon_0}$, we have

$$\mathbb{E}[\mathbb{W}_2^2(q_0(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\# \hat{q}_{T-T_0}^R(\cdot|\mathbf{y}))] \leq C' \varepsilon^{\frac{1}{2}} \log \varepsilon^{-1},$$

where C' is a constant only depending on $d, \kappa_{\mathbf{y}}, B, H_{\mathbf{y}}, V_{\text{SG}}$, and C_{SG} .

The convergence bound in Theorem 5.9 is fundamentally composed of two competing terms, both dependent on the early-stopping time T_0 .

- (i) The early-stopping error: This error is characterized by the term $\mathcal{O}(\sigma_{T_0}^2)$. As noted, $\sigma_{T_0} = 1 - \exp(-2T_0) \approx 2T_0$, which strictly converges to zero as $T_0 \rightarrow 0$. This error arises from the inherent approximation made by stopping the reverse diffusion process at $T - T_0$ instead of the terminal time T . A smaller T_0 is desirable to minimize this early-stopping error.
- (ii) The posterior score estimation error: Our analysis reveals that posterior score estimation error is magnified by a prefactor $\mathcal{O}(\sigma_{T_0}^{-1})$, which diverges as T_0 approaches zero. This theoretical finding provides rigorous justification for the early-stopping strategy commonly employed in practice: terminating the time-reversal diffusion process at time $T - T_0$ rather than T is not merely a computational convenience but a theoretical necessity to ensure bounded estimation errors.

Theorem 5.9 offers theoretical guidance for selecting the early-stopping time T_0 by balancing the trade-off between posterior score estimation error and early-stopping error.

Discussions with related work. Theorem 5.9 also provides insights into related work, such as the closely connected method of Grenioux et al. (2024). Specifically, by setting the prior score error $\varepsilon_{\text{prior}}$ (Assumption 5) to zero, our theorem provides a non-asymptotic convergence analysis for their approach.

6 Numerical Experiments

In this section, we conduct numerical experiments to evaluate the performance of the proposed PDPS method. We begin by establishing the experimental setup, describing the forward measurement models in Section 6.1 and the datasets and baselines in Section 6.2. The experimental results are then presented in three parts: Section 6.3 demonstrates the sampling efficacy and uncertainty quantification capabilities of PDPS on both linear and nonlinear

inverse problems; Section 6.4 investigates the robustness of our method under prior mismatch scenarios; and Section 6.5 provides an ablation study on the diffusion terminal time to validate the theoretical trade-off discussed in Section 4.3. To facilitate reproducibility, complete experimental details (e.g., pretrained models and method-specific hyperparameters) are deferred to Appendix B. The code is available at <https://github.com/Ruoxuan0077/PDPS>.

6.1 Measurement models. To begin with, we describe the forward observation operators employed in our experiments, encompassing linear deblurring tasks (Gaussian and motion blur) and a more challenging nonlinear deblurring scenario. All experiments are conducted on 64×64 images corrupted by additive Gaussian noise $\mathbf{n} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ with $\sigma = 0.05$.

Linear forward operator. For linear deblurring, the measurement $\mathbf{Y} \in \mathbb{R}^n$ is modeled as the convolution of the unknown image $\mathbf{X}_0 \in \mathbb{R}^d$ with a blur kernel ψ , corrupted by additive Gaussian noise:

$$\mathbf{Y} = \mathcal{F}_\psi(\mathbf{X}_0) + \mathbf{n} = \psi * \mathbf{X}_0 + \mathbf{n},$$

where $*$ denotes the convolution. We evaluate our method on two distinct scenarios: **Gaussian deblurring**, which employs a standard Gaussian kernel; and **motion deblurring**, where the kernel is synthesized using publicly available code¹ to simulate realistic camera shake effects.

Nonlinear forward operator. As to the nonlinear deblurring scenario, we consider a blurring process inspired by the GOPRO dataset (Nah et al., 2017). This process models blur formation not as a simple convolution but as a temporal aggregation of multiple sharp frames, followed by a nonlinear camera response function $\mathcal{R}(\cdot)$. A discrete representation is:

$$\mathbf{Y} = \mathcal{R}(\bar{\mathbf{Y}}), \quad \bar{\mathbf{Y}} = \frac{1}{M} \sum_{i=1}^M \mathbf{X}_0^i,$$

where $\{\mathbf{X}_0^i\}_{i=1}^M$ are sharp frames of $\mathbf{x}_0$ within the exposure time, M is the number of frames, and $\mathcal{R}(\mathbf{z}) = \mathbf{z}^{1/2.2}$ represents a typical gamma correction.

In practice, we employ the pretrained distillation model $\mathcal{F}_{\tau^*}$ from Tran et al. (2021) to emulate the complex forward degradation process. Accordingly, our nonlinear measurement model takes the following form:

$$\mathbf{Y} = \mathcal{F}_{\tau^*}(\mathbf{X}_0, \mathbf{c}^*) + \mathbf{n},$$

where $\mathbf{c}^*$ is a fixed latent vector conditioning the operator.

6.2 Dataset and baseline. We employ the FFHQ dataset (Karras et al., 2019) at a resolution of 64×64 as the source of prior samples. For a comprehensive comparative analysis, we benchmark our proposed method against two baselines: Diffusion Posterior Sampling (DPS) (Chung et al., 2023), a state-of-the-art diffusion-based inverse problem solver, and the classical Total Variation (TV) regularization. To ensure a fair comparison, we explicitly align the prior information used by both diffusion-based methods. Specifically, both PDPS and the DPS baseline employ the same pre-trained prior score network based on the EDM framework (Karras et al., 2022), ensuring that any performance differences are attributable to the sampling algorithms rather than the prior models.

¹<https://github.com/LeviBorodenko/motionblur>

6.3 Sampling Efficacy and Uncertainty Quantification. As the first example, we assess the sampling effectiveness of PDPS on the previously defined tasks, which include two linear deblurring problems (Gaussian and motion) and one complex nonlinear deblurring scenario. We compare its performance against the baselines using both average quantitative metrics and case studies with uncertainty quantification.

The average quantitative performance of PDPS and the two baselines are summarized in Table 1, where we report the average PSNR and SSIM computed over 128 randomly selected images from the FFHQ64 dataset. As shown in the table, PDPS consistently outperforms the competing approaches across all tasks, with particularly pronounced gains in the challenging nonlinear deblurring setting, demonstrating its superior effectiveness in complex scenarios.

Table 1: Comparison of PSNR and SSIM on Different Methods

	Gaussian Deblur			Motion Deblur			Nonlinear Deblur		
	TV	DPS	PDPS	TV	DPS	PDPS	TV	DPS	PDPS
PSNR	23.95	24.15	26.42	24.65	26.66	28.86	19.70	20.93	28.44
SSIM	0.81	0.81	0.87	0.80	0.88	0.92	0.53	0.68	0.91

For in-depth case studies, we selected six representative images (two per task) to evaluate the reconstruction quality of the different methods. For each image and each method, we performed 24 independent sampling runs. The primary visual result shown for each method corresponds to the single run achieving the highest combined PSNR and SSIM. To enable qualitative comparison and uncertainty quantification, we further report (i) the pixel-wise mean absolute error (MAE) with respect to the ground truth (labeled “mean err.”) and (ii) the pixel-wise standard deviation (labeled “std. dev.”) computed across the 24 runs.

Motion Deblurring. Figure 2 illustrates the results for motion deblurring task, comparing our PDPS method against the TV and DPS baselines.

As clearly demonstrated, our method achieves superior performance in the motion deblurring task. Quantitatively, PDPS achieves higher performance in both PSNR and SSIM metrics. Qualitatively, while the TV method produces overly smoothed results that fail to recover fine details, such as individual hair strands and textures, PDPS effectively removes motion blur while faithfully restoring intricate details and sharp edges. In comparison, although the DPS method shows notable improvements in PSNR over TV and recovers a considerable amount of detail, it still introduces some unnatural artifacts in facial regions. For instance, in the top example (“face 1”), DPS introduces visible reconstruction artifacts along the subject’s right arm, whereas PDPS renders this area cleanly and naturally. Similarly, in the bottom example (“face 2”), the subject’s right eye appears distorted and lacks coherence in the DPS result, while PDPS provides a clear and natural rendering.

Furthermore, the pixel-wise mean absolute error (MAE, denoted as “mean err.”) and pixel-wise standard deviation (“std. dev.”) shown in Figure 2 enable a visual assessment of the model’s capability for uncertainty quantification (UQ). Ideally, a higher standard deviation should correlate with higher reconstruction error, reflecting the model’s confidence in its predictions. The TV baseline, however, exhibits negligible standard deviation despite notable reconstruction errors, thereby failing to provide meaningful uncertainty estimates. In contrast, both PDPS and DPS demonstrate the desired correlation: their respective “std. dev.” maps highlight areas of uncertainty—such as facial features and hair edges—that visually align with high-error regions in their “mean err.” maps, indicating potential for effective UQ.

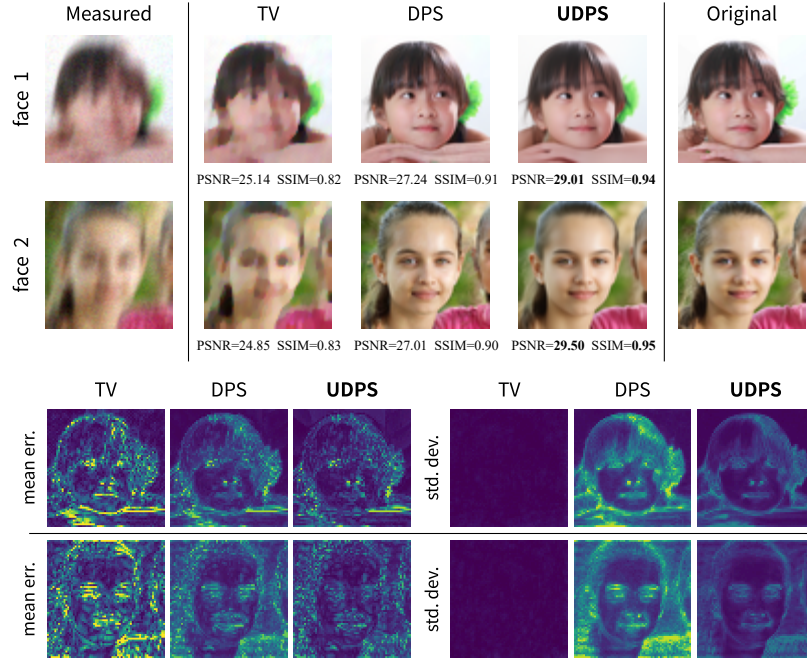


Figure 2: Result of motion deblurring task (face 1-2). Comparison of naive input, TV, DPS, PDPS (Ours), and original ground truth are shown. Bottom rows show mean absolute error (mean err.) and standard deviation (std. dev.) maps computed over 24 runs for all methods.

Comparing with DPS, PDPS offers clear advantages: it yields evidently lower mean error (superior accuracy), and its standard deviation map is comparatively smoother and less noisy. This suggests PDPS provides not only accurate reconstructions but also more stable and reliable uncertainty estimates, capturing inherent ambiguities without excessive sampling variance.

Gaussian Deblurring. For the Gaussian deblurring examples shown in Figure 3, PDPS (PSNR ≈ 26.3 -28.3) again provides visually superior reconstructions compared to both the over-smoothed TV results (PSNR ≈ 23.8 -24.1) and DPS outputs (PSNR ≈ 23.7 -26.8). For “face 3”, PDPS renders a more natural face shape and eye expression compared to the slightly distorted DPS result. For “face 4”, PDPS demonstrates significantly better recovery of the fine texture on the subject’s clothing, which appears blurred or lost in the DPS reconstruction. The UQ analysis exhibits similar characteristics to those observed in motion deblurring. PDPS again delivers reliable uncertainty quantification: its standard deviation map closely aligns with the mean error distribution, providing trustworthy uncertainty estimates alongside high-accuracy reconstructions.

Nonlinear Deblurring. Figure 4 displays the results for the nonlinear deblurring task, starkly demonstrating PDPS’s superiority in handling nonlinearity. This is reflected in the substantial quantitative lead reported in Table 1, with PDPS achieving significantly higher PSNR (≈ 29.0 -31.8) compared to both DPS (≈ 23.8 -27.6) and the largely unsuccessful TV baseline (≈ 18.7 -22.1). Visually, while TV fails to produce clear results and DPS introduces artifacts with erroneous details (affecting facial clarity in “face 5” and the leafy decoration’s texture in face 6), PDPS delivers clear, detailed, and artifact-free reconstructions faithful to the ground truth. The UQ analysis further underscores PDPS’s robustness in this challenging

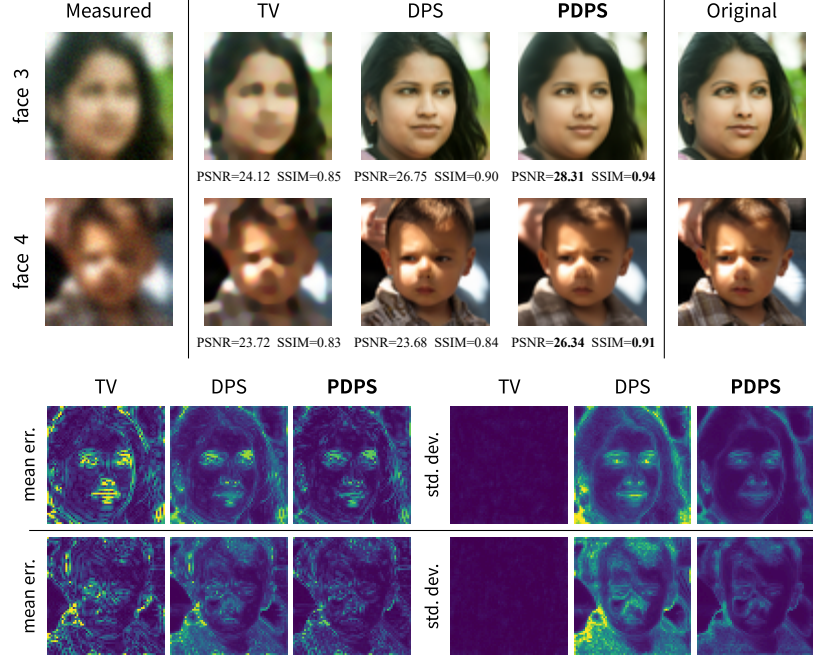


Figure 3: Gaussian deblurring examples (face 3-4). Comparison of naive input, TV, DPS, PDPS (Ours), and original ground truth. Bottom rows show mean absolute error (mean err.) and standard deviation (std. dev.) maps computed over 24 runs for TV, DPS, and PDPS.

nonlinear regime. The “mean err.” map confirms high reconstruction accuracy, showing visibly lower error levels compared to the baselines. Coupled with this accuracy, PDPS provides reliable spatial uncertainty estimates: its “std. dev.” map correlates well with error patterns. This demonstrates PDPS’s effectiveness in delivering both accurate results and trustworthy uncertainty quantification for complex nonlinear problems where baseline methods struggle significantly.

6.4 Robustness to Prior Mismatch. Next, we investigate the robustness of PDPS to prior mismatch, a common challenge where the pre-trained prior may not perfectly align with the target data domain. We specifically evaluate performance using the prior score model trained on human faces (FFHQ64) to restore images of animal faces (AFHQ dataset (Choi et al., 2020)) degraded by motion and nonlinear blurring. This cross-dataset setup tests whether PDPS’s unbiased posterior score estimation, leveraging the RGO mechanism, can effectively compensate for inaccuracies stemming from the mismatched prior score. We compare PDPS against DPS using the same prior score pretrained from FFHQ64, and the prior-agnostic TV baseline.

The average quantitative performance over 128 randomly selected AFHQ images is summarized in Table 2. The results highlight the impact of prior mismatch: compared to the in-domain evaluations reported in Table 1, all three methods exhibit a notable decline in PSNR and SSIM scores, with the degradation being particularly pronounced in the nonlinear deblurring case. Nevertheless, while the performance of DPS becomes comparable to that of the TV baseline in the motion deblurring task, and even inferior in the nonlinear case, PDPS continues to demonstrate a clear advantage. This observation suggests that PDPS exhibits notable robustness under distribution shift conditions.

These quantitative findings are further supported by the visual results on representative

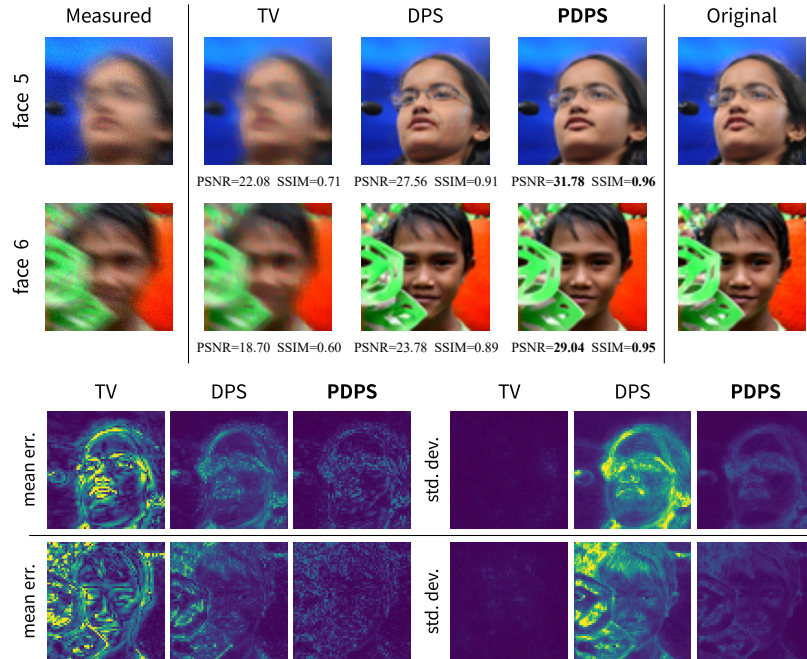


Figure 4: Nonlinear deblurring examples (face 5-6). Comparison of naive input, TV, DPS, PDPS (Ours), and original ground truth. Bottom rows show mean absolute error (mean err.) and standard deviation (std. dev.) maps computed over 24 runs for TV, DPS, and PDPS.

examples shown in Figure 5. In the motion deblurring cases, while PDPS yields plausible and coherent reconstructions, DPS exhibits noticeable difficulty in handling the prior mismatch, producing outputs with cluttered artifacts and unnatural textures, which appears even less consistent than those generated by the TV baseline in certain regions. The nonlinear deblurring examples reveal even more pronounced failures for DPS under the mismatched prior. For instance, the reconstruction of “cat 2” is almost entirely misrepresented, preserving only a faint outline and heavily distorted internal features. In contrast, PDPS demonstrates strong resilience in these challenging scenarios, delivering reconstructions that remain faithful to the underlying content despite the distribution shift. Notably, in the case of “cat 2”, PDPS successfully reconstructs plausible details such as cat whiskers (features absent in both the corrupted input and the human-face prior), highlighting its capacity to exploit the likelihood information via the RGO step to produce accurate, domain-consistent structures.

Table 2: Comparison for various methods under cross-dataset evaluation.

	Motion Deblur			Nonlinear Deblur		
	TV	DPS	PDPS	TV	DPS	PDPS
PSNR	23.35	23.54	25.45	19.31	17.20	24.45
SSIM	0.75	0.77	0.83	0.45	0.40	0.80

6.5 Ablation study: impact of terminal time. Last but not least, we investigate the impact of the terminal time T on sampling performance, aiming to validate the theoretical trade-off discussed in Section 4.3. Theorem 4.16 demonstrates the importance of the choice of the terminal time T . Specifically, the terminal time T must be large enough ($T > \underline{t}$) to ensure

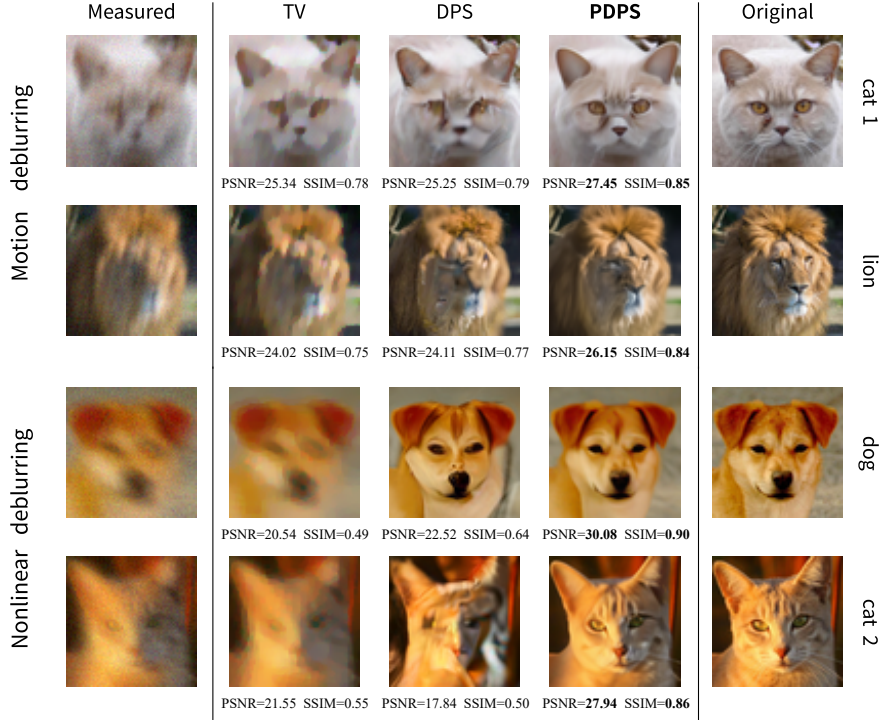


Figure 5: Cross-dataset deblurring results on AFHQ animal faces (‘cat 1’, ‘lion’, ‘dog’, ‘cat 2’) using an FFHQ human face prior. Top two rows: Motion Deblurring. Bottom two rows: Nonlinear Deblurring. Comparison of Naive input, TV, DPS, PDPS (Ours), and original ground truth.

the terminal posterior density $q_T(\cdot|\mathbf{y})$ satisfies log-Sobolev inequality, which is necessary for the convergence of the warm-start Langevin sampler (4.11). Simultaneously, it must be small enough ($T < \bar{t}$) to guarantee the log-concavity of the posterior denoising density $p_t(\cdot|\mathbf{x}, \mathbf{y})$ for all $t \in (0, T)$, which is crucial for the convergence of our Monte Carlo score estimator (4.7).

To demonstrate this trade-off, we perform an ablation study on the Gaussian deblurring task, with results for two representative images shown in Figure 6. The plots clearly show that for very small T (e.g., $T < 0.05$), both PSNR and SSIM metrics are poor, which aligns with the theoretical difficulty of sampling from a terminal posterior density $q_T(\cdot|\mathbf{y})$ that either possesses a large log-Sobolev constant or fails to satisfy the inequality entirely. As T increases, performance rapidly improves and enters a stable plateau of near-optimal results for T within an intermediate range (approximately 0.05 to 1.0). This region represents the practical “sweet point” where both theoretical conditions are met. However, if T becomes too large ($T > 1.0$), performance begins to degrade, which can be attributed to the loss of log-concavity in the posterior denoising density that compromises the score estimation accuracy. These empirical findings provide strong evidence for our theoretical analysis and underscore the importance of selecting a well-balanced T .

7 Conclusions

In this work, we introduced a novel diffusion-based posterior sampling method. Its first key feature is a plug-and-play (PnP) architecture, achieved by learning a prior score that is independent of the measurement model. This allows flexible application to various

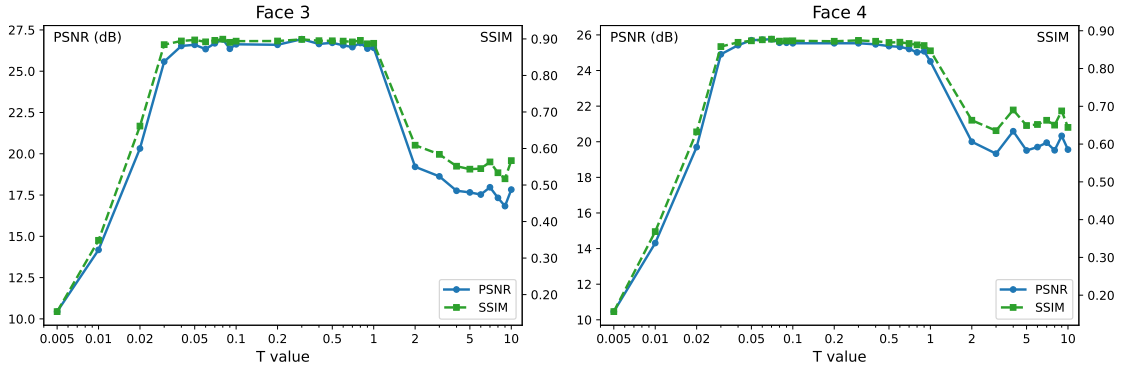


Figure 6: Ablation study on the terminal time T for the Gaussian deblurring task on two representative images, face 3 (left) and face 4 (right). The plots illustrate the final reconstruction PSNR and SSIM as a function of T . The results demonstrate a clear performance peak for T in an intermediate range (e.g., approximately 0.05 to 1.0), validating the theoretical trade-off between ensuring convergence of the warm-start (requiring larger T) and of the posterior score estimation (requiring smaller T).

inverse problems without retraining. The second key feature of our work is it provide a provable estimator to the posterior score, without heuristic approximation. Specifically, our approach estimates the posterior score via a Monte Carlo procedure using Langevin dynamics, supported by an efficient warm-start mechanism. Theoretically, we provide the first non-asymptotic convergence guarantees for a diffusion-based posterior score estimator, establishing 2-Wasserstein error bounds that notably hold for multi-modal distributions. Our analysis bounds all key error sources and offers practical guidance for hyperparameter selection. Extensive experiments on challenging imaging inverse problems confirm our method’s strong performance, achieving competitive results and demonstrating its utility for uncertainty quantification. This research provides a robust, flexible, and theoretically-grounded tool for Bayesian inference.

This work also opens several avenues for future investigation. First, extending the current method to infinite-dimensional inverse problems (Stuart, 2010) is an important direction. In this context, evaluating the forward model requires solving computationally expensive PDEs, making the development of accurate deep learning-based surrogate models, like operator learning (Li et al., 2024c; Bachmayr et al., 2025), essential. Second, our method currently depends on the log-likelihood gradient, which is unavailable in many scientific applications. Therefore, extending our method to solve derivative-free Bayesian inference problems is crucial (Chen et al., 2024b; Zheng et al., 2025b). Third, our framework could be extended to dynamic inverse problems, such as data assimilation (Sanz-Alonso et al., 2023; Bach et al., 2025; Reich and Cotter, 2015; Reich, 2019). Finally, from a theoretical perspective, establishing convergence analyses for other types of multi-modal target posterior distributions remains an important area for future work.

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Outline of Appendices

The appendices are organized as follows:

- (i) **Appendix A** provides a comprehensive summary of notations used throughout the paper, categorized by the Introduction, Method, and Convergence Analysis sections.
- (ii) **Appendix B** presents detailed experimental settings, including mathematical formulations of measurement operators, implementation details of the proposed PDPS algorithm and baseline methods, and a complete list of hyperparameters.
- (iii) **Appendix C** discusses the theoretical properties of the concrete examples used in our analysis, specifically verifying the assumptions for Gaussian mixture and Gaussian convolution priors.
- (iv) **Appendix D** provides proofs for the lemmas in Section 4, covering the properties of the posterior denoiser, the log-concavity of the RGO, and the log-Sobolev inequality for the terminal posterior.
- (v) **Appendix E** establishes the error decomposition for the posterior sampling, analyzing both the early-stopping error and the decomposition of the total variation distance.

- (vi) **Appendix F** derives the rigorous error bounds for the posterior score estimation, accounting for the errors from Langevin dynamics, Monte Carlo approximation, and prior score matching.
- (vii) **Appendix G** provides the convergence analysis and error bounds for the warm-start strategy, ensuring the validity of the initialization for the reverse diffusion.
- (viii) **Appendix H** lists auxiliary lemmas regarding sub-Gaussian properties of the forward process and standard convergence results for Langevin dynamics used in our proofs.

A A Summary of Notations

Tables 3, 4 and 5 summarize the notations used in Sections 1, 4 and 5, respectively, for easy reference and cross-checking.

Table 3: The list of notations defined in Section 1.

Symbols	Description
$\mathbf{X}_0$	A random variable representing the unknown signal.
π_0	The prior density of the signal $\mathbf{X}_0$.
$\mathbf{Y}$	A random variable representing the measurement of the signal.
$p_{\mathbf{Y} \mathbf{X}_0}(\cdot \mathbf{x}_0)$	The likelihood of $\mathbf{Y}$ given $\mathbf{X}_0 = \mathbf{x}_0$.
$\ell_{\mathbf{Y}}(\mathbf{x}_0)$	The negative log-likelihood of $\mathbf{Y} = \mathbf{y}$ given $\mathbf{X}_0 = \mathbf{x}_0$.
$q_0(\cdot \mathbf{y}) = p_{\mathbf{X}_0 \mathbf{Y}}(\cdot \mathbf{y})$	The conditional density of $\mathbf{X}_0$ given $\mathbf{Y} = \mathbf{y}$.

B Experimental Details

In this appendix, we provide the specific parameter settings, mathematical formulations for the measurement operators, and detailed implementation steps for the algorithms and baseline methods used in Section 6.

B.1 Details of Measurement Operators.

B.1.1 Linear Blur Kernels. For the linear deblurring tasks, the forward operator is a convolution with a kernel $\psi \in \mathbb{R}^{15 \times 15}$.

- **Gaussian Deblurring:** The kernel ψ is a Gaussian filter with a standard deviation of $\sigma_{\text{blur}} = 2.0$ and a kernel size of 15×15 .
- **Motion Deblurring:** The motion blur kernel is synthesized with an intensity value of 0.5 and a kernel size of 15×15 , simulating random camera trajectories.

B.1.2 Nonlinear Operator Details. The nonlinear forward operator $\mathcal{F}_{\tau^*}$ approximates the complex physical blurring process involving temporal aggregation and gamma correction. In our experiments, we directly utilized the pre-trained model provided by [Tran et al. \(2021\)](#). For completeness, we note that this network was originally trained using a dataset of sharp-blur image pairs $\{(\mathbf{X}_j, \mathbf{Y}_j)\}_{j=1}^N$ to minimize the Charbonnier loss ([Lai et al., 2017](#)):

$$(B.1) \quad \tau^*, \phi^* = \arg \min_{\tau, \phi} \sum_{j=1}^N \rho\left(\mathbf{Y}_j, \mathcal{F}_{\tau}(\mathbf{X}_j, \mathcal{C}_{\phi}(\mathbf{X}_j, \mathbf{Y}_j))\right),$$

where $\mathcal{C}_{\phi}$ is an auxiliary network used to extract blur information during training, and $\rho(u)$ is the Charbonnier loss function. To establish the deterministic operator used in our

Table 4: The list of notations defined in Section 4.

Symbols	Description
$\mathbf{X}_t$	The unconditional forward process (2.1).
$p_{\mathbf{X}_t \mathbf{X}_0}(\cdot \mathbf{x}_0)$	The conditional density of $\mathbf{X}_t$ given $\mathbf{X}_0 = \mathbf{x}_0$.
π_t	The marginal density of the forward process $\mathbf{X}_t$ (2.3).
T	The terminal time of the forward process.
$\mathbf{X}_t^y$	The posterior forward process given $\mathbf{Y} = \mathbf{y}$ (2.4).
$q_t(\cdot \mathbf{y})$	The marginal density of the posterior forward process $\mathbf{X}_t^y$ (2.5).
$p_{\mathbf{X}_t \mathbf{Y}}(\cdot \mathbf{y})$	The conditional density of $\mathbf{X}_t$ given $\mathbf{Y} = \mathbf{y}$.
$\mathbf{X}_t^y$	The posterior time-reversal process (2.7).
$\mathbf{D}(t, \mathbf{x}, \mathbf{y})$	The posterior denoiser (4.2), i.e., the conditional expectation of $\mathbf{X}_0$ given $\mathbf{X}_t = \mathbf{x}$ and $\mathbf{Y} = \mathbf{y}$.
$p_t(\cdot \mathbf{x}, \mathbf{y})$	The posterior denoising density (4.3), i.e., the conditional density of $\mathbf{X}_0$ given $\mathbf{X}_t = \mathbf{x}$ and $\mathbf{Y} = \mathbf{y}$.
$\mathbf{X}_{0,s}^{\mathbf{x}, \mathbf{y}, t}$	The Langevin dynamics with invariant density $p_t(\cdot \mathbf{x}, \mathbf{y})$ (4.4)
$p_t^s(\cdot \mathbf{x}, \mathbf{y})$	The marginal density of $\mathbf{X}_{0,s}^{\mathbf{x}, \mathbf{y}, t}$.
$\hat{\mathbf{X}}_{0,s}^{\mathbf{x}, \mathbf{y}, t}$	The Langevin dynamics with estimated prior score (4.5)
$\hat{p}_t^s(\cdot \mathbf{x}, \mathbf{y})$	The marginal density of $\hat{\mathbf{X}}_{0,s}^{\mathbf{x}, \mathbf{y}, t}$.
S	The simulation horizon of (4.5).
m	The number of particles used in Monte Carlo approximation.
$\hat{\mathbf{X}}_{0,S,1}^{\mathbf{x}, \mathbf{y}, t}, \dots, \hat{\mathbf{X}}_{0,S,m}^{\mathbf{x}, \mathbf{y}, t}$	Particles generated by (4.5), following the density $\hat{p}_t^s(\cdot \mathbf{x}, \mathbf{y})$.
$\hat{\mathbf{D}}_m^S(t, \mathbf{x}, \mathbf{y})$	The Monte Carlo estimator of the posterior denoiser (4.6).
$\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$	The Monte Carlo estimator of the posterior score (4.7).
$q_T(\cdot \mathbf{y})$	The terminal posterior density, i.e., the marginal density of $\mathbf{X}_T^y$.
$\mathbf{X}_{T,u}^y$	The Langevin dynamics with invariant density $q_T(\cdot \mathbf{y})$ (4.10).
$q_T^u(\cdot \mathbf{y})$	The marginal density of $\mathbf{X}_{T,u}^y$.
$\hat{\mathbf{X}}_{T,u}^y$	The Langevin dynamics with estimated score $\hat{\mathbf{s}}_m^S(T, \cdot, \mathbf{y})$ (4.11).
$\hat{q}_T^u(\cdot \mathbf{y})$	The marginal density of $\hat{\mathbf{X}}_{T,u}^y$.
U	The simulation horizon of (4.11).
C_{SG} and V_{SG}	The parameters of sub-Gaussian tails defined in Assumption 2.
$\kappa_{\mathbf{y}}$	The condition number of the posterior sampling (4.12).
$C_{\text{LSI}}(\mu)$	The log-Sobolev inequality constant of a distribution μ .
$\hat{\mathbf{X}}_t^y$	The approximate posterior time-reversal process (4.13).
T_0	The early-stopping time defined in (4.13).

Table 5: The list of notations defined in Section 5.

Symbols	Description
$\mathcal{M}$	The scaling operator, i.e., $\mathcal{M}(\xi) : \mathbf{x} \mapsto \xi \mathbf{x}$.
B	The prior potential $\log \pi_0$ is B -smooth (Assumption 3).
L	The negative log-likelihood $\ell_{\mathbf{y}}$ is L -smooth (Assumption 3).
$\varepsilon_{\text{prior}}$	The prior score matching error defined in Assumption 5.
$\eta_{\mathbf{y}}$	The initial discrepancy of (4.5) defined in Lemma 5.5.
$\varepsilon_{\text{post}}$	The posterior score estimation error, defined in Corollary 5.6.
$\zeta_{\mathbf{y}}$	The initial discrepancy of (4.11) defined in Lemma 5.8.
Δ	The error of the warm-start, defined in Lemma 5.8.
$\varepsilon_{\text{post}}$	The terminal posterior score estimation error, defined in Lemma 5.8.

experiments, we replaced the output of $\mathcal{C}_\phi$ with a fixed latent vector $\mathbf{c}^*$. This vector was instantiated by sampling from a normal distribution:

$$(B.2) \quad \mathbf{c}^* \sim N(\mathbf{0}, \sigma_c^2 \mathbf{I}_c), \quad \text{with } \sigma_c = 0.3.$$

The resulting operator $\mathcal{F}_{\tau^*}(\cdot, \mathbf{c}^*)$ serves as our fixed nonlinear forward model.

B.2 Implementation of PDPS.

B.2.1 Prior Score Approximation. Our method requires a neural network estimator, $\hat{\mathbf{s}}_{\text{prior}}$, for the prior score, $\nabla_{\mathbf{x}_0} \log \pi_0(\mathbf{x}_0)$, and we employ a publicly available, pre-trained model based on the well-known EDM framework (Karras et al., 2022). Specifically, it provides a neural network estimate for the unconditional denoiser $\mathbf{D}(\mathbf{x}, \sigma)$, denoted as $\hat{\mathbf{D}}(\mathbf{x}, \sigma)$, where

$$\mathbf{D}(\mathbf{x}, \sigma) := \mathbb{E}[\mathbf{X}_0 | \mathbf{X}_0 + \sigma \mathbf{Z} = \mathbf{x}], \quad \mathbf{X}_0 \sim \pi_0, \quad \mathbf{Z} \sim N(\mathbf{0}, \mathbf{I}_d), \quad \mathbf{X}_0 \perp \mathbf{Z}.$$

Furthermore, the denoiser and score function are related by Tweedie’s formula (Robbins, 1992):

$$\nabla_{\mathbf{x}} \log \pi_\sigma(\mathbf{x}) = \frac{\mathbf{D}(\mathbf{x}, \sigma) - \mathbf{x}}{\sigma^2},$$

where π_σ is the density of $\mathbf{X}_0 + \sigma \mathbf{Z}$. Thus, $(\hat{\mathbf{D}}(\mathbf{x}, \sigma) - \mathbf{x})/\sigma^2$ serves to approximate the σ -smoothed prior score, $\nabla_{\mathbf{x}} \log \pi_\sigma(\mathbf{x})$. Given that $\pi_\sigma \rightarrow \pi$ as $\sigma \rightarrow 0$, achieving a precise approximation of the true prior score $\nabla_{\mathbf{x}} \log \pi(\mathbf{x})$ theoretically necessitates a sufficiently small σ for the denoiser $\hat{\mathbf{D}}(\mathbf{x}, \sigma)$. However, practical implementation revealed significant numerical instability when using very small σ values, likely due to the inherent approximation errors of the neural network $\hat{\mathbf{D}}$. After empirical investigation to balance accuracy and stability, we selected $\sigma_d = 0.09$. Consequently, the prior score estimator employed consistently throughout our numerical experiments is fixed as $\hat{\mathbf{s}}_{\text{prior}}(\mathbf{x}) \triangleq (\hat{\mathbf{D}}(\mathbf{x}, \sigma_d) - \mathbf{x})/\sigma_d^2$.

B.2.2 Posterior Score Estimation (Algorithm 1). The core of our method is the RGO-based score estimation, which we implement using a Langevin Monte Carlo (LMC) sampler. Specifically, we discretize the RGO process (4.5) through Euler-Maruyama method: Let $\xi_k \sim N(\mathbf{0}, \mathbf{I}_n)$. For $k = 0, \dots, N_{\text{in}} - 1$, we start with $\hat{\mathbf{X}}_{0,0}^{\mathbf{x}, \mathbf{y}, t} \sim N(\mathbf{0}, \mathbf{I}_n)$ and iterate as

$$\hat{\mathbf{X}}_{0,k+1}^{\mathbf{x}, \mathbf{y}, t} - \hat{\mathbf{X}}_{0,k}^{\mathbf{x}, \mathbf{y}, t} = \left(\hat{\mathbf{s}}_{\text{prior}}(\hat{\mathbf{X}}_{0,k}^{\mathbf{x}, \mathbf{y}, t}) + \frac{\mu_t}{\sigma_t^2} (\mathbf{x} - \mu_t \hat{\mathbf{X}}_{0,k}^{\mathbf{x}, \mathbf{y}, t}) - \nabla \ell_{\mathbf{y}}(\hat{\mathbf{X}}_{0,k}^{\mathbf{x}, \mathbf{y}, t}) \right) \Delta t_{\text{in}} + \sqrt{2 \Delta t_{\text{in}}} \xi_k.$$

In practice, we use $M = 20$ parallel chains to generate samples. The LMC step size Δt_{in} is determined adaptively based on a fixed signal-to-noise ratio (SNR) of $r_{\text{in}} = 0.075$, following the strategy in Song et al. (2021). To compute the MC average for the posterior denoiser (4.6), we use samples from the latter half of the LMC iterations (a burn-in factor of $\rho = 0.5$). Thus, $m = \rho M N_{\text{in}}$. The LMC iteration count, N_{in} , varies depending on the stage, as detailed below.

B.2.3 Warm-Start Stage (Algorithm 2). The warm-start procedure consists of an outer Langevin loop designed to sample from the initial posterior density $q_T(\cdot|\mathbf{y})$: Let $\boldsymbol{\xi}_k \sim N(\mathbf{0}, \mathbf{I}_n)$. For $k = 0, \dots, N_{\text{out}} - 1$, we start with $\hat{\mathbf{X}}_{T,0}^{\mathbf{y}} \sim N(\mathbf{0}, \mathbf{I}_n)$ and discretize (4.11) as

$$\hat{\mathbf{X}}_{T,k+1}^{\mathbf{y}} - \hat{\mathbf{X}}_{T,k}^{\mathbf{y}} = \hat{\mathbf{s}}_m(T, \hat{\mathbf{X}}_{T,k}^{\mathbf{y}}, \mathbf{y})\Delta t_{\text{out}} + \sqrt{2\Delta t_{\text{out}}} \boldsymbol{\xi}_k.$$

This outer loop runs for $N_{\text{out}} = 400$ iterations, and its adaptive step size Δt_{out} is governed by an SNR of $r_{\text{out}} = 0.16$. We invoke Algorithm 1 to compute the score $\hat{\mathbf{s}}_m(\mathbf{x}, \mathbf{y}, t)$, setting the inner LMC iteration count to $N_{\text{in}} = 50$. To accelerate convergence, the final states of the $M = 20$ inner chains from one outer step are reused to initialize the next. The choice of the warm-start time T is critical and is tuned for each problem.

B.2.4 Reverse Diffusion Stage (Algorithm 3). The main sampling process simulates the reverse SDE from the warm-start time T down to a fixed early-stopping time of $T_0 = 0.05$. In practice, an Euler-Maruyama discretization for (4.13) with step size $\Delta t_{\text{rev}} = (T - T_0)/N_{\text{rev}}$ is implemented: For $k = 0, \dots, N_{\text{rev}} - 1$, we have

$$\hat{\mathbf{X}}_{k+1}^{\mathbf{y}} - \hat{\mathbf{X}}_k^{\mathbf{y}} = (\hat{\mathbf{X}}_k^{\mathbf{y}} + 2\hat{\mathbf{s}}_m(T - k\Delta t_{\text{rev}}, \hat{\mathbf{X}}_k^{\mathbf{y}}, \mathbf{y}))\Delta t_{\text{rev}} + \sqrt{2\Delta t_{\text{rev}}} \boldsymbol{\xi}_k,$$

where $\hat{\mathbf{X}}_0^{\mathbf{y}} = \hat{\mathbf{X}}_{T, N_{\text{out}}}^{\mathbf{y}}$ from the warm-start phase and $\boldsymbol{\xi}_k \sim N(\mathbf{0}, \mathbf{I}_n)$. The number of discretization steps is set to $N_{\text{rev}} = 1200T$. At each step of this reverse process, we again call upon Algorithm 1 for score estimation $\hat{\mathbf{s}}_m(\mathbf{x}, \mathbf{y}, t)$, this time with an inner LMC iteration count of $N_{\text{in}} = 20$. Following Song et al. (2021), upon reaching T_0 , a deterministic step using only the drift term is executed to map $\hat{\mathbf{X}}_{N_{\text{rev}}}^{\mathbf{y}}$ directly from T_0 to 0. We denote the output as $\hat{\mathbf{X}}^{\mathbf{y}}$.

B.2.5 Final Denoising Step. Since our prior score estimator approximates a smoothed prior (with $\sigma_d = 0.09$), the raw output $\hat{\mathbf{X}}^{\mathbf{y}}$ may contain minor residual noise. To mitigate this, we apply a final, unconditional denoising step using the pre-trained denoiser network $\hat{\mathbf{D}}$ with an empirically chosen noise level of $\sigma'_d = 0.03$. The final reported samples are thus $\hat{\mathbf{D}}(\hat{\mathbf{X}}^{\mathbf{y}}, \sigma'_d)$, which enhances the perceptual quality by reducing noise without overly smoothing fine details.

B.3 Details of Comparison Methods.

B.3.1 Diffusion Posterior Sampling (DPS). In essence, the DPS method proposed in Chung et al. (2023) relies on a standard DDPM Ho et al. (2020) forward noise addition process, namely:

$$\mathbf{X}_t^{\text{DDPM}} = \sqrt{\bar{\alpha}_t} \mathbf{X}_0 + \sqrt{1 - \bar{\alpha}_t} \mathbf{Z}, \quad \mathbf{Z} \sim N(\mathbf{0}, \mathbf{I}_n), \quad \mathbf{X}_0 \perp\!\!\!\perp \mathbf{Z}.$$

For brevity, we abbreviate $\mathbf{X}_t^{\text{DDPM}}$ as $\mathbf{X}_t$. DPS further uses (2.8), while approximates $p_t(\mathbf{x}_0|\mathbf{x})$ with $\delta_{\mathbb{E}[\mathbf{X}_0|\mathbf{X}_t=\mathbf{x}]}(\mathbf{x}_0)$. It then controls the generation process by adding a guidance term $-\zeta \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{F}(\mathbb{E}[\mathbf{X}_0|\mathbf{X}_t = \mathbf{x}_t])\|_2$ to the unconditional DDPM iterative scheme. Here, ζ represents a critical hyperparameter that controls the strength of conditional guidance. To ensure a fair comparison where both PDPS and DPS leverage the same prior knowledge, we fully reused the original codebase of Chung et al. (2023), while employing the score transformation techniques presented in Karras et al. (2022) to reconstruct the DDPM-type score from the pre-trained EDM model D_θ , and then use it to replace the original DDPM-type score in Chung et al. (2023).

B.3.2 Total Variation (TV). This classical approach solves the optimization problem:

$$(B.3) \quad \min_{\mathbf{x} \in \mathbb{R}^d} \frac{1}{2} \|\mathbf{y} - \mathcal{F}(\mathbf{x})\|_2^2 + \lambda \|\nabla \mathbf{x}\|_{2,1},$$

balancing data fidelity against isotropic Total Variation (TV) regularization, controlled by $\lambda > 0$. We solved (B.3) using a gradient descent based algorithm implemented via the `deepinv` library². The regularization weight λ was optimized using grid search, setting $\lambda = 0.01$ for the linear deblurring experiments and $\lambda = 0.05$ for the nonlinear deblurring experiment.

B.4 Hyperparameter Settings. The key hyperparameters for PDPS (warm-start time T) and DPS (guidance scale ζ) used in our experiments are summarized in Table 6. We present the settings for both the quantitative evaluations on the full test sets (FFHQ and AFHQ) and the specific parameters used for the qualitative case studies, where fine-tuning was applied to exploit the potential of each method.

Table 6: Hyperparameters T (for PDPS) and ζ (for DPS) used in different experimental settings.

Task	Subset / Image	PDPS (T)	DPS (ζ)
1. Quantitative Evaluation (Batch Average)			
FFHQ Dataset (128 images)			
Gaussian Deblur	full batch	0.2	0.9
Motion Deblur	full batch	0.5	1.2
Nonlinear Deblur	full batch	20.0	0.2
AFHQ Dataset (Cross-Dataset, 128 images)			
Motion Deblur	full batch	0.5	1.3
Nonlinear Deblur	full batch	20.0	0.2
2. Qualitative Case Studies (Specific Images)			
FFHQ (In-Distribution)			
Gaussian Deblur	face 3	0.2	0.9
	face 4	0.2	0.8
Motion Deblur	face 1-2	0.5	1.3
Nonlinear Deblur	face 5	3.5	0.5
	face 6	3.5	0.9
AFHQ (Out-of-Distribution)			
Motion Deblur	Cat 1 & Lion	0.5	1.3
Nonlinear Deblur	Dog	5.0	0.4
	Cat 2	9.0	0.3

C More Details of Concrete Examples

C.1 Gaussian mixture. Recall the Gaussian mixture introduced in Example 4.4:

$$\pi_0(\mathbf{x}_0) = \sum_{k=1}^K w_k \gamma_{d, \sigma_k^2}(\mathbf{x}_0 - \mathbf{m}_k).$$

²<https://github.com/deepinv/deepinv>

It is straightforward that

$$\nabla \pi_0(\mathbf{x}_0) = \sum_{k=1}^K w_k \nabla \gamma_{d, \sigma_k^2}(\mathbf{x}_0 - \mathbf{m}_k) = \sum_{k=1}^K w_k \gamma_{d, \sigma_k^2}(\mathbf{x}_0 - \mathbf{m}_k) \mathbf{s}_k(\mathbf{x}_0),$$

where $\mathbf{s}_k$ is the score function of the k -th Gaussian, defined as

$$\mathbf{s}_k(\mathbf{x}_0) = \nabla \log \gamma_{d, \sigma_k^2}(\mathbf{x}_0 - \mathbf{m}_k) = -\frac{\mathbf{x}_0 - \mathbf{m}_k}{\sigma_k^2}.$$

Consequently, the score of the Gaussian mixture can be expressed as a weighted sum of the score of each Gaussian component:

$$(C.1) \quad \mathbf{s}(\mathbf{x}_0) := \nabla \log \pi_0(\mathbf{x}_0) = \frac{\nabla \pi_0(\mathbf{x}_0)}{\pi_0(\mathbf{x}_0)} = \sum_{k=1}^K \xi_k(\mathbf{x}_0) \mathbf{s}_k(\mathbf{x}_0),$$

where the weights are given as

$$(C.2) \quad \xi_k(\mathbf{x}_0) = \frac{w_k \gamma_{d, \sigma_k^2}(\mathbf{x}_0 - \mathbf{m}_k)}{\sum_{k=1}^K w_k \gamma_{d, \sigma_k^2}(\mathbf{x}_0 - \mathbf{m}_k)} = \frac{w_k}{\pi_0(\mathbf{x}_0)} \gamma_{d, \sigma_k^2}(\mathbf{x}_0 - \mathbf{m}_k).$$

Linear growth of the Gaussian mixture score. Applying (C.1) implies

$$(C.3) \quad \|\mathbf{s}(\mathbf{x}_0)\|_2 \leq \sum_{k=1}^K \frac{\xi_k(\mathbf{x}_0)}{\sigma_k^2} \|\mathbf{m}_k\|_2 + \sum_{k=1}^K \frac{\xi_k(\mathbf{x}_0)}{\sigma_k^2} \|\mathbf{x}_0\|_2,$$

where we used the triangular inequality. Since $\xi_k(\mathbf{x}_0) \in (0, 1)$ for each $1 \leq k \leq K$ and $\mathbf{x}_0 \in \mathbb{R}^d$, the right-hand side of the inequality is uniformly bounded. Hence the Gaussian mixture π_0 satisfies Assumption 3 with $r = 1$.

Lipschitz continuity of the Gaussian mixture score. Taking gradient on both sides of (C.1) with respect to $\mathbf{x}_0$ yields

$$(C.4) \quad \begin{aligned} \nabla \mathbf{s}(\mathbf{x}_0) &= \nabla \left(\sum_{k=1}^K \xi_k(\mathbf{x}_0) \mathbf{s}_k(\mathbf{x}_0) \right) \\ &= \sum_{k=1}^K \nabla \xi_k(\mathbf{x}_0) \mathbf{s}_k(\mathbf{x}_0)^\top + \sum_{k=1}^K \xi_k(\mathbf{x}_0) \nabla \mathbf{s}_k(\mathbf{x}_0). \end{aligned}$$

For the first term in (C.4), we have

$$(C.5) \quad \nabla \xi_k(\mathbf{x}_0) = \xi_k(\mathbf{x}_0) \nabla \log \xi_k(\mathbf{x}_0) = \xi_k(\mathbf{x}_0) (\mathbf{s}_k(\mathbf{x}_0) - \mathbf{s}(\mathbf{x}_0)).$$

For the second term in (C.4), it holds that

$$(C.6) \quad \nabla \mathbf{s}_k(\mathbf{x}_0) = \nabla \left(-\frac{\mathbf{x}_0 - \mathbf{m}_k}{\sigma_k^2} \right) = -\frac{1}{\sigma_k^2} \mathbf{I}_d.$$

Substituting (C.5) and (C.6) into (C.4) yields

$$(C.7) \quad \nabla \mathbf{s}(\mathbf{x}_0) = \underbrace{\sum_{k=1}^K \xi_k(\mathbf{x}_0) (\mathbf{s}_k(\mathbf{x}_0) - \mathbf{s}(\mathbf{x}_0)) \mathbf{s}_k(\mathbf{x}_0)^\top}_{(i)} - \underbrace{\sum_{k=1}^K \frac{\xi_k(\mathbf{x}_0)}{\sigma_k^2} \mathbf{I}_d}_{(ii)}.$$

For a fixed $\mathbf{x}_0 \in \mathbb{R}^d$, define its dominated component index ℓ as:

$$\ell \in \arg \max_{1 \leq k \leq K} \gamma_{d, \sigma_k^2}(\mathbf{x}_0 - \mathbf{m}_k) = \arg \min_{1 \leq k \leq K} \frac{\|\mathbf{x}_0 - \mathbf{m}_k\|_2^2}{\sigma_k^2}.$$

It is apparent that π_0 is dominated by $N(\mathbf{m}_\ell, \sigma_\ell^2 \mathbf{I}_d)$.

We first consider the term (i) in (C.7). For $k = \ell$, we have $\mathbf{s}_k(\mathbf{x}_0) - \mathbf{s}(\mathbf{x}_0) \rightarrow 0$ as $\|\mathbf{x}_0\|_2$ goes to infinity. For $k \neq \ell$, according to (C.3), both $\|\mathbf{s}(\mathbf{x}_0)\|_2$ and $\|\mathbf{s}_k(\mathbf{x}_0)\|_2$ increase linearly with respect to the growth of $\|\mathbf{x}_0\|_2$, while $\xi_k(\mathbf{x}_0)$ decays exponentially with respect to the growth of $\|\mathbf{x}_0\|_2$. Therefore, the k -th term in (i) of (C.7) converges to zero as $\|\mathbf{x}_0\|_2$ goes to infinity.

We then consider the term (ii) in (C.7). Note that $\xi_k(\mathbf{x}_0) \rightarrow 0$ as $\|\mathbf{x}_0\|_2$ goes to infinity for $k \neq \ell$, while $\xi_k(\mathbf{x}_0) \rightarrow 1$ as $\|\mathbf{x}_0\|_2$ goes to infinity for $k = \ell$. Therefore, term (ii) converges to $-\sigma_\ell^{-2} \mathbf{I}_d$.

In summary, the Hessian of log-prior $\log \pi_0$ is bounded for $\|\mathbf{x}_0\|_2 \rightarrow \infty$. Since a continuous function that is bounded at infinity must be uniformly bounded, $\|\nabla \mathbf{s}\|_{\text{op}}$ is uniformly bounded on $\mathbb{R}^d$. Hence, $\mathbf{s}$ is uniformly Lipschitz, satisfying Assumption 1.

Sub-Gaussian tails. It is straightforward that

$$\begin{aligned} & \int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) \pi_0(\mathbf{x}_0) d\mathbf{x}_0 \\ &= \int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) \sum_{k=1}^K w_k \gamma_{d, \sigma_k^2}(\mathbf{x}_0 - \mathbf{m}_k) d\mathbf{x}_0 \\ &= \sum_{k=1}^K w_k \int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) \gamma_{d, \sigma_k^2}(\mathbf{x}_0 - \mathbf{m}_k) d\mathbf{x}_0 \\ &= \sum_{k=1}^K w_k (2\pi\sigma_k^2)^{-\frac{d}{2}} \int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2} - \frac{\|\mathbf{x}_0 - \mathbf{m}_k\|_2^2}{2\sigma_k^2}\right) d\mathbf{x}_0 \\ &= \sum_{k=1}^K w_k (2\pi\sigma_k^2)^{-\frac{d}{2}} \int \exp\left\{-\left(\frac{1}{2\sigma_k^2} - \frac{1}{V_{\text{SG}}^2}\right)\|\mathbf{x}_0\|_2^2 + \frac{1}{\sigma_k^2}\langle \mathbf{x}_0, \mathbf{m}_k \rangle - \frac{1}{2\sigma_k^2}\|\mathbf{m}_k\|_2^2\right\} d\mathbf{x}_0. \end{aligned}$$

So this integral is finite, provided that for each $1 \leq k \leq K$,

$$\frac{1}{2\sigma_k^2} - \frac{1}{V_{\text{SG}}^2} > 0,$$

which means $V_{\text{SG}}^2 > 2\sigma_k^2$ for each $1 \leq k \leq K$.

C.2 Gaussian convolution. The Lipschitz continuity and linear growth of the score can be found as Ding et al. (2024a, Propositions 3.2 and 3.5).

Sub-Gaussian tails. It is straightforward that

$$\begin{aligned}
 & \int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) \pi_0(\mathbf{x}_0) d\mathbf{x}_0 \\
 &= \int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) \left\{ \int \gamma_{d,\sigma^2}(\mathbf{x}_0 - \mathbf{z}) d\nu(\mathbf{z}) \right\} d\mathbf{x}_0 \\
 &= \int \left\{ \int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) \gamma_{d,\sigma^2}(\mathbf{x}_0 - \mathbf{z}) d\mathbf{x}_0 \right\} d\nu(\mathbf{z}) \\
 &= \int (2\pi\sigma^2)^{-\frac{d}{2}} \left\{ \int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2} - \frac{\|\mathbf{x}_0 - \mathbf{z}\|_2^2}{2\sigma^2}\right) d\mathbf{x}_0 \right\} d\nu(\mathbf{z}) \\
 &= \int (2\pi\sigma_k^2)^{-\frac{d}{2}} \left[\int \exp\left\{ -\left(\frac{1}{2\sigma^2} - \frac{1}{V_{\text{SG}}^2}\right) \|\mathbf{x}_0\|_2^2 + \frac{1}{\sigma^2} \langle \mathbf{x}_0, \mathbf{z} \rangle - \frac{1}{2\sigma^2} \|\mathbf{z}\|_2^2 \right\} d\mathbf{x}_0 \right] d\nu(\mathbf{z}),
 \end{aligned}$$

where the second equality holds from Fubini's theorem. If $V_{\text{SG}}^2 > 2\sigma^2$, the inner integral is finite for each $\mathbf{z}$. This means inner integral is a continuous function of $\mathbf{z}$, and thus it is uniformly bounded on $\text{supp}(\nu)$. Since $\text{supp}(\nu)$ is compact, the outer integral is finite.

D Proofs of Lemmas in Section 4

In this section, we provide proofs of Lemmas in Section 4.

D.1 Proof of Lemma 4.1.

Proof of Lemma 4.1. According to the solution of the forward process (2.2), we have

$$p_{\mathbf{X}_t|\mathbf{X}_0}(\mathbf{x}|\mathbf{x}_0) \propto \exp\left(-\frac{\|\mathbf{x} - \mu_t \mathbf{x}_0\|_2^2}{2\sigma_t^2}\right),$$

which implies

$$(D.1) \quad \nabla_{\mathbf{x}} p_{\mathbf{X}_t|\mathbf{X}_0}(\mathbf{x}|\mathbf{x}_0) = \frac{\mu_t \mathbf{x}_0 - \mathbf{x}}{\sigma_t^2} p_{\mathbf{X}_t|\mathbf{X}_0}(\mathbf{x}|\mathbf{x}_0).$$

Using Chapman-Kolmogorov identity yields

$$\begin{aligned}
 (D.2) \quad q_t(\mathbf{x}|\mathbf{y}) &= \int p_{\mathbf{X}_t|\mathbf{X}_0, \mathbf{Y}}(\mathbf{x}|\mathbf{x}_0, \mathbf{y}) p_{\mathbf{X}_0|\mathbf{Y}}(\mathbf{x}_0|\mathbf{y}) d\mathbf{x}_0 \\
 &= \int p_{\mathbf{X}_t|\mathbf{X}_0}(\mathbf{x}|\mathbf{x}_0) p_{\mathbf{X}_0|\mathbf{Y}}(\mathbf{x}_0|\mathbf{y}) d\mathbf{x}_0,
 \end{aligned}$$

where the second equality holds from the fact that $\mathbf{X}_t$ is conditional independent of $\mathbf{Y}$ given $\mathbf{X}_0$. Then we find

$$\begin{aligned}
 \nabla \log q_t(\mathbf{x}|\mathbf{y}) &= \frac{1}{q_t(\mathbf{x}|\mathbf{y})} \int \nabla p_{\mathbf{X}_t|\mathbf{X}_0}(\mathbf{x}|\mathbf{x}_0) p_{\mathbf{X}_0|\mathbf{Y}}(\mathbf{x}_0|\mathbf{y}) d\mathbf{x}_0 \\
 &= \frac{1}{q_t(\mathbf{x}|\mathbf{y})} \int \frac{\mu_t \mathbf{x}_0 - \mathbf{x}}{\sigma_t^2} p_{\mathbf{X}_t|\mathbf{X}_0}(\mathbf{x}|\mathbf{x}_0) p_{\mathbf{X}_0|\mathbf{Y}}(\mathbf{x}_0|\mathbf{y}) d\mathbf{x}_0 \\
 &= \frac{1}{q_t(\mathbf{x}|\mathbf{y})} \int \frac{\mu_t \mathbf{x}_0 - \mathbf{x}}{\sigma_t^2} p_{\mathbf{X}_t, \mathbf{X}_0|\mathbf{Y}}(\mathbf{x}, \mathbf{x}_0|\mathbf{y}) d\mathbf{x}_0 \\
 &= \int \frac{\mu_t \mathbf{x}_0 - \mathbf{x}}{\sigma_t^2} p_{\mathbf{X}_0|\mathbf{X}_t, \mathbf{Y}}(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) d\mathbf{x}_0,
 \end{aligned}$$

where the first equality holds from (D.2), the second equality invokes (D.1), the third equality uses the fact that $\mathbf{X}_t$ is conditional independent of $\mathbf{Y}$ given $\mathbf{X}_0$, and the last equality is due to the definition of the conditional density. This completes the proof. $\square$

D.2 Proof of Lemma 4.6.

Proof of Lemma 4.6. According to Bayes' rule, we have

$$\begin{aligned} p_{\mathbf{X}_0|\mathbf{X}_t, \mathbf{Y}}(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) &= \frac{1}{p_{\mathbf{X}_t|\mathbf{Y}}(\mathbf{x}|\mathbf{y})} p_{\mathbf{X}_t|\mathbf{X}_0, \mathbf{Y}}(\mathbf{x}|\mathbf{x}_0, \mathbf{y}) p_{\mathbf{X}_0|\mathbf{Y}}(\mathbf{x}_0|\mathbf{y}) \\ &= \frac{1}{p_{\mathbf{X}_t|\mathbf{Y}}(\mathbf{x}|\mathbf{y})} p_{\mathbf{X}_t|\mathbf{X}_0}(\mathbf{x}|\mathbf{x}_0) p_{\mathbf{X}_0|\mathbf{Y}}(\mathbf{x}_0|\mathbf{y}), \end{aligned}$$

where the last equality invokes the fact that $\mathbf{X}_t$ is conditional independent of $\mathbf{Y}$ given $\mathbf{X}_0$. As a consequence,

$$p_{\mathbf{X}_0|\mathbf{X}_t, \mathbf{Y}}(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) \propto \exp\left(-\frac{\|\mathbf{x} - \mu_t \mathbf{x}_0\|_2^2}{2\sigma_t^2}\right) p_{\mathbf{X}_0|\mathbf{Y}}(\mathbf{x}_0|\mathbf{y}),$$

which implies

$$\nabla_{\mathbf{x}_0}^2 \log p_{\mathbf{X}_0|\mathbf{X}_t, \mathbf{Y}}(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) = -\frac{\mu_t^2}{\sigma_t^2} \mathbf{I}_d + \nabla_{\mathbf{x}_0}^2 \log p_{\mathbf{X}_0|\mathbf{Y}}(\mathbf{x}_0|\mathbf{y}).$$

Combining the above equality with Assumption 1 completes the proof. $\square$

D.3 Proof of Lemma 4.10.

Proof of Lemma 4.10. It is straightforward that

$$\begin{aligned} &\int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) q_0(\mathbf{x}_0|\mathbf{y}) d\mathbf{x}_0 \\ &= \int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) \frac{\exp(-\ell_{\mathbf{y}}(\mathbf{x}_0))}{\int \exp(-\ell_{\mathbf{y}}(\mathbf{x}_0)) \pi_0(\mathbf{x}_0) d\mathbf{x}_0} \pi_0(\mathbf{x}_0) d\mathbf{x}_0 \\ &\leq \frac{\sup_{\mathbf{x}_0} \exp(-\ell_{\mathbf{y}}(\mathbf{x}_0))}{\int \exp(-\ell_{\mathbf{y}}(\mathbf{x}_0)) \pi_0(\mathbf{x}_0) d\mathbf{x}_0} \int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) \pi_0(\mathbf{x}_0) d\mathbf{x}_0 \leq \kappa_{\mathbf{y}} C_{\text{SG}}. \end{aligned}$$

where the first equality follows from the Bayes' rule, and the second inequality holds from Assumption 2. This completes the proof. $\square$

D.4 Proof of Lemma 4.12. Before proceeding, we introduce two auxiliary lemmas.

Lemma D.1. *Let μ be a probability distribution on $\mathbb{R}^d$ having sub-Gaussian tails, that is, there exist constants $V_{\text{SG}} > 0$ and $C_{\text{SG}} > 0$, such that*

$$\int \exp\left(\frac{\|\mathbf{x}\|_2^2}{V_{\text{SG}}^2}\right) d\mu(\mathbf{x}) \leq C_{\text{SG}}.$$

Then it holds that

$$\iint \exp\left(\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{2V_{\text{SG}}^2}\right) d\mu(\mathbf{x}) d\mu(\mathbf{x}') \leq C_{\text{SG}}^2.$$

Proof of Lemma D.1. It is straightforward that

$$\begin{aligned} &\iint \exp\left(\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{2V_{\text{SG}}^2}\right) d\mu(\mathbf{x}) d\mu(\mathbf{x}') \\ &\leq \iint \exp\left(\frac{\|\mathbf{x}\|_2^2 + \|\mathbf{x}'\|_2^2}{V_{\text{SG}}^2}\right) d\mu(\mathbf{x}) d\mu(\mathbf{x}') \\ &= \left\{ \int \exp\left(\frac{\|\mathbf{x}\|_2^2}{V_{\text{SG}}^2}\right) d\mu(\mathbf{x}) \right\} \left\{ \int \exp\left(\frac{\|\mathbf{x}'\|_2^2}{V_{\text{SG}}^2}\right) d\mu(\mathbf{x}') \right\} \leq C_{\text{SG}}^2, \end{aligned}$$

which completes the proof. $\square$

Lemma D.2 (Chen et al. (2021, Theorem 2)). *Let μ be a probability distribution on $\mathbb{R}^d$ having sub-Gaussian tails, that is, there exist constants $V_{\text{SG}} > 0$ and $C_{\text{SG}} > 0$, such that*

$$\iint \exp\left(\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{2V_{\text{SG}}^2}\right) d\mu(\mathbf{x}) d\mu(\mathbf{x}') \leq C_{\text{SG}}^2.$$

Then for $\sigma^2 > 2V_{\text{SG}}^2$, the convolution distribution

$$\mu_\sigma(\mathbf{z}) := \int \gamma_{d, \sigma^2}(\mathbf{z} - \mathbf{x}) d\mu(\mathbf{x})$$

satisfies the log-Sobolev inequality with

$$C_{\text{LSI}}(\mu_\sigma) \leq 3\sigma^2 \left\{ \frac{\sigma^2}{\sigma^2 - 2V_{\text{SG}}^2} + \exp\left(\frac{2V_{\text{SG}}^2}{\sigma^2 - 2V_{\text{SG}}^2} \log C_{\text{SG}}^2\right) \right\} \left\{ 1 + \frac{2V_{\text{SG}}^2}{\sigma^2 - 2V_{\text{SG}}^2} \log C_{\text{SG}}^2 \right\}.$$

Proof of Lemma 4.12. Define an auxiliary distribution $\mathcal{M}(\mu_t) \sharp q_0(\mathbf{z}_t | \mathbf{y}) := \mu_t^{-d} q_0(\mu_t^{-1} \mathbf{z}_t | \mathbf{y})$. We first show this distribution has sub-Gaussian tails, given $q_0(\cdot | \mathbf{y})$ is sub-Gaussian as shown by Lemma 4.10. Indeed,

$$\begin{aligned} & \iint \exp\left(\frac{\|\mathbf{z}_t - \mathbf{z}'_t\|_2^2}{2\mu_t^2 V_{\text{SG}}^2}\right) \mathcal{M}(\mu_t) \sharp q_0(\mathbf{z}_t | \mathbf{y}) \mathcal{M}(\mu_t) \sharp q_0(\mathbf{z}'_t | \mathbf{y}) d\mathbf{z}_t d\mathbf{z}'_t \\ (D.3) \quad &= \iint \exp\left(\frac{\|\mathbf{x}_0 - \mathbf{x}'_0\|_2^2}{2V_{\text{SG}}^2}\right) q_0(\mathbf{x}_0 | \mathbf{y}) q_0(\mathbf{x}'_0 | \mathbf{y}) d\mathbf{x}_0 d\mathbf{x}'_0 \leq \kappa_{\mathbf{y}}^2 C_{\text{SG}}^2, \end{aligned}$$

where the inequality holds from Lemmas 4.10 and D.1. We next turn to verify the log-Sobolev inequality of $q_t(\cdot | \mathbf{y})$. By recalling (2.5), we find

$$q_t(\mathbf{x}_t | \mathbf{y}) = \int \gamma_{d, \sigma_t^2}(\mathbf{x}_t - \mu_t \mathbf{x}_0) q_0(\mathbf{x}_0 | \mathbf{y}) d\mathbf{x}_0 = \int \gamma_{d, \sigma_t^2}(\mathbf{x}_t - \mathbf{z}_t) \mathcal{M}(\mu_t) \sharp q_0(\mathbf{z}_t | \mathbf{y}) d\mathbf{z}_t,$$

where we used a change of variable $\mathbf{z}_t := \mu_t \mathbf{x}_0$. Substituting (D.3) into Lemma D.2, it follows that

$$C_{\text{LSI}}(q_t(\cdot | \mathbf{y})) \leq 3\sigma_t^2 \left\{ \frac{\sigma_t^2}{\sigma_t^2 - 2\mu_t^2 V_{\text{SG}}^2} + \exp\left(\frac{2\mu_t^2 V_{\text{SG}}^2}{\sigma_t^2 - 2\mu_t^2 V_{\text{SG}}^2} \log \kappa_{\mathbf{y}}^2 C_{\text{SG}}^2\right) \right\}^2,$$

where we used the fact

$$1 + \frac{2\mu_t^2 V_{\text{SG}}^2}{\sigma_t^2 - 2\mu_t^2 V_{\text{SG}}^2} \log \kappa_{\mathbf{y}}^2 C_{\text{SG}}^2 \leq \frac{\sigma_t^2}{\sigma_t^2 - 2\mu_t^2 V_{\text{SG}}^2} + \exp\left(\frac{2\mu_t^2 V_{\text{SG}}^2}{\sigma_t^2 - 2\mu_t^2 V_{\text{SG}}^2} \log \kappa_{\mathbf{y}}^2 C_{\text{SG}}^2\right).$$

Here we used $\sigma_t^2 - 2\mu_t^2 V_{\text{SG}}^2 > 0$ for $t > \underline{t} := \frac{1}{2} \log(1 + 2V_{\text{SG}}^2)$. Further, note that

$$\frac{\sigma_t^2}{\sigma_t^2 - 2\mu_t^2 V_{\text{SG}}^2} \leq \frac{\sigma_t^2 + 2\mu_t^2 V_{\text{SG}}^2}{\sigma_t^2 - 2\mu_t^2 V_{\text{SG}}^2} \leq \exp\left(\frac{\sigma_t^2 + 2\mu_t^2 V_{\text{SG}}^2}{\sigma_t^2 - 2\mu_t^2 V_{\text{SG}}^2} \log \kappa_{\mathbf{y}}^2 C_{\text{SG}}^2\right).$$

This completes the proof. □

D.5 Proof of Theorem 4.16.

Proof of Theorem 4.16. A direct conclusion of Lemmas 4.6 and 4.12. □

E Proof of the Error Decomposition

In this section, we aim to provide a proof of the error decomposition of the posterior estimation (Lemma 5.4). Recall the time-reversal process (2.7):

$$d\bar{\mathbf{X}}_t^{\mathbf{y}} = (\bar{\mathbf{X}}_t^{\mathbf{y}} + 2\nabla_{\mathbf{x}} \log \bar{q}_t(\bar{\mathbf{X}}_t^{\mathbf{y}}|\mathbf{y})) dt + \sqrt{2} d\mathbf{B}_t, \quad \bar{\mathbf{X}}_0^{\mathbf{y}} \sim q_T(\cdot|\mathbf{y}), \quad t \in (0, T - T_0),$$

where $T_0 \in (0, T)$ is the early-stopping time, and $\bar{q}_t(\cdot|\mathbf{y}) := q_{T-t}(\cdot|\mathbf{y})$. In practical applications, the posterior score $\nabla_{\mathbf{x}} \log \bar{q}_t(\cdot|\mathbf{y})$ is intractable, and we only have an estimator $\hat{\mathbf{s}}_m^S(t, \cdot, \mathbf{y}) \approx \nabla \log \bar{q}_t(\cdot|\mathbf{y})$ (4.1). The time-reversal process with this estimated score is given as

$$d\hat{\mathbf{X}}_t^{\mathbf{y}} = (\hat{\mathbf{X}}_t^{\mathbf{y}} + 2\hat{\mathbf{s}}_m^S(T - t, \hat{\mathbf{X}}_t^{\mathbf{y}}, \mathbf{y})) dt + \sqrt{2} d\mathbf{B}_t, \quad \hat{\mathbf{X}}_0^{\mathbf{y}} \sim \hat{q}_T^U(\cdot|\mathbf{y}), \quad t \in (0, T - T_0).$$

Denote by $\hat{q}_t(\cdot|\mathbf{y})$ the marginal density of $\hat{\mathbf{X}}_t^{\mathbf{y}}$ for each $t \in (0, T - T_0)$.

E.1 Error of the early-stopping.

Lemma E.1. *Suppose Assumption 2 holds. Let $T_0 \in (0, \frac{1}{2})$ and $\varepsilon_{T_0} \in (0, 1)$. Then*

$$\mathbb{W}_2^2(q_0(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\# \hat{q}_{T-T_0}^R(\cdot|\mathbf{y})) \leq C \left\{ \frac{\sigma_{T_0}^2}{\mu_{T_0}^2} + \varepsilon_{T_0} \log \left(\frac{\kappa_{\mathbf{y}}}{\varepsilon_{T_0}} \right) \right\},$$

provided that

$$\|q_{T_0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0}(\cdot|\mathbf{y})\|_{\text{TV}} = \varepsilon_{T_0}.$$

Here C is a constant only depending on d , V_{SG} , and C_{SG} .

Proof of Lemma E.1. By applying Lemmas E.2, E.3, and E.4, we have

$$\begin{aligned} & \mathbb{W}_2^2(q_0(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\# \hat{q}_{T-T_0}^R(\cdot|\mathbf{y})) \\ & \leq 3\mathbb{W}_2^2(q_0(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\# q_{T_0}(\cdot|\mathbf{y})) + 3\mathbb{W}_2^2(\mathcal{M}(\mu_{T_0}^{-1})\# q_{T_0}(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\# q_{T_0}^R(\cdot|\mathbf{y})) \\ & \quad + 3\mathbb{W}_2^2(\mathcal{M}(\mu_{T_0}^{-1})\# q_{T_0}^R(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\# \hat{q}_{T-T_0}^R(\cdot|\mathbf{y})) \\ & \leq \frac{3d\sigma_{T_0}^2}{\mu_{T_0}^2} + \frac{3D}{\mu_{T_0}^2} \kappa_{\mathbf{y}} \exp \left(- \frac{R^2}{4\mu_{T_0}^2 V_{\text{SG}}^2 + 16\sigma_{T_0}^2} \right) + \frac{2R^2}{\mu_{T_0}^2} \|q_{T_0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0}(\cdot|\mathbf{y})\|_{\text{TV}} \\ & \leq C \left\{ \frac{\sigma_{T_0}^2}{\mu_{T_0}^2} + \kappa_{\mathbf{y}} \exp \left(- \frac{R^2}{4\mu_{T_0}^2 V_{\text{SG}}^2 + 16\sigma_{T_0}^2} \right) + R^2 \|q_{T_0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0}(\cdot|\mathbf{y})\|_{\text{TV}} \right\}, \end{aligned}$$

where the first inequality is owing to the triangular inequality of 2-Wasserstein distance, and the last inequality used the fact that $\mu_{T_0}^{-1} \leq e$ for $T_0 \leq \frac{1}{2}$. Here the constant C only depends on d , V_{SG} , and C_{SG} . By setting

$$\varepsilon_{T_0} = \kappa_{\mathbf{y}} \exp \left(- \frac{R^2}{4\mu_{T_0}^2 V_{\text{SG}}^2 + 16\sigma_{T_0}^2} \right),$$

we have

$$R^2 = (4\mu_{T_0}^2 V_{\text{SG}}^2 + 16\sigma_{T_0}^2) \log \left(\frac{\kappa_{\mathbf{y}}}{\varepsilon_{T_0}} \right).$$

This completes the proof. $\square$

Lemma E.2. *For each $T_0 > 0$, it follows that*

$$\mathbb{W}_2^2(q_0(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\# q_{T_0}(\cdot|\mathbf{y})) \leq \frac{d\sigma_{T_0}^2}{\mu_{T_0}^2}.$$

Proof of Lemma E.2. We first produce a coupling of $q_0(\cdot|\mathbf{y})$ and $\mathcal{M}(\mu_{T_0}^{-1})\sharp q_{T_0}(\cdot|\mathbf{y})$ as follows. Let $\mathbf{X}_0^{\mathbf{y}} \sim q_0(\cdot|\mathbf{y})$, and let $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{I}_d)$ be independent of $\mathbf{X}_0^{\mathbf{y}}$. Then

$$\mathbf{Z}_{T_0}^{\mathbf{y}} := \mathbf{X}_0^{\mathbf{y}} + \frac{\sigma_{T_0}}{\mu_{T_0}} \boldsymbol{\varepsilon} \sim \mathcal{M}(\mu_{T_0}^{-1})\sharp q_{T_0}(\cdot|\mathbf{y}).$$

As a consequence,

$$\mathbb{W}_2^2(q_0(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\sharp q_{T_0}(\cdot|\mathbf{y})) \leq \mathbb{E}[\|\mathbf{X}_0^{\mathbf{y}} - \mathbf{Z}_{T_0}^{\mathbf{y}}\|_2^2] = \frac{\sigma_{T_0}^2}{\mu_{T_0}^2} \mathbb{E}[\|\boldsymbol{\varepsilon}\|_2^2] = \frac{d\sigma_{T_0}^2}{\mu_{T_0}^2},$$

which completes the proof. $\square$

Lemma E.3. Suppose Assumption 2 holds. For each $T_0 > 0$, it follows that

$$\mathbb{W}_2(\mathcal{M}(\mu_{T_0}^{-1})\sharp q_{T_0}(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\sharp q_{T_0}^R(\cdot|\mathbf{y})) \leq \frac{D}{\mu_{T_0}^2} \kappa_{\mathbf{y}} \exp\left(-\frac{R^2}{4\mu_{T_0}^2 V_{\text{SG}}^2 + 16\sigma_{T_0}^2}\right),$$

where D is a constant only depending on d , V_{SG} , and C_{SG} .

Proof of Lemma E.3. Let $\mathbf{X}_{T_0}^{\mathbf{y}} \sim q_{T_0}(\cdot|\mathbf{y})$. Then $(\mu_{T_0}^{-1}\mathbf{X}_{T_0}^{\mathbf{y}}, \mu_{T_0}^{-1}\mathbf{X}_{T_0}^{\mathbf{y}} \mathbb{1}\{\|\mathbf{X}_{T_0}^{\mathbf{y}}\|_2 \leq R\})$ is a coupling of $\mathcal{M}(\mu_{T_0}^{-1})\sharp q_{T_0}(\cdot|\mathbf{y})$ and $\mathcal{M}(\mu_{T_0}^{-1})\sharp q_{T_0}^R(\cdot|\mathbf{y})$. Therefore,

$$\begin{aligned} & \mathbb{W}_2^2(\mathcal{M}(\mu_{T_0}^{-1})\sharp q_{T_0}(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\sharp q_{T_0}(\cdot|\mathbf{y})) \\ & \leq \mathbb{E}[\|\mu_{T_0}^{-1}\mathbf{X}_{T_0}^{\mathbf{y}} - \mu_{T_0}^{-1}\mathbf{X}_{T_0}^{\mathbf{y}} \mathbb{1}\{\|\mathbf{X}_{T_0}^{\mathbf{y}}\|_2 \leq R\}\|_2^2] \\ & = \frac{1}{\mu_{T_0}^2} \int \|\mathbf{x} - \mathbf{x} \mathbb{1}\{\|\mathbf{x}\|_2 \leq R\}\|_2^2 q_{T_0}(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \\ & = \frac{1}{\mu_{T_0}^2} \int \|\mathbf{x}\|_2^2 \mathbb{1}\{\|\mathbf{x}\|_2 > R\} q_{T_0}(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \\ & \leq \frac{1}{\mu_{T_0}^2} \mathbb{E}^{\frac{1}{2}}[\|\mathbf{X}_{T_0}^{\mathbf{y}}\|_2^4] \mathbb{P}^{\frac{1}{2}}\{\|\mathbf{X}_{T_0}^{\mathbf{y}}\|_2 > R\} \\ & \leq \frac{D}{\mu_{T_0}^2} \kappa_{\mathbf{y}} \exp\left(-\frac{R^2}{4\mu_{T_0}^2 V_{\text{SG}}^2 + 16\sigma_{T_0}^2}\right), \end{aligned}$$

where the first inequality holds from the definition of 2-Wasserstein distance, the second inequality follows from Cauchy-Schwarz inequality, and the last inequality is due to Lemmas H.3 and H.4. This completes the proof. $\square$

Lemma E.4. For each $T_0 > 0$ and $R \geq 1$, it follows that

$$\mathbb{W}_2^2(\mathcal{M}(\mu_{T_0}^{-1})\sharp q_{T_0}^R(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1})\sharp \hat{q}_{T-T_0}^R(\cdot|\mathbf{y})) \leq \frac{2R^2}{\mu_{T_0}^2} \|q_{T_0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0}(\cdot|\mathbf{y})\|_{\text{TV}}.$$

Proof of Lemma E.4. Let $\mathbf{X}_{T_0}^{\mathbf{y},R} \sim q_{T_0}^R(\cdot|\mathbf{y})$ and $\hat{\mathbf{X}}_{T-T_0}^{\mathbf{y},R} \sim \hat{q}_{T-T_0}^R(\cdot|\mathbf{y})$ be optimal coupled. This means

$$(E.1) \quad \mathbb{W}_2^2(q_{T_0}^R(\cdot|\mathbf{y}), \hat{q}_{T-T_0}^R(\cdot|\mathbf{y})) = \mathbb{E}[\|\mathbf{X}_{T_0}^{\mathbf{y},R} - \hat{\mathbf{X}}_{T-T_0}^{\mathbf{y},R}\|_2^2].$$

It is apparent that $\mu_{T_0}^{-1} \mathbf{X}_{T_0}^{\mathbf{y}, R} \sim \mathcal{M}(\mu_{T_0}^{-1}) \# q_{T_0}^R(\cdot | \mathbf{y})$ and $\mu_{T_0}^{-1} \hat{\mathbf{X}}_{T-T_0}^{\mathbf{y}, R} \sim \mathcal{M}(\mu_{T_0}^{-1}) \# \hat{q}_{T-T_0}^R(\cdot | \mathbf{y})$. Hence,

$$(E.2) \quad \begin{aligned} & \mathbb{W}_2^2(\mathcal{M}(\mu_{T_0}^{-1}) \# q_{T_0}^R(\cdot | \mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1}) \# \hat{q}_{T-T_0}^R(\cdot | \mathbf{y})) \\ & \leq \mathbb{E}[\|\mu_{T_0}^{-1} \mathbf{X}_{T_0}^{\mathbf{y}, R} - \mu_{T_0}^{-1} \hat{\mathbf{X}}_{T-T_0}^{\mathbf{y}, R}\|_2^2] = \frac{1}{\mu_{T_0}^2} \mathbb{W}_2^2(q_{T_0}^R(\cdot | \mathbf{y}), \hat{q}_{T-T_0}^R(\cdot | \mathbf{y})), \end{aligned}$$

where the equality holds from (E.1). On the other hand,

$$(E.3) \quad \begin{aligned} \mathbb{W}_2^2(q_{T_0}^R(\cdot | \mathbf{y}), \hat{q}_{T-T_0}^R(\cdot | \mathbf{y})) &= 2R^2 \|q_{T_0}^R(\cdot | \mathbf{y}) - \hat{q}_{T-T_0}^R(\cdot | \mathbf{y})\|_{\text{TV}} \\ &\leq 2R^2 \|q_{T_0}(\cdot | \mathbf{y}) - \hat{q}_{T-T_0}(\cdot | \mathbf{y})\|_{\text{TV}}, \end{aligned}$$

where the first inequality is due to Villani (2009, Theorem 6.15), and the second inequality follows from the data processing inequality (Duchi, 2025, Proposition 2.2.13). Combining (E.2) and (E.3) completes the proof. $\square$

E.2 Error decomposition for TV-distance. The following lemma indicates that the TV-error in time $T - T_0$ consists of two parts: the posterior score estimation error and the warm-start error.

Lemma E.5. *For each $0 < T_0 < T$, the following inequality holds:*

$$\begin{aligned} & \|q_{T_0}(\cdot | \mathbf{y}) - \hat{q}_{T-T_0}(\cdot | \mathbf{y})\|_{\text{TV}}^2 \\ & \leq \underbrace{\int_{T_0}^T \mathbb{E}[\|\nabla \log q_t(\mathbf{X}_t^{\mathbf{y}} | \mathbf{y}) - \hat{\mathbf{s}}_m^S(t, \mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|_2^2] dt}_{\text{posterior score estimation}} + \underbrace{2\|q_T(\cdot | \mathbf{y}) - \hat{q}_0^U(\cdot | \mathbf{y})\|_{\text{TV}}^2}_{\text{warm-start}}, \end{aligned}$$

where the expectation is taken with respect to $\mathbf{X}_t^{\mathbf{y}} \sim q_t(\cdot | \mathbf{y})$.

Proof of Lemma E.5. Combining Lemmas E.6 and E.7 completes the proof. $\square$

Recall that the time-reversal process of the posterior forward process (2.4) is defined as:

$$(E.4) \quad d\bar{\mathbf{X}}_t^{\mathbf{y}} = (\bar{\mathbf{X}}_t^{\mathbf{y}} + 2\nabla \log q_{T-t}(\bar{\mathbf{X}}_t^{\mathbf{y}} | \mathbf{y})) dt + \sqrt{2} d\mathbf{B}_t, \quad \bar{\mathbf{X}}_0^{\mathbf{y}} \sim q_T(\cdot | \mathbf{y}), \quad t \in (0, T).$$

and the approximate reverse process is defined as:

$$d\hat{\mathbf{X}}_t^{\mathbf{y}} = (\hat{\mathbf{X}}_t^{\mathbf{y}} + 2\hat{\mathbf{s}}_m^S(T-t, \hat{\mathbf{X}}_t^{\mathbf{y}}, \mathbf{y})) dt + \sqrt{2} d\mathbf{B}_t, \quad \hat{\mathbf{X}}_0^{\mathbf{y}} \sim \hat{q}_T^U(\cdot | \mathbf{y}), \quad t \in (0, T-T_0),$$

Let $\bar{q}_{s|t}(\cdot | \mathbf{x}, \mathbf{y})$ be the conditional density of $\bar{\mathbf{X}}_s^{\mathbf{y}}$ given $\bar{\mathbf{X}}_t^{\mathbf{y}} = \mathbf{x}$. Denote by $\bar{q}_{s,t}(\cdot | \mathbf{y})$ the joint density of $\bar{\mathbf{X}}_s^{\mathbf{y}}$ and $\bar{\mathbf{X}}_t^{\mathbf{y}}$. Let $\hat{q}_{s|t}(\cdot | \mathbf{x}, \mathbf{y})$ be the conditional density of $\hat{\mathbf{X}}_s^{\mathbf{y}}$ given $\hat{\mathbf{X}}_t^{\mathbf{y}} = \mathbf{x}$. Denote by $\hat{q}_{s,t}(\cdot | \mathbf{y})$ the joint density of $\hat{\mathbf{X}}_s^{\mathbf{y}}$ and $\hat{\mathbf{X}}_t^{\mathbf{y}}$.

Lemma E.6. *For each $0 < T_0 < T$, the following inequality holds:*

$$\begin{aligned} & \|q_{T_0}(\cdot | \mathbf{y}) - \hat{q}_{T-T_0}(\cdot | \mathbf{y})\|_{\text{TV}}^2 \\ & \leq \int \text{KL}(\bar{q}_{T-T_0|0}(\cdot | \bar{\mathbf{x}}_0, \mathbf{y}) \| \hat{q}_{T-T_0|0}(\cdot | \bar{\mathbf{x}}_0, \mathbf{y})) \bar{q}_0(\bar{\mathbf{x}}_0 | \mathbf{y}) d\bar{\mathbf{x}}_0 + 2\|q_T(\cdot | \mathbf{y}) - \hat{q}_0^U(\cdot | \mathbf{y})\|_{\text{TV}}^2, \end{aligned}$$

where the expectation is taken with respect to $\bar{\mathbf{X}}_0^{\mathbf{y}} \sim \bar{q}_0(\cdot | \mathbf{y})$.

Proof of Lemma E.6. It is straightforward that

$$\begin{aligned}
 & \|\bar{q}_{T-T_0,0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0,0}(\cdot|\mathbf{y})\|_{\text{TV}} \\
 &= \frac{1}{2} \iint |\bar{q}_{T-T_0,0}(\bar{\mathbf{x}}_{T-T_0}, \bar{\mathbf{x}}_0|\mathbf{y}) - \hat{q}_{T-T_0,0}(\bar{\mathbf{x}}_{T-T_0}, \bar{\mathbf{x}}_0|\mathbf{y})| d\bar{\mathbf{x}}_{T-T_0} d\bar{\mathbf{x}}_0 \\
 &\geq \frac{1}{2} \int \left| \int \bar{q}_{T-T_0,0}(\bar{\mathbf{x}}_{T-T_0}, \bar{\mathbf{x}}_0|\mathbf{y}) d\bar{\mathbf{x}}_0 - \int \hat{q}_{T-T_0,0}(\bar{\mathbf{x}}_{T-T_0}, \bar{\mathbf{x}}_0|\mathbf{y}) d\bar{\mathbf{x}}_0 \right| d\bar{\mathbf{x}}_{T-T_0} \\
 &= \frac{1}{2} \int |\bar{q}_{T-T_0}(\bar{\mathbf{x}}_{T-T_0}|\mathbf{y}) - \hat{q}_{T-T_0}(\bar{\mathbf{x}}_{T-T_0}|\mathbf{y})| d\bar{\mathbf{x}}_{T-T_0} \\
 (E.5) \quad &= \|q_{T_0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0}(\cdot|\mathbf{y})\|_{\text{TV}},
 \end{aligned}$$

where the inequality holds from Jensen's inequality. On the other hand, we have

$$\begin{aligned}
 & \|\bar{q}_{T-T_0,0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0,0}(\cdot|\mathbf{y})\|_{\text{TV}} \\
 &= \frac{1}{2} \iint |\bar{q}_{T-T_0|0}(\bar{\mathbf{x}}_{T-T_0}|\bar{\mathbf{x}}_0, \mathbf{y}) \bar{q}_0(\bar{\mathbf{x}}_0|\mathbf{y}) - \hat{q}_{T-T_0|0}(\bar{\mathbf{x}}_{T-T_0}|\bar{\mathbf{x}}_0, \mathbf{y}) \hat{q}_0^U(\bar{\mathbf{x}}_0|\mathbf{y})| d\bar{\mathbf{x}}_{T-T_0} d\bar{\mathbf{x}}_0 \\
 &\leq \frac{1}{2} \iint |\bar{q}_{T-T_0|0}(\bar{\mathbf{x}}_{T-T_0}|\bar{\mathbf{x}}_0, \mathbf{y}) - \hat{q}_{T-T_0|0}(\bar{\mathbf{x}}_{T-T_0}|\bar{\mathbf{x}}_0, \mathbf{y})| \bar{q}_0(\bar{\mathbf{x}}_0|\mathbf{y}) d\bar{\mathbf{x}}_{T-T_0} d\bar{\mathbf{x}}_0 \\
 &\quad + \frac{1}{2} \iint \hat{q}_{T-T_0|0}(\bar{\mathbf{x}}_{T-T_0}|\bar{\mathbf{x}}_0, \mathbf{y}) |\bar{q}_0(\bar{\mathbf{x}}_0|\mathbf{y}) - \hat{q}_0^U(\bar{\mathbf{x}}_0|\mathbf{y})| d\bar{\mathbf{x}}_{T-T_0} d\bar{\mathbf{x}}_0 \\
 (E.6) \quad &= \int \|\bar{q}_{T-T_0|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y}) - \hat{q}_{T-T_0|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y})\|_{\text{TV}} \bar{q}_0(\bar{\mathbf{x}}_0|\mathbf{y}) d\bar{\mathbf{x}}_0 + \|q_T(\cdot|\mathbf{y}) - \hat{q}_0^U(\cdot|\mathbf{y})\|_{\text{TV}},
 \end{aligned}$$

where the inequality invokes the triangular inequality. Combining (E.5) and (E.6) implies

$$\begin{aligned}
 & \|q_{T_0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0}(\cdot|\mathbf{y})\|_{\text{TV}}^2 \\
 &\leq 2 \int \|\bar{q}_{T-T_0|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y}) - \hat{q}_{T-T_0|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y})\|_{\text{TV}}^2 \bar{q}_0(\bar{\mathbf{x}}_0|\mathbf{y}) d\bar{\mathbf{x}}_0 + 2 \|q_T(\cdot|\mathbf{y}) - \hat{q}_0^U(\cdot|\mathbf{y})\|_{\text{TV}}^2 \\
 &\leq \int \text{KL}(\bar{q}_{T-T_0|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y}) \|\hat{q}_{T-T_0|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y})) \bar{q}_0(\bar{\mathbf{x}}_0|\mathbf{y}) d\bar{\mathbf{x}}_0 + 2 \|q_T(\cdot|\mathbf{y}) - \hat{q}_0^U(\cdot|\mathbf{y})\|_{\text{TV}}^2,
 \end{aligned}$$

where the first inequality used Jensen's inequality, and the second inequality holds from Pinsker's inequality (Tsybakov, 2009, Lemma 2.5). This completes the proof. $\square$

Lemma E.7. For each $0 < T_0 < T$, the following inequality holds:

$$\begin{aligned}
 & \int \text{KL}(\bar{q}_{T-T_0|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y}) \|\hat{q}_{T-T_0|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y})) \bar{q}_0(\bar{\mathbf{x}}_0|\mathbf{y}) d\bar{\mathbf{x}}_0 \\
 &\leq \frac{1}{2} \int_{T_0}^T \mathbb{E} [\|\nabla \log q_t(\mathbf{X}_t^{\mathbf{y}}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(t, \mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|_2^2] dt,
 \end{aligned}$$

where the expectation is taken with respect to $\mathbf{X}_t^{\mathbf{y}} \sim q_t(\cdot|\mathbf{y})$.

Proof of Lemma E.7. According to Chen et al. (2023a, Lemmas C.1), we find for a fixed $\bar{\mathbf{x}}_0$,

$$\begin{aligned}
 & \frac{d}{dt} \text{KL}(\bar{q}_{t|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y}) \|\hat{q}_{t|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y})) \\
 &= -2 \int \left\| \nabla \log \frac{\bar{q}_{t|0}(\bar{\mathbf{x}}_t|\bar{\mathbf{x}}_0, \mathbf{y})}{\hat{q}_{t|0}(\bar{\mathbf{x}}_t|\bar{\mathbf{x}}_0, \mathbf{y})} \right\|_2^2 \bar{q}_{t|0}(\bar{\mathbf{x}}_t|\bar{\mathbf{x}}_0, \mathbf{y}) d\bar{\mathbf{x}}_t \\
 &\quad + 2 \int \left\langle \nabla \log \bar{q}_t(\bar{\mathbf{x}}_t|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T-t, \bar{\mathbf{x}}_t, \mathbf{y}), \nabla \log \frac{\bar{q}_{t|0}(\bar{\mathbf{x}}_t|\bar{\mathbf{x}}_0, \mathbf{y})}{\hat{q}_{t|0}(\bar{\mathbf{x}}_t|\bar{\mathbf{x}}_0, \mathbf{y})} \right\rangle \bar{q}_{t|0}(\bar{\mathbf{x}}_t|\bar{\mathbf{x}}_0, \mathbf{y}) d\bar{\mathbf{x}}_t \\
 (E.7) \quad &\leq \frac{1}{2} \int \left\| \nabla \log \bar{q}_t(\bar{\mathbf{x}}_t|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T-t, \bar{\mathbf{x}}_t, \mathbf{y}) \right\|_2^2 \bar{q}_{t|0}(\bar{\mathbf{x}}_t|\bar{\mathbf{x}}_0, \mathbf{y}) d\bar{\mathbf{x}}_t,
 \end{aligned}$$

where we used the inequality $\langle \mathbf{a}, \mathbf{b} \rangle \leq \frac{1}{4} \|\mathbf{a}\|_2^2 + \|\mathbf{b}\|_2^2$ with $\mathbf{a} = \nabla \log \bar{q}_t - \hat{\mathbf{s}}_m^S$ and $\mathbf{b} = \nabla \log(\bar{q}_{t|0}/\hat{q}_{t|0})$. Using [Chen et al. \(2023a, Lemma C.2\)](#) implies

$$(E.8) \quad \lim_{t \downarrow 0} \text{KL}(\bar{q}_{t|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y}) \|\hat{q}_{t|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y})) = 0.$$

Combining (E.7) with (E.8), and integrating with respect to $t \in (0, T - T_0)$ yield

$$\begin{aligned} & \text{KL}(\bar{q}_{t|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y}) \|\hat{q}_{t|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y})) \\ & \leq \frac{1}{2} \int_0^{T-T_0} \left(\int \|\nabla \log \bar{q}_t(\bar{\mathbf{x}}_t|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T-t, \bar{\mathbf{x}}_t, \mathbf{y})\|_2^2 \bar{q}_{t|0}(\bar{\mathbf{x}}_t|\bar{\mathbf{x}}_0, \mathbf{y}) d\bar{\mathbf{x}}_t \right) dt. \end{aligned}$$

Integrating both sides of the inequality with respect to $\bar{\mathbf{x}}_0 \sim \bar{q}_0(\cdot|\mathbf{y})$ deduces

$$\begin{aligned} & \int \text{KL}(\bar{q}_{T-T_0|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y}) \|\hat{q}_{T-T_0|0}(\cdot|\bar{\mathbf{x}}_0, \mathbf{y})) \bar{q}_0(\bar{\mathbf{x}}_0|\mathbf{y}) d\bar{\mathbf{x}}_0 \\ & \leq \frac{1}{2} \int_0^{T-T_0} \left(\iint \|\nabla \log \bar{q}_t(\bar{\mathbf{x}}_t|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T-t, \bar{\mathbf{x}}_t, \mathbf{y})\|_2^2 \bar{q}_{t,0}(\bar{\mathbf{x}}_t, \bar{\mathbf{x}}_0|\mathbf{y}) d\bar{\mathbf{x}}_t d\bar{\mathbf{x}}_0 \right) dt \\ & = \frac{1}{2} \int_0^{T-T_0} \left(\int \|\nabla \log \bar{q}_t(\bar{\mathbf{x}}_t|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T-t, \bar{\mathbf{x}}_t, \mathbf{y})\|_2^2 \bar{q}_t(\bar{\mathbf{x}}_t|\mathbf{y}) d\bar{\mathbf{x}}_t \right) dt \\ & = \frac{1}{2} \int_{T_0}^T \left(\int \|\nabla \log q_t(\mathbf{x}_t|\mathbf{y}) - \hat{\mathbf{s}}_m^S(t, \mathbf{x}_t, \mathbf{y})\|_2^2 q_t(\mathbf{x}_t|\mathbf{y}) d\mathbf{x}_t \right) dt, \end{aligned}$$

This completes the proof. $\square$

E.3 Proof of Lemma 5.4.

Proof of Lemma 5.4. Combining Lemmas E.1 and E.5 completes the proof. $\square$

F Error Bounds of the Posterior Score Estimation

In the section, we provide a proof of Lemma 5.5, which propose error bounds for the posterior score estimation.

F.1 Posterior score estimation error decomposition. We begin by introducing an error decomposition as the following lemma, which divides the posterior score estimation error into the error of the Langevin dynamics with estimated error (4.5), and the error of Monte Carlo approximation (4.6). The proof of this lemma is inspired by [He et al. \(2024, Proposition 1\)](#).

Lemma F.1. *Suppose Assumption 2 holds. Then the following inequality holds:*

$$\begin{aligned} & \mathbb{E}[\|\nabla \log q_t(\mathbf{X}_t^{\mathbf{y}}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(t, \mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|_2^2] \\ & \leq \frac{2\mu_t^2}{\sigma_t^4} \left\{ \underbrace{\mathbb{E}[\mathbf{W}_2^2(\hat{p}_t^S(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}), p_t(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))]}_{\text{error of Langevin dynamics}} + \underbrace{\frac{\kappa_{\mathbf{y}} C_{\text{SG}} V_{\text{SG}}^2}{m}}_{\text{Monte Carlo}} \right\}, \end{aligned}$$

where the expectations are taken with respect to $\hat{\mathbf{X}}_{0,S,1}^{\mathbf{x},\mathbf{y},t}, \dots, \hat{\mathbf{X}}_{0,S,m}^{\mathbf{x},\mathbf{y},t} \sim \hat{p}_t^S(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y})$ and $\mathbf{X}_t^{\mathbf{y}} \sim q_t(\cdot|\mathbf{y})$.

Proof of Lemma F.1. It follows from (4.1) that for each $t > 0$,

$$(F.1) \quad \|\nabla \log q_t(\mathbf{x}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})\|_2^2 = \frac{\mu_t^2}{\sigma_t^4} \|\mathbf{D}(t, \mathbf{x}, \mathbf{y}) - \hat{\mathbf{D}}_m^S(t, \mathbf{x}, \mathbf{y})\|_2^2.$$

It remains to estimate the error of the estimated posterior denoiser.

Step 1. Construct a Wasserstein coupling. Let $\mathbf{X}_{0,1}^{\mathbf{x},\mathbf{y},t}, \dots, \mathbf{X}_{0,m}^{\mathbf{x},\mathbf{y},t}$ be a sequence of i.i.d. samples drawn from $p_t(\cdot|\mathbf{x}, \mathbf{y})$. Further, suppose that $\mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t} \sim p_t(\cdot|\mathbf{x}, \mathbf{y})$ and $\hat{\mathbf{X}}_{0,S,i}^{\mathbf{x},\mathbf{y},t} \sim \hat{p}_t^S(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y})$ are optimally coupled for each $1 \leq i \leq m$, which means,

$$(F.2) \quad \mathbb{E}[\|\mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t} - \hat{\mathbf{X}}_{0,S,i}^{\mathbf{x},\mathbf{y},t}\|_2^2] = W_2^2(\hat{p}_t^S(\cdot|\mathbf{x}, \mathbf{y}), p_t(\cdot|\mathbf{x}, \mathbf{y})).$$

By the definition of the estimated posterior denoiser (4.6), we find

$$\begin{aligned} \|\mathbf{D}(t, \mathbf{x}, \mathbf{y}) - \hat{\mathbf{D}}_m^S(t, \mathbf{x}, \mathbf{y})\|_2^2 &= \|\mathbf{D}(t, \mathbf{x}, \mathbf{y}) - \frac{1}{m} \sum_{i=1}^m \mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t} + \frac{1}{m} \sum_{i=1}^m \mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t} - \hat{\mathbf{D}}_m^S(t, \mathbf{x}, \mathbf{y})\|_2^2 \\ &\leq 2\|\mathbf{D}(t, \mathbf{x}, \mathbf{y}) - \frac{1}{m} \sum_{i=1}^m \mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t}\|_2^2 + 2\|\frac{1}{m} \sum_{i=1}^m (\mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t} - \hat{\mathbf{X}}_{0,S,i}^{\mathbf{x},\mathbf{y},t})\|_2^2. \end{aligned}$$

Taking expectation with respect to $\mathbf{X}_{0,1}^{\mathbf{x},\mathbf{y},t}, \dots, \mathbf{X}_{0,m}^{\mathbf{x},\mathbf{y},t}$ on both sides of the above equality yields

$$(F.3) \quad \begin{aligned} &\mathbb{E}[\|\mathbf{D}(t, \mathbf{x}, \mathbf{y}) - \hat{\mathbf{D}}_m^S(t, \mathbf{x}, \mathbf{y})\|_2^2] \\ &\leq \underbrace{2\mathbb{E}\left[\|\mathbf{D}(t, \mathbf{x}, \mathbf{y}) - \frac{1}{m} \sum_{i=1}^m \mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t}\|_2^2\right]}_{(i)} + \underbrace{2\mathbb{E}\left[\|\frac{1}{m} \sum_{i=1}^m (\mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t} - \hat{\mathbf{X}}_{0,S,i}^{\mathbf{x},\mathbf{y},t})\|_2^2\right]}_{(ii)}. \end{aligned}$$

Here the term (i) measures the error of the Monte Carlo approximation (4.6), while the term (ii) reveals the error of the Langevin dynamics with estimated score (4.5).

Step 2. Bound the error of the Monte Carlo approximation. Let $\mathbf{X}_0^{\mathbf{x},\mathbf{y},t}$ be a random copy of $\mathbf{X}_{0,1}^{\mathbf{x},\mathbf{y},t}$ and independent of $\mathbf{X}_{0,1:m}^{\mathbf{x},\mathbf{y},t}$. It is apparent that $\mathbf{D}(t, \mathbf{x}, \mathbf{y}) = \mathbb{E}[\mathbf{X}_0^{\mathbf{x},\mathbf{y},t}]$. Consequently,

$$(F.4) \quad \begin{aligned} &\mathbb{E}\left[\|\mathbf{D}(t, \mathbf{x}, \mathbf{y}) - \frac{1}{m} \sum_{i=1}^m \mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t}\|_2^2\right] \\ &= \frac{1}{m^2} \mathbb{E}\left[\left\|\sum_{i=1}^m (\mathbb{E}[\mathbf{X}_0^{\mathbf{x},\mathbf{y},t}] - \mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t})\right\|_2^2\right] \\ &= \frac{1}{m^2} \sum_{i=1}^m \sum_{j=1}^m \mathbb{E}[\langle \mathbb{E}[\mathbf{X}_0^{\mathbf{x},\mathbf{y},t}] - \mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t}, \mathbb{E}[\mathbf{X}_0^{\mathbf{x},\mathbf{y},t}] - \mathbf{X}_{0,j}^{\mathbf{x},\mathbf{y},t} \rangle] \\ &= \frac{1}{m^2} \sum_{i=1}^m \mathbb{E}[\|\mathbb{E}[\mathbf{X}_0^{\mathbf{x},\mathbf{y},t}] - \mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t}\|_2^2] \\ &= \frac{1}{m} \text{trace}(\text{Cov}(\mathbf{X}_{0,1:m}^{\mathbf{x},\mathbf{y},t})) = \frac{1}{m} \text{trace}(\text{Cov}(\mathbf{X}_0|\mathbf{X}_t = \mathbf{x}, \mathbf{Y} = \mathbf{y})), \end{aligned}$$

where the third equality is due to $\mathbb{E}[\mathbf{X}_0^{\mathbf{x},\mathbf{y},t}] = \mathbb{E}[\mathbf{X}_{0,i}^{\mathbf{x},\mathbf{y},t}]$ for each $1 \leq i \leq m$. Note that

$$(F.5) \quad \begin{aligned} &\int \text{Cov}(\mathbf{X}_0|\mathbf{X}_t = \mathbf{x}, \mathbf{Y} = \mathbf{y}) q_t(\mathbf{x}|\mathbf{y}) d\mathbf{x} \\ &= \text{Cov}(\mathbf{X}_0|\mathbf{Y} = \mathbf{y}) - \text{Cov}(\mathbb{E}[\mathbf{X}_0|\mathbf{X}_t = \mathbf{x}, \mathbf{Y} = \mathbf{y}]) \\ &\preceq \text{Cov}(\mathbf{X}_0|\mathbf{Y} = \mathbf{y}) \preceq \mathbb{E}[\mathbf{X}_0 \mathbf{X}_0^\top | \mathbf{Y} = \mathbf{y}], \end{aligned}$$

where the equality holds from the law of total variance, and the first inequality used the fact that covariance matrix is semi-positive definite. As a consequence,

$$\begin{aligned}
& \int \text{trace}(\text{Cov}(\mathbf{X}_0 | \mathbf{X}_t = \mathbf{x}, \mathbf{Y} = \mathbf{y})) q_t(\mathbf{x} | \mathbf{y}) d\mathbf{x} \\
& \leq \text{trace}(\mathbb{E}[\mathbf{X}_0 \mathbf{X}_0^\top | \mathbf{Y} = \mathbf{y}]) = \int \|\mathbf{x}_0\|_2^2 q_0(\mathbf{x}_0 | \mathbf{y}) d\mathbf{x}_0 \\
& = V_{\text{SG}}^2 \int \left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2} + 1 \right) q_0(\mathbf{x}_0 | \mathbf{y}) d\mathbf{x}_0 - V_{\text{SG}}^2 \\
& \leq V_{\text{SG}}^2 \int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) q_0(\mathbf{x}_0 | \mathbf{y}) d\mathbf{x}_0 \leq \kappa_{\mathbf{y}} C_{\text{SG}} V_{\text{SG}}^2,
\end{aligned}
\tag{F.6}$$

where the first inequality is due to (F.5), the second inequality follows from $1 + z \leq \exp(z)$ for each $z \in \mathbb{R}$, and the last inequality invokes Lemma 4.10. Multiplying both sides of (F.4) by $q_t(\mathbf{x} | \mathbf{y})$, integrating with respect to $\mathbf{x}$, and substituting (F.6) imply

$$\mathbb{E} \left[\|\mathbf{D}(t, \mathbf{x}, \mathbf{y}) - \frac{1}{m} \sum_{i=1}^m \mathbf{X}_{0,i}^{\mathbf{x}, \mathbf{y}, t}\|_2^2 \right] q_t(\mathbf{x} | \mathbf{y}) d\mathbf{x} \leq \frac{\kappa_{\mathbf{y}} C_{\text{SG}} V_{\text{SG}}^2}{m}.
\tag{F.7}$$

Step 3. Bound the error of the Langevin dynamics with estimated score. For the term (ii) in (F.3), we have

$$\begin{aligned}
& \mathbb{E} \left[\frac{1}{m} \sum_{i=1}^m (\mathbf{X}_{0,i}^{\mathbf{x}, \mathbf{y}, t} - \widehat{\mathbf{X}}_{0,S,i}^{\mathbf{x}, \mathbf{y}, t})^2 \right] \\
& = \frac{1}{m^2} \sum_{i=1}^m \sum_{j=1}^m \mathbb{E} [\langle \mathbf{X}_{0,i}^{\mathbf{x}, \mathbf{y}, t} - \widehat{\mathbf{X}}_{0,S,i}^{\mathbf{x}, \mathbf{y}, t}, \mathbf{X}_{0,j}^{\mathbf{x}, \mathbf{y}, t} - \widehat{\mathbf{X}}_{0,S,j}^{\mathbf{x}, \mathbf{y}, t} \rangle] \\
& \leq \frac{1}{2m^2} \sum_{i=1}^m \sum_{j=1}^m \mathbb{E} [\|\mathbf{X}_{0,i}^{\mathbf{x}, \mathbf{y}, t} - \widehat{\mathbf{X}}_{0,S,i}^{\mathbf{x}, \mathbf{y}, t}\|_2^2 + \|\mathbf{X}_{0,j}^{\mathbf{x}, \mathbf{y}, t} - \widehat{\mathbf{X}}_{0,S,j}^{\mathbf{x}, \mathbf{y}, t}\|_2^2] \\
& = \mathbb{W}_2^2(\widehat{p}_t^S(\cdot | \mathbf{x}, \mathbf{y}), p_t(\cdot | \mathbf{x}, \mathbf{y})),
\end{aligned}
\tag{F.8}$$

where the first inequality holds from the Cauchy-Schwarz inequality, and the last equality is due to (F.2).

Step 4. Conclusion. Substituting (F.7) and (F.8) into (F.3) implies

$$\mathbb{E} [\|\mathbf{D}(t, \mathbf{X}_t^{\mathbf{y}}, \mathbf{y}) - \widehat{\mathbf{D}}_m^S(t, \mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|_2^2] \leq \mathbb{E} [\mathbb{W}_2^2(\widehat{p}_t^S(\cdot | \mathbf{X}_t^{\mathbf{y}}, \mathbf{y}), p_t(\cdot | \mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))] + \frac{\kappa_{\mathbf{y}} C_{\text{SG}} V_{\text{SG}}^2}{m},$$

where the expectation is taken with respect to $\mathbf{X}_{0,1:m}^{\mathbf{x}, \mathbf{y}, t}$, and $\mathbf{X}_t^{\mathbf{y}} \sim q_t(\cdot | \mathbf{y})$. Combining the above inequality with (F.1) completes the proof. $\square$

F.2 Convergence of the score-based Langevin in Wasserstein distance. In this subsection, we aim to offer a bound for the first term in Lemma F.1, which is the Wasserstein error of Langevin dynamics with estimated score (4.5).

Before presenting, we propose the following auxiliary lemma, which connects the Wasserstein error between the laws of two SDEs to the L^2 -error between their drift terms. The proof of this lemma is based on the technique of Wasserstein coupling.

Lemma F.2. *Let $\mathbf{Z}_s$ and $\bar{\mathbf{Z}}_s$ be two stochastic processes defined, respectively, by the following SDEs:*

$$(F.9) \quad \begin{aligned} d\mathbf{Z}_s &= \mathbf{b}(\mathbf{Z}_s) ds + \sqrt{2} d\mathbf{B}_s, & \mathbf{Z}_0 &\sim \mu_0, \\ d\bar{\mathbf{Z}}_s &= \bar{\mathbf{b}}(\bar{\mathbf{Z}}_s) ds + \sqrt{2} d\mathbf{B}_s, & \bar{\mathbf{Z}}_0 &\sim \mu_0. \end{aligned}$$

Denote by μ_s the distribution of $\mathbf{Z}_s$, and denote by $\bar{\mu}_s$ the distribution of $\bar{\mathbf{Z}}_s$. Assume further that $\bar{\mathbf{b}}$ is β -Lipschitz. Then it holds that

$$W_2^2(\mu_S, \bar{\mu}_S) \leq \exp(2\beta S) \int_0^S \mathbb{E}_{\mathbf{Z}_s \sim \mu_s} [\|\mathbf{b}(\mathbf{Z}_s) - \bar{\mathbf{b}}(\mathbf{Z}_s)\|_2^2] ds,$$

where $S > 0$ is the time horizon.

Proof of Lemma F.2. We evolve these two SDEs (F.9) from the same initial particle $\bar{\mathbf{Z}}_0 = \mathbf{Z}_0 \sim \mu_0$, and use the same Brownian motion. Denote the joint distribution $\gamma_0(\mathbf{z}_0, \bar{\mathbf{z}}_0) = \delta\{\mathbf{z}_0 = \bar{\mathbf{z}}_0\}\mu_0(\mathbf{z}_0)$, which is a coupling of (μ_0, μ_0) . Let γ_s denote the law of $(\mathbf{Z}_s, \bar{\mathbf{Z}}_s)$ (F.9). Notice that γ_s is a coupling of $(\mu_s, \bar{\mu}_s)$. We next aim to bound $\mathbb{E}[\|\mathbf{Z}_S - \bar{\mathbf{Z}}_S\|_2^2]$. First, it follows from Itô's formula that

$$\begin{aligned} d\|\mathbf{Z}_s - \bar{\mathbf{Z}}_s\|_2^2 &= 2\langle \mathbf{Z}_s - \bar{\mathbf{Z}}_s, d\mathbf{Z}_s - d\bar{\mathbf{Z}}_s \rangle \\ &= 2\langle \mathbf{Z}_s - \bar{\mathbf{Z}}_s, \mathbf{b}(\mathbf{Z}_s) - \bar{\mathbf{b}}(\bar{\mathbf{Z}}_s) \rangle ds \\ &\leq 2\|\mathbf{Z}_s - \bar{\mathbf{Z}}_s\|_2 \|\mathbf{b}(\mathbf{Z}_s) - \bar{\mathbf{b}}(\bar{\mathbf{Z}}_s)\|_2 ds, \end{aligned}$$

where the second equality used (F.9) and the fact that the Brownian motions in two SDEs are the same, and the inequality holds from Cauchy-Schwarz inequality. On the other hand,

$$d\|\mathbf{Z}_s - \bar{\mathbf{Z}}_s\|_2^2 = 2\|\mathbf{Z}_s - \bar{\mathbf{Z}}_s\|_2 d\|\mathbf{Z}_s - \bar{\mathbf{Z}}_s\|_2.$$

Consequently,

$$\begin{aligned} d\|\mathbf{Z}_s - \bar{\mathbf{Z}}_s\|_2 &\leq \|\mathbf{b}(\mathbf{Z}_s) - \bar{\mathbf{b}}(\bar{\mathbf{Z}}_s)\|_2 ds \\ &\leq \|\mathbf{b}(\mathbf{Z}_s) - \bar{\mathbf{b}}(\mathbf{Z}_s)\|_2 ds + \|\bar{\mathbf{b}}(\mathbf{Z}_s) - \bar{\mathbf{b}}(\bar{\mathbf{Z}}_s)\|_2 ds \\ &\leq \|\mathbf{b}(\mathbf{Z}_s) - \bar{\mathbf{b}}(\mathbf{Z}_s)\|_2 ds + \beta\|\mathbf{Z}_s - \bar{\mathbf{Z}}_s\|_2 ds, \end{aligned}$$

where the second inequality follows from the triangular inequality, and the last inequality is due to that $\bar{\mathbf{b}}$ is β -Lipschitz. Applying Gronwall's inequality (Evans, 2010, Section B.2) yields

$$\|\mathbf{Z}_S - \bar{\mathbf{Z}}_S\|_2 \leq \exp(\beta S) \int_0^S \|\mathbf{b}(\mathbf{Z}_s) - \bar{\mathbf{b}}(\mathbf{Z}_s)\|_2 ds,$$

for each $S > 0$. Here we used the fact that $\mathbf{Z}_0 = \bar{\mathbf{Z}}_0$. Taking expectation on both sides of the inequality with respect to $(\mathbf{Z}_S, \bar{\mathbf{Z}}_S) \sim \gamma_S$, and using Jensen's inequality yields

$$W_2^2(\mu_S, \bar{\mu}_S) \leq \mathbb{E}_{(\mathbf{Z}_S, \bar{\mathbf{Z}}_S) \sim \gamma_S} [\|\mathbf{Z}_S - \bar{\mathbf{Z}}_S\|_2^2] \leq \exp(2\beta S) \int_0^S \mathbb{E}_{\mathbf{Z}_s \sim \mu_s} [\|\mathbf{b}(\mathbf{Z}_s) - \bar{\mathbf{b}}(\mathbf{Z}_s)\|_2^2] ds,$$

where the first inequality follows from the definition of the Wasserstein distance. This completes the proof. $\square$

With the aid of Lemma F.2, we propose a error analysis in Wasserstein distance for Langevin dynamics with estimated score (4.5).

Lemma F.3. Suppose Assumptions 1, 3, 5, and 6 hold. Let $0 < T_0 < T < \frac{1}{2} \log(1 + \alpha^{-1})$. For each time $t \in (T_0, T)$,

$$\begin{aligned} & \mathbb{E}[\mathbb{W}_2^2(p_t(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}), \hat{p}_t^S(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))] \\ & \leq 2 \exp\left(-\frac{2(\mu_t^2 - \alpha\sigma_t^2)S}{\sigma_t^2}\right) \eta_{\mathbf{y}}^2 \\ & \quad + 2 \exp\left\{2\left(G + \frac{\mu_t^2}{\sigma_t^2}\right)S\right\} \int_0^S \int \mathbb{E}[\|\nabla \log \pi_0(\mathbf{X}_{0,s}^{\mathbf{x}, \mathbf{y}, t}) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{X}_{0,s}^{\mathbf{x}, \mathbf{y}, t})\|_2^2] q_t(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \, ds, \end{aligned}$$

where the initial discrepancy $\eta_{\mathbf{y}}$ satisfies

$$\eta_{\mathbf{y}}^2 \geq \sup_{t \in (T_0, T)} \mathbb{E}[\mathbb{W}_2^2(\hat{p}_t^0(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}), p_t(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))].$$

Proof of Lemma F.3. By the triangular inequality, we find that for each fixed $\mathbf{x} \in \mathbb{R}^d$,

$$\begin{aligned} & \mathbb{W}_2^2(p_t(\cdot|\mathbf{x}, \mathbf{y}), \hat{p}_t^S(\cdot|\mathbf{x}, \mathbf{y})) \\ (F.10) \quad & \leq 2 \underbrace{\mathbb{W}_2^2(p_t(\cdot|\mathbf{x}, \mathbf{y}), p_t^S(\cdot|\mathbf{x}, \mathbf{y}))}_{\text{convergence of Langevin dynamics}} + 2 \underbrace{\mathbb{W}_2^2(p_t^S(\cdot|\mathbf{x}, \mathbf{y}), \hat{p}_t^S(\cdot|\mathbf{x}, \mathbf{y}))}_{\text{score estimation error}}, \end{aligned}$$

where $p_t^S(\cdot|\mathbf{x}, \mathbf{y})$ is the marginal density of $\mathbf{X}_{0,S}^{\mathbf{x}, \mathbf{y}, t}$ defined in (4.4).

Step 1. The convergence of Langevin dynamics. For the first term in the right-hand side of (F.10),

$$\mathbb{W}_2^2(p_t(\cdot|\mathbf{x}, \mathbf{y}), p_t^S(\cdot|\mathbf{x}, \mathbf{y})) \leq \exp\left(-\frac{2(\mu_t^2 - \alpha\sigma_t^2)S}{\sigma_t^2}\right) \mathbb{W}_2^2(p_t(\cdot|\mathbf{x}, \mathbf{y}), \hat{p}_t^0(\cdot|\mathbf{x}, \mathbf{y})),$$

where the inequality holds from Lemmas 4.6 and H.5. Multiplying both sides of the inequality by $q_t(\mathbf{x}|\mathbf{y})$, and integrating with respect to $\mathbf{x}$ yield

$$(F.11) \quad \mathbb{E}[\mathbb{W}_2^2(p_t(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}), p_t^S(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))] \leq \exp\left(-\frac{2(\mu_t^2 - \alpha\sigma_t^2)S}{\sigma_t^2}\right) \eta_{\mathbf{y}}^2.$$

Step 2. The error of the score matching. We turn to consider the second term in the right-hand side of (F.10). Denote by $\bar{\mathbf{b}}$ the drift term of (4.5), that is,

$$\bar{\mathbf{b}}(t, \mathbf{x}_0) := \hat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0) + \frac{\mu_t}{\sigma_t^2}(\mathbf{x} - \mu_t \mathbf{x}_0) - \nabla \ell_{\mathbf{y}}(\mathbf{x}_0), \quad \mathbf{x}_0 \in \mathbb{R}^d.$$

It follows from Assumption 6 that

$$\|\bar{\mathbf{b}}(t, \mathbf{x}_0) - \bar{\mathbf{b}}(t, \mathbf{x}'_0)\|_2 \leq \left(G + \frac{\mu_t^2}{\sigma_t^2}\right) \|\mathbf{x}_0 - \mathbf{x}'_0\|_2.$$

Combining this Lipschitz continuity with Lemma F.2 implies

$$\begin{aligned} & \mathbb{E}[\mathbb{W}_2^2(p_t^S(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}), \hat{p}_t^S(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))] \\ & = \int \mathbb{W}_2^2(p_t^S(\cdot|\mathbf{x}, \mathbf{y}), \hat{p}_t^S(\cdot|\mathbf{x}, \mathbf{y})) q_t(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \\ (F.12) \quad & \leq \exp\left\{2\left(G + \frac{\mu_t^2}{\sigma_t^2}\right)S\right\} \int_0^S \int \mathbb{E}[\|\nabla \log \pi_0(\mathbf{X}_{0,s}^{\mathbf{x}, \mathbf{y}, t}) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{X}_{0,s}^{\mathbf{x}, \mathbf{y}, t})\|_2^2] q_t(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \, ds, \end{aligned}$$

where the expectation is taken with respect to $\mathbf{X}_{0,s}^{\mathbf{x}, \mathbf{y}, t} \sim p_t^S(\cdot|\mathbf{x}, \mathbf{y})$.

Step 3. Conclusion. Plugging (F.11) and (F.12) into (F.10) completes the proof. $\square$

The following lemma offers a bound for the L^2 -error of the prior score estimator, which is inspired by [Tang and Yang \(2024, Theorem 1\)](#), [Jiang et al. \(2025\)](#) and [Ding et al. \(2024b, Theorem 3.2\)](#).

Lemma F.4 (Score estimation error). *Suppose Assumptions 1, 2, 3, and 5 hold. Assume further $\|\hat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2 \leq B(1 + \|\mathbf{x}_0\|_2^r)$. Let $0 < T_0 < T < \frac{1}{2} \log(1 + \alpha^{-1})$. For each time $t \in (T_0, T)$, it holds that*

$$\begin{aligned} & \int_0^S \int \mathbb{E}[\|\nabla \log \pi_0(\mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t}) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t})\|_2^2] q_t(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \, ds \\ & \leq C \left\{ \frac{\sigma_t^2 \eta_{\mathbf{y}}^2}{2(\mu_t^2 - \alpha \sigma_t^2)} + S \right\}^{\frac{1}{2}} \kappa_{\mathbf{y}}^{\frac{1}{2}} \varepsilon_{\text{prior}}^{\frac{1}{2}}, \end{aligned}$$

where C is a constant only depending on B , r , V_{SG} and C_{SG} , and the initial discrepancy $\eta_{\mathbf{y}}$ satisfies

$$\eta_{\mathbf{y}}^2 \geq \sup_{t \in (T_0, T)} \mathbb{E}[\chi^2(\hat{p}_t^0(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}) \| p_t(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))].$$

Proof of Lemma F.4. It is straightforward that

$$\begin{aligned} & \int_0^S \int \mathbb{E}[\|\nabla \log \pi_0(\mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t}) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t})\|_2^2] q_t(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \, ds \\ & = \int_0^S \left(\iint \|\nabla \log \pi_0(\mathbf{x}_0) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2^2 \frac{p_t^s(\mathbf{x}_0|\mathbf{x}, \mathbf{y})}{p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y})} p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) \, d\mathbf{x}_0 q_t(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \right) ds \\ & \leq \underbrace{\left(\iint \|\nabla \log \pi_0(\mathbf{x}_0) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2^2 p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) \, d\mathbf{x}_0 q_t(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \right)^{\frac{1}{4}}}_{(i)} \\ & \quad \times \underbrace{\left(\iint \|\nabla \log \pi_0(\mathbf{x}_0) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2^6 p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) \, d\mathbf{x}_0 q_t(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \right)^{\frac{1}{4}}}_{(ii)} \\ (F.13) \quad & \times \underbrace{\int_0^S \left(\iint \left(\frac{p_t^s(\mathbf{x}_0|\mathbf{x}, \mathbf{y})}{p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y})} \right)^2 p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) \, d\mathbf{x}_0 q_t(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \right)^{\frac{1}{2}} ds}_{(iii)}, \end{aligned}$$

where the inequality holds from Hölder's inequality.

Step 1. Bound the term (i) in (F.13). It follows from Assumption 5 that

$$\begin{aligned} & \iint \|\nabla \log \pi_0(\mathbf{x}_0) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2^2 p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) \, d\mathbf{x}_0 q_t(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \\ & = \int \|\nabla \log \pi_0(\mathbf{x}_0) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2^2 \left(\frac{q_0(\mathbf{x}_0|\mathbf{y})}{\pi_0(\mathbf{x}_0)} \right) \pi_0(\mathbf{x}_0) \, d\mathbf{x}_0 \\ (F.14) \quad & \leq \sup_{\mathbf{x}_0} \left(\frac{q_0(\mathbf{x}_0|\mathbf{y})}{\pi_0(\mathbf{x}_0)} \right) \int \|\nabla \log \pi_0(\mathbf{x}_0) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2^2 \pi_0(\mathbf{x}_0) \, d\mathbf{x}_0 \leq \kappa_{\mathbf{y}} \varepsilon_{\text{prior}}^2, \end{aligned}$$

where the first inequality invokes Hölder's inequality, and the second inequality is due to Assumption 5.

Step 2. Bound the term (ii) in (F.13). By an argument similar to (F.14), we have

$$\begin{aligned}
 & \iint \|\nabla \log \pi_0(\mathbf{x}_0) - \widehat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2^6 p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) d\mathbf{x}_0 q_t(\mathbf{x}|\mathbf{y}) d\mathbf{x} \\
 & \leq \kappa_{\mathbf{y}} \int \|\nabla \log \pi_0(\mathbf{x}_0) - \widehat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2^6 \pi_0(\mathbf{x}_0) d\mathbf{x}_0 \\
 & \leq 2048B^6 \kappa_{\mathbf{y}} \int (1 + \|\mathbf{x}_0\|_2^{6r}) \pi_0(\mathbf{x}_0) d\mathbf{x}_0 \\
 (F.15) \quad & = 2048B^6 \kappa_{\mathbf{y}} + 2048B^6 \kappa_{\mathbf{y}} \mathbb{E}[\|\mathbf{X}_0\|_2^{6r}],
 \end{aligned}$$

where the second inequality follows from Assumptions 3 and $\|\widehat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2 \leq B(1 + \|\mathbf{x}_0\|_2^r)$. Using the same argument as Lemma H.2, we have

$$(F.16) \quad \mathbb{E}[\|\mathbf{X}_0\|_2^{6r}] \leq 2592C_{\text{SG}}V_{\text{SG}}^{6r},$$

Substituting (F.16) into (F.15) implies

$$(F.17) \quad \iint \|\nabla \log \pi_0(\mathbf{x}_0) - \widehat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2^6 p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) d\mathbf{x}_0 q_t(\mathbf{x}|\mathbf{y}) d\mathbf{x} \lesssim B^6 \kappa_{\mathbf{y}} (1 + C_{\text{SG}}V_{\text{SG}}^{6r}).$$

Step 3. Bound the term (iii) in (F.13). By the definition of the χ^2 -divergence, we find

$$\begin{aligned}
 & \int_0^S \left(\iint \left(\frac{p_t^s(\mathbf{x}_0|\mathbf{x}, \mathbf{y})}{p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y})} \right)^2 p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y}) d\mathbf{x}_0 q_t(\mathbf{x}|\mathbf{y}) d\mathbf{x} \right)^{\frac{1}{2}} ds \\
 & = \int_0^S \left(\mathbb{E}[\chi^2(p_t^s(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|p_t(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))] + 1 \right)^{\frac{1}{2}} ds \\
 & \leq \left(\int_0^S \mathbb{E}[\chi^2(p_t^s(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|p_t(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))] ds + S \right)^{\frac{1}{2}} \\
 & \leq \left\{ \mathbb{E}[\chi^2(\widehat{p}_t^0(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|p_t(\cdot|\mathbf{X}_t^{\mathbf{y}}, \mathbf{y}))] \int_0^S \exp\left(-\frac{2(\mu_t^2 - \alpha\sigma_t^2)s}{\sigma_t^2}\right) ds + S \right\}^{\frac{1}{2}} \\
 (F.18) \quad & \leq \left\{ \frac{\sigma_t^2 \eta_{\mathbf{y}}^2}{2(\mu_t^2 - \alpha\sigma_t^2)} + S \right\}^{\frac{1}{2}},
 \end{aligned}$$

where the expectation is taken with respect to $\mathbf{X}_t^{\mathbf{y}} \sim q_t(\cdot|\mathbf{y})$, the first inequality invokes Jensen's inequality, and the second inequality holds from Lemmas 4.6 and H.6.

Step 4. Conclusion. Substituting (F.14), (F.17) and (F.18) into (F.13) yields the desired results. $\square$

F.3 Proof of Lemma 5.5 and Corollary 5.6.

Proof of Lemma 5.5. Combining Lemmas F.1 and F.3, F.4 implies

$$\begin{aligned}
 & \mathbb{E}[\|\nabla \log q_t(\mathbf{X}_t^{\mathbf{y}}|\mathbf{y}) - \widehat{\mathbf{s}}_m^S(t, \mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|_2^2] \\
 & \leq C \frac{\mu_t^2}{\sigma_t^4} \left\{ \frac{\kappa_{\mathbf{y}}}{m} + \exp\left(-\frac{2(\mu_t^2 - \alpha\sigma_t^2)}{\sigma_t^2} S\right) \eta_{\mathbf{y}}^2 \right. \\
 & \quad \left. + \left(\frac{\sigma_t^2 \eta_{\mathbf{y}}^2}{\mu_t^2 - \alpha\sigma_t^2} + S \right)^{\frac{1}{2}} \kappa_{\mathbf{y}}^{\frac{1}{2}} \exp\left(2\left(G + \frac{\mu_t^2}{\sigma_t^2}\right) S\right) \varepsilon_{\text{prior}}^{\frac{1}{2}} \right\},
 \end{aligned}$$

where $t \in (0, \frac{1}{2} \log(1 + \alpha^{-1}))$, and C is a constant only depending on B , V_{SG} and C_{SG} . Note that $\mu_t = \exp(-t)$ is decreaseing and $\sigma_t^2 = 1 - \exp(-2t)$ is increases as t grows. As a consequence,

$$\begin{aligned} & \mathbb{E}[\|\nabla \log q_t(\mathbf{X}_t^{\mathbf{y}}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(t, \mathbf{X}_t^{\mathbf{y}}, \mathbf{y})\|_2^2] \\ & \leq C \frac{\mu_{T_0}^2}{\sigma_{T_0}^4} \left\{ \frac{\kappa_{\mathbf{y}}}{m} + \exp\left(-\frac{2(\mu_T^2 - \alpha\sigma_T^2)}{\sigma_T^2} S\right) \eta_{\mathbf{y}}^2 \right. \\ & \quad \left. + \left(\frac{\sigma_T^2 \eta_{\mathbf{y}}^2}{\mu_T^2 - \alpha\sigma_T^2} + S\right)^{\frac{1}{2}} \kappa_{\mathbf{y}}^{\frac{1}{2}} \exp\left(2\left(G + \frac{\mu_{T_0}^2}{\sigma_{T_0}^2}\right) S\right) \varepsilon_{\text{prior}}^{\frac{1}{2}} \right\}, \end{aligned}$$

for each $t \in (T_0, T)$ with $0 < T_0 < T < \frac{1}{2} \log(1 + \alpha^{-1})$. This completes the proof. $\square$

Proof of Corollary 5.6. A direct conclusion of Lemma 5.5. $\square$

G Error Bounds of the Warm-Start

In this section, we provide a proof of Lemma 5.4, which proposes error bounds for the warm-start strategy.

G.1 Boundedness of the terminal posterior score. Before proceeding, we introduce an auxiliary lemma, showing the uniform boundedness of the score function of the terminal posterior density $q_T(\cdot|\mathbf{y})$.

Lemma G.1. *Let ν be an m -strongly log-concave distribution on $\mathbb{R}^d$. Suppose $d\nu(\mathbf{z}) = \exp(-U(\mathbf{z})) d\mathbf{z}$. Let $\mathbf{z}^* \in \mathbb{R}^d$ be the unique minimizer of U . Then we have*

$$\mathbb{E}_{\mathbf{Z} \sim \nu}[\|\mathbf{Z}\|_2^2] \leq \frac{2d}{m} + \frac{\|\nabla U(\mathbf{0})\|_2^2}{m^2}.$$

Proof of Lemma G.1. First, we apply the Gauss-Green divergence theorem (Evans, 2010, Theorem 1 in Section C2) to the vector field $(\mathbf{z} - \mathbf{z}^*) \exp(-U(\mathbf{z}))$, for which the integral of its divergence over $\mathbb{R}^d$ vanishes due to rapid decay at infinity,

$$(G.1) \quad \int_{\mathbb{R}^d} \nabla \cdot [(\mathbf{z} - \mathbf{z}^*) \exp(-U(\mathbf{z}))] d\mathbf{z} = 0.$$

On the other hand,

$$\begin{aligned} \nabla \cdot [(\mathbf{z} - \mathbf{z}^*) \exp(-U(\mathbf{z}))] &= d \exp(-U(\mathbf{z})) - \langle \mathbf{z} - \mathbf{z}^*, \nabla U(\mathbf{z}) \rangle \exp(-U(\mathbf{z})) \\ &= d \exp(-U(\mathbf{z})) - \langle \mathbf{z} - \mathbf{z}^*, \nabla U(\mathbf{z}) - \nabla U(\mathbf{z}^*) \rangle \exp(-U(\mathbf{z})), \end{aligned}$$

where the second equality is owing to $\nabla U(\mathbf{z}^*) = 0$. Since ν is m -strongly log-concave,

$$(G.2) \quad \langle \mathbf{z} - \mathbf{z}^*, \nabla U(\mathbf{z}) - \nabla U(\mathbf{z}^*) \rangle \geq m \|\mathbf{z} - \mathbf{z}^*\|_2^2.$$

Therefore,

$$\nabla \cdot [(\mathbf{z} - \mathbf{z}^*) \exp(-U(\mathbf{z}))] \leq d \exp(-U(\mathbf{z})) - m \|\mathbf{z} - \mathbf{z}^*\|_2^2 \exp(-U(\mathbf{z})),$$

Integrating both sides of the inequality and using (G.1) implies

$$\mathbb{E}_{\mathbf{Z} \sim \nu}[\|\mathbf{Z} - \mathbf{z}^*\|_2^2] \leq \frac{d}{m}.$$

Then it follows from the triangular inequality that

$$(G.3) \quad \mathbb{E}_{\mathbf{Z} \sim \nu} [\|\mathbf{Z}\|_2^2] \leq 2\mathbb{E}_{\mathbf{Z} \sim \nu} [\|\mathbf{Z} - \mathbf{z}^*\|_2^2] + 2\|\mathbf{z}^*\|_2^2 \leq \frac{2d}{m} + 2\|\mathbf{z}^*\|_2^2.$$

By setting $\mathbf{z} = \mathbf{0}$ in (G.2), we have

$$\|\mathbf{z}^*\|_2^2 \leq \frac{1}{m} \|\nabla U(\mathbf{0})\|_2 \|\mathbf{z}^*\|_2,$$

which implies $\|\mathbf{z}^*\|_2 \leq \frac{1}{m} \|\nabla U(\mathbf{0})\|_2$. Substituting this inequality into (G.3) completes the proof. $\square$

Lemma G.2. Suppose Assumptions 1 and 4 hold. Let $0 < T < \frac{1}{2} \log(1 + \alpha^{-1})$. Then

$$\|\mathbf{D}(T, \mathbf{x}, \mathbf{y})\|_2^2 \leq \frac{2d\sigma_T^2}{\mu_T^2 - \alpha\sigma_T^2} + \frac{6\sigma_T^4 H_{\mathbf{y}}^2}{(\mu_T^2 - \alpha\sigma_T^2)^2} + \frac{6\mu_T^2}{(\mu_T^2 - \alpha\sigma_T^2)^2} \|\mathbf{x}\|_2^2.$$

Proof of Lemma G.2. By the definition of the denoiser (4.2), we have

$$\begin{aligned} \|\mathbf{D}(T, \mathbf{x}, \mathbf{y})\|_2^2 &= \|\mathbb{E}[\mathbf{X}_0 | \mathbf{X}_T = \mathbf{x}, \mathbf{Y} = \mathbf{y}]\|_2^2 \\ &\leq \mathbb{E}[\|\mathbf{X}_0\|_2^2 | \mathbf{X}_T = \mathbf{x}, \mathbf{Y} = \mathbf{y}] \\ &\leq \frac{2d\sigma_T^2}{\mu_T^2 - \alpha\sigma_T^2} + \frac{6\sigma_T^4}{(\mu_T^2 - \alpha\sigma_T^2)^2} \left(H_{\mathbf{y}}^2 + \frac{\mu_T^2}{\sigma_T^4} \|\mathbf{x}\|_2^2 \right) \\ &\leq \frac{2d\sigma_T^2}{\mu_T^2 - \alpha\sigma_T^2} + \frac{6\sigma_T^4 H_{\mathbf{y}}^2}{(\mu_T^2 - \alpha\sigma_T^2)^2} + \frac{6\mu_T^2}{(\mu_T^2 - \alpha\sigma_T^2)^2} \|\mathbf{x}\|_2^2, \end{aligned}$$

where the first inequality holds from Jensen's inequality, the second inequality invokes Lemmas 4.6 and G.1 and Assumption 4. This completes the proof. $\square$

Lemma G.3. Suppose Assumptions 1, and 3 hold. Let $0 < T < \frac{1}{2} \log(1 + \alpha^{-1})$. Then

$$\|\nabla \log q_T(\mathbf{x} | \mathbf{y})\|_2^2 \lesssim \frac{H_{\mathbf{y}}^2 + \|\mathbf{x}\|_2^2}{\sigma_T^4 (\mu_T^2 - \alpha\sigma_T^2)^2}.$$

Proof of Lemma G.3. It follows from Lemma 4.1 that

$$\nabla \log q_T(\mathbf{x} | \mathbf{y}) = -\frac{\mathbf{x}}{\sigma_T^2} + \frac{\mu_T}{\sigma_T^2} \mathbf{D}(T, \mathbf{x}, \mathbf{y}).$$

Then using the triangular inequality yields

$$\begin{aligned} \|\nabla \log q_T(\mathbf{x} | \mathbf{y})\|_2^2 &\leq \frac{2}{\sigma_T^4} \|\mathbf{x}\|_2^2 + \frac{2\mu_T^2}{\sigma_T^4} \|\mathbf{D}(T, \mathbf{x}, \mathbf{y})\|_2^2 \\ &\leq \frac{2}{\sigma_T^4} \|\mathbf{x}\|_2^2 + \frac{2\mu_T^2}{\sigma_T^4} \left(\frac{2d\sigma_T^2}{\mu_T^2 - \alpha\sigma_T^2} + \frac{6\sigma_T^4 H_{\mathbf{y}}^2}{(\mu_T^2 - \alpha\sigma_T^2)^2} + \frac{6\mu_T^2}{(\mu_T^2 - \alpha\sigma_T^2)^2} \|\mathbf{x}\|_2^2 \right) \\ &= \frac{4d\mu_T^2}{\sigma_T^2 (\mu_T^2 - \alpha\sigma_T^2)} + \frac{12\mu_T^2 H_{\mathbf{y}}^2}{(\mu_T^2 - \alpha\sigma_T^2)^2} + \frac{2(\mu_T^2 - \alpha\sigma_T^2)^2 + 12\mu_T^4}{\sigma_T^4 (\mu_T^2 - \alpha\sigma_T^2)^2} \|\mathbf{x}\|_2^2 \\ &\lesssim \frac{H_{\mathbf{y}}^2 + \|\mathbf{x}\|_2^2}{\sigma_T^4 (\mu_T^2 - \alpha\sigma_T^2)^2}, \end{aligned}$$

where the second inequality is due to Lemma G.2, and we used the fact that $\mu_T^2 - \alpha\sigma_T^2 \in (0, 1)$. This completes the proof. $\square$

G.2 Proof of Lemma 5.4. We next provide error analysis for the warm-start, which uses a similar argument as the proof of Lemma 5.5.

Proof of Lemma 5.4. According to the triangular inequality, we find

$$(G.4) \quad \|q_T(\cdot|\mathbf{y}) - \hat{q}_T^U(\cdot|\mathbf{y})\|_{\text{TV}}^2 \leq 2 \underbrace{\|q_T(\cdot|\mathbf{y}) - q_T^U(\cdot|\mathbf{y})\|_{\text{TV}}^2}_{\text{convergence of Langevin}} + 2 \underbrace{\|q_T^U(\cdot|\mathbf{y}) - \hat{q}_T^U(\cdot|\mathbf{y})\|_{\text{TV}}^2}_{\text{score estimation error}},$$

where $q_T^u(\cdot|\mathbf{y})$ is the marginal density of $\mathbf{X}_{T,u}^{\mathbf{y}}$ defined in (4.10) with $q_T^0(\cdot|\mathbf{y}) = \hat{q}_T^0(\cdot|\mathbf{y})$.

Step 1. The convergence of Langevin dynamics. Recall that $\zeta_{\mathbf{y}}^2 := \chi^2(\hat{q}_T^0(\cdot|\mathbf{y})\|q_T(\cdot|\mathbf{y}))$. For the first term in the right-hand side of (G.4),

$$(G.5) \quad \|q_T(\cdot|\mathbf{y}) - q_T^U(\cdot|\mathbf{y})\|_{\text{TV}}^2 \leq \frac{1}{2} \chi^2(q_T^U(\cdot|\mathbf{y})\|q_T(\cdot|\mathbf{y})) \leq \frac{1}{2} \exp\left(-\frac{2U}{C_{\text{LSI}}(q_T(\cdot|\mathbf{y}))}\right) \zeta_{\mathbf{y}}^2,$$

where the first inequality is due to Pinsker's inequality (Tsybakov, 2009, Lemma 2.5) and Tsybakov (2009, Lemma 2.7), and the second inequality holds from Lemmas 4.12 and H.6.

Step 2. The error of the score estimation. For the second term in the right-hand side of (G.4),

$$(G.6) \quad \begin{aligned} & \|q_T^U(\cdot|\mathbf{y}) - \hat{q}_T^U(\cdot|\mathbf{y})\|_{\text{TV}}^2 \leq \frac{1}{2} \text{KL}(q_T^U(\cdot|\mathbf{y})\|\hat{q}_T^U(\cdot|\mathbf{y})) \\ & \leq \int_0^U \int \|\nabla \log q_T(\mathbf{x}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T, \mathbf{x}, \mathbf{y})\|_2^2 q_T^u(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \, du \\ & = \int_0^U \int \|\nabla \log q_T(\mathbf{x}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T, \mathbf{x}, \mathbf{y})\|_2^2 \frac{q_T^u(\mathbf{x}|\mathbf{y})}{q_T(\mathbf{x}|\mathbf{y})} q_T(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \, du \\ & \leq \underbrace{\left(\int \|\nabla \log q_T(\mathbf{x}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T, \mathbf{x}, \mathbf{y})\|_2^2 q_T(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \right)^{\frac{1}{4}}}_{(i)} \\ & \quad \times \underbrace{\left(\int \|\nabla \log q_T(\mathbf{x}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T, \mathbf{x}, \mathbf{y})\|_2^6 q_T(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \right)^{\frac{1}{4}}}_{(ii)} \\ & \quad \times \underbrace{\int_0^U \left(\int \left(\frac{q_T^u(\mathbf{x}|\mathbf{y})}{q_T(\mathbf{x}|\mathbf{y})} \right)^2 q_T(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \right)^{\frac{1}{2}} du}_{(iii)} \end{aligned}$$

where the first inequality is due to Pinsker's inequality (Tsybakov, 2009, Lemma 2.5), the second inequality invokes Girsanov theorem. See [Oko et al. \(2023, Proposition D.1\)](#) or [Chen et al. \(2023c, Theorem 2\)](#) for details. The second inequality is owing to Hölder's inequality. For the term (i) in (G.6), we find

$$(G.7) \quad \int \|\nabla \log q_T(\mathbf{x}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T, \mathbf{x}, \mathbf{y})\|_2^2 q_T(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \leq \varepsilon_{\text{post}}^2.$$

For the term (ii) in (G.6), by the same argument as (F.17), we have

$$(G.8) \quad \begin{aligned} & \int \|\nabla \log q_T(\mathbf{x}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T, \mathbf{x}, \mathbf{y})\|_2^6 q_T(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \\ & \lesssim \frac{1}{\sigma_T^{12}(\mu_T^2 - \alpha\sigma_T^2)^6} \int (H^6 + \|\mathbf{x}\|_2^6) q_T(\mathbf{x}|\mathbf{y}) \, d\mathbf{x} \\ & \lesssim \frac{H^6}{\sigma_t^{12}(\mu_T^2 - \alpha\sigma_T^2)^6} + \frac{\kappa_{\mathbf{y}} C_{\text{SG}} V_{\text{SG}}^6 + d^3}{\sigma_t^{12}(\mu_T^2 - \alpha\sigma_T^2)^6} \leq C \frac{\kappa_{\mathbf{y}}}{(\mu_T^2 - \alpha\sigma_T^2)^6}, \end{aligned}$$

where the first inequality follows from Lemma G.3, the second inequality is due to Lemma H.3 and used the fact that $T \in (\underline{t}, \bar{t})$, and C is a constant only depending on d , V_{SG} , and C_{SG} . For the term (iii) in (G.6), by an argument similar to (F.18), we have

$$(G.9) \quad \int_0^U \left(\int \left(\frac{q_T^u(\mathbf{x}|\mathbf{y})}{q_T(\mathbf{x}|\mathbf{y})} \right)^2 q_T(\mathbf{x}|\mathbf{y}) d\mathbf{x} \right)^{\frac{1}{2}} du \lesssim \{C_{\text{LSI}}(q_T(\cdot|\mathbf{y}))\zeta_{\mathbf{y}}^2 + U\}^{\frac{1}{2}},$$

where $C_{\text{LSI}}(q_T(\cdot|\mathbf{y}))$ is specified as Lemma 4.12. Substituting (G.7), (G.8), and (G.9) into (G.6) yields

$$(G.10) \quad \|q_T^U(\cdot|\mathbf{y}) - \hat{q}_T^U(\cdot|\mathbf{y})\|_{\text{TV}}^2 \leq C \{C_{\text{LSI}}(q_T(\cdot|\mathbf{y}))\zeta_{\mathbf{y}}^2 + U\}^{\frac{1}{2}} (\mu_T^2 - \alpha\sigma_T^2)^{-\frac{3}{2}} \kappa_{\mathbf{y}}^{\frac{1}{4}} \varepsilon_{\text{post}}^{\frac{1}{2}},$$

where C is a constant only depending on d , V_{SG} , and C_{SG} .

Step 3. Conclusion. According to (G.5), to satisfy $\|q_T(\cdot|\mathbf{y}) - \hat{q}_T^U(\cdot|\mathbf{y})\|_{\text{TV}}^2 = \varepsilon$, we set

$$(G.11) \quad U = \Theta \left(C_{\text{LSI}}(q_T(\cdot|\mathbf{y})) \log \left(\frac{\zeta_{\mathbf{y}}^2}{\varepsilon} \right) \right).$$

To bound (G.10) by the accuracy tolerance ε , we set

$$(G.12) \quad \varepsilon_{\text{post}} = \Theta \left(\frac{(\mu_T^2 - \alpha\sigma_T^2)^3}{C_{\text{LSI}}(q_T(\cdot|\mathbf{y}))\zeta_{\mathbf{y}}^2 + U} \frac{\varepsilon^2}{\kappa_{\mathbf{y}}^{1/2}} \right).$$

Substituting (G.11) into (G.12) yields

$$\varepsilon_{\text{post}} = \Theta \left(\frac{(\mu_T^2 - \alpha\sigma_T^2)^3}{C_{\text{LSI}}(q_T(\cdot|\mathbf{y}))} \left(\zeta_{\mathbf{y}}^2 + \log \left(\frac{\zeta_{\mathbf{y}}^2}{\varepsilon} \right) \right)^{-1} \frac{\varepsilon^2}{\kappa_{\mathbf{y}}^{1/2}} \right).$$

This completes the proof. $\square$

H Auxiliary Lemmas

H.1 Sub-Gaussian Properties of the Forward Process. In this subsection, we establish some sub-Gaussian properties of $\mathbf{X}_0^{\mathbf{y}}$ and $\mathbf{X}_t^{\mathbf{y}}$. These results are inspired by Vershynin (2018, Proposition 2.6.1)

Lemma H.1. *Suppose Assumption 2 holds. Let $\mathbf{X}_0^{\mathbf{y}} \sim q_0(\cdot|\mathbf{y})$. Then for each $\xi > 0$,*

$$\mathbb{P}\{\|\mathbf{X}_0^{\mathbf{y}}\|_2 \geq \xi\} \leq \kappa_{\mathbf{y}} C_{\text{SG}} \exp \left(- \frac{\xi^2}{V_{\text{SG}}^2} \right).$$

Proof of Lemma H.1. It is straightforward that

$$\begin{aligned} \mathbb{P}\{\|\mathbf{X}_0^{\mathbf{y}}\|_2 \geq \xi\} &= \mathbb{P}\left\{ \frac{\|\mathbf{X}_0^{\mathbf{y}}\|_2^2}{V_{\text{SG}}^2} \geq \frac{\xi^2}{V_{\text{SG}}^2} \right\} = \mathbb{P}\left\{ \exp \left(\frac{\|\mathbf{X}_0^{\mathbf{y}}\|_2^2}{V_{\text{SG}}^2} \right) \geq \exp \left(\frac{\xi^2}{V_{\text{SG}}^2} \right) \right\} \\ &\leq \exp \left(- \frac{\xi^2}{V_{\text{SG}}^2} \right) \mathbb{E} \left[\exp \left(\frac{\|\mathbf{X}_0^{\mathbf{y}}\|_2^2}{V_{\text{SG}}^2} \right) \right] \leq \kappa_{\mathbf{y}} C_{\text{SG}} \exp \left(- \frac{\xi^2}{V_{\text{SG}}^2} \right), \end{aligned}$$

where the first inequality holds from Markov's inequality, and the last inequality invokes Assumption 2 and Lemma 4.10. This completes the proof. $\square$

Lemma H.2. *Suppose Assumption 2 holds. Let $\mathbf{X}_0^{\mathbf{y}} \sim q_0(\cdot|\mathbf{y})$. Then*

$$\mathbb{E}[\|\mathbf{X}_0^{\mathbf{y}}\|_2^m] \leq 2m^{\frac{m}{2}+1} \kappa_{\mathbf{y}} C_{\text{SG}} V_{\text{SG}}^m.$$

Proof of Lemma H.2. It follows from Lemma H.1 that

$$\begin{aligned}
 \mathbb{E}[\|\mathbf{X}_0^y\|_2^m] &= \int_0^\infty \mathbb{P}\{\|\mathbf{X}_0^y\|_2^m \geq \xi\} d\xi \\
 &= m \int_0^\infty \mathbb{P}\{\|\mathbf{X}_0^y\|_2 \geq \eta\} \eta^{m-1} d\eta \\
 &\leq m\kappa_y C_{\text{SG}} \int_0^\infty \exp\left(-\frac{\eta^2}{V_{\text{SG}}^2}\right) \eta^{m-1} d\eta \\
 &= \frac{1}{2} m\kappa_y C_{\text{SG}} V_{\text{SG}}^m \int_0^\infty \exp\left(-\frac{\eta^2}{V_{\text{SG}}^2}\right) \left(\frac{\eta^2}{V_{\text{SG}}^2}\right)^{\frac{m}{2}-1} d\frac{\eta^2}{V_{\text{SG}}^2} \\
 &= \frac{1}{2} m\kappa_y C_{\text{SG}} V_{\text{SG}}^m \int_0^\infty \exp(-\zeta) \zeta^{\frac{m}{2}-1} d\zeta \\
 &= \frac{1}{2} m\Gamma\left(\frac{m}{2}\right) \kappa_y C_{\text{SG}} V_{\text{SG}}^m \leq 2m^{\frac{m}{2}+1} \kappa_y C_{\text{SG}} V_{\text{SG}}^m,
 \end{aligned}$$

where the second equality used the change of variables $\xi = \eta^m$, the first inequality invokes Lemma H.1, the third equality used the change of variable $\eta = \sqrt{\xi}$, the fourth equality follows from a change of variables $\zeta = \eta^2 V_{\text{SG}}^{-2}$. This completes the proof. $\square$

Lemma H.3. Suppose Assumption 2 holds. Let $\mathbf{X}_t^y \sim q_t(\cdot|\mathbf{y})$. Then

$$\mathbb{E}[\|\mathbf{X}_t^y\|_2^m] \leq 2^{m-1} \{2m^{\frac{m}{2}+1} \kappa_y C_{\text{SG}} V_{\text{SG}}^m + (d+m)^{\frac{m}{2}}\}.$$

Proof of Lemma H.3. Let $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{I}_d)$. It is straightforward that

$$\mathbb{E}[\|\boldsymbol{\varepsilon}\|_2^m] = 2^{\frac{m}{2}} \Gamma\left(\frac{d+m}{2}\right) \Gamma\left(\frac{d}{2}\right) \leq (d+m)^{\frac{m}{2}}.$$

Since $\mathbf{X}_t^y \stackrel{d}{=} \mu_t \mathbf{X}_0^y + \sigma_t \boldsymbol{\varepsilon}$ with $\mathbf{X}_0^y \sim q_0(\cdot|\mathbf{y})$ independent of $\boldsymbol{\varepsilon}$, it follows from the triangular inequality that

$$\begin{aligned}
 \mathbb{E}[\|\mathbf{X}_t^y\|_2^m] &\leq 2^{m-1} \mu_t^m \mathbb{E}[\|\mathbf{X}_0^y\|_2^m] + 2^{m-1} \sigma_t^m \mathbb{E}[\|\boldsymbol{\varepsilon}\|_2^m] \\
 &\leq 2^{m-1} \{2m^{\frac{m}{2}+1} \kappa_y C_{\text{SG}} V_{\text{SG}}^m + (d+m)^{\frac{m}{2}}\},
 \end{aligned}$$

where we used the fact that $\mu_t, \sigma_t \leq 1$. This completes the proof. $\square$

Lemma H.4. Suppose Assumption 2 holds. Let $\mathbf{X}_t^y \sim q_t(\cdot|\mathbf{y})$ be a random variable defined by (2.4). Then for each $R > 0$,

$$\mathbb{P}\{\|\mathbf{X}_t^y\|_2 \geq \xi\} \leq 2^d \kappa_y C_{\text{SG}} \exp\left(-\frac{\xi^2}{2\mu_t^2 V_{\text{SG}}^2 + 8\sigma_t^2}\right).$$

Proof of Lemma H.1. According to Assumption 2 and Lemma 4.10, we have

$$(H.1) \quad \mathbb{E}\left[\exp\left(\frac{\|\mu_t \mathbf{X}_0^y\|_2^2}{\mu_t^2 V_{\text{SG}}^2}\right)\right] = \mathbb{E}\left[\exp\left(\frac{\|\mathbf{X}_0^y\|_2^2}{V_{\text{SG}}^2}\right)\right] \leq \kappa_y C_{\text{SG}}.$$

Let $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{I}_d)$. Then it follows that

$$\begin{aligned}
 (H.2) \quad \mathbb{E}\left[\exp\left(\frac{\|\sigma_t \boldsymbol{\varepsilon}\|_2^2}{4\sigma_t^2}\right)\right] &= \mathbb{E}\left[\exp\left(\frac{\|\boldsymbol{\varepsilon}\|_2^2}{4}\right)\right] \\
 &= (2\pi)^{-\frac{d}{2}} \int \exp\left(\frac{\|\boldsymbol{\varepsilon}\|_2^2}{4}\right) \exp\left(-\frac{\|\boldsymbol{\varepsilon}\|_2^2}{2}\right) d\boldsymbol{\varepsilon} \\
 &= (2\pi)^{-\frac{d}{2}} \int \exp\left(-\frac{\|\boldsymbol{\varepsilon}\|_2^2}{4}\right) d\boldsymbol{\varepsilon} \leq 2^d.
 \end{aligned}$$

Notice that $\mathbf{X}_t^y \stackrel{d}{=} \mu_t \mathbf{X}_0^y + \sigma_t \boldsymbol{\varepsilon}$, where $\mathbf{X}_0^y \sim q_0(\cdot | \mathbf{y})$ and $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{I}_d)$ are independent. Therefore,

$$\begin{aligned}
 \mathbb{E} \left[\exp \left(\frac{\|\mathbf{X}_t^y\|_2^2}{2\mu_t^2 V_{SG}^2 + 8\sigma_t^2} \right) \right] &= \mathbb{E} \left[\exp \left(\frac{\|\mu_t \mathbf{X}_0^y + \sigma_t \boldsymbol{\varepsilon}\|_2^2}{2\mu_t^2 V_{SG}^2 + 8\sigma_t^2} \right) \right] \\
 &\leq \mathbb{E} \left[\exp \left(\frac{\|\mu_t \mathbf{X}_0^y\|_2^2}{\mu_t^2 V_{SG}^2 + 4\sigma_t^2} + \frac{\|\sigma_t \boldsymbol{\varepsilon}\|_2^2}{\mu_t^2 V_{SG}^2 + 4\sigma_t^2} \right) \right] \\
 &\leq \mathbb{E} \left[\exp \left(\frac{\|\mu_t \mathbf{X}_0^y\|_2^2}{\mu_t^2 V_{SG}^2 + 4\sigma_t^2} \right) \right] \mathbb{E} \left[\exp \left(\frac{\|\sigma_t \boldsymbol{\varepsilon}\|_2^2}{\mu_t^2 V_{SG}^2 + 4\sigma_t^2} \right) \right] \\
 (H.3) \quad &\leq \mathbb{E} \left[\exp \left(\frac{\|\mu_t \mathbf{X}_0^y\|_2^2}{\mu_t^2 V_{SG}^2} \right) \right] \mathbb{E} \left[\exp \left(\frac{\|\sigma_t \boldsymbol{\varepsilon}\|_2^2}{4\sigma_t^2} \right) \right] \leq 2^d \kappa_y C_{SG},
 \end{aligned}$$

where the first inequality follows from Cauchy-Schwarz inequality, the second inequality holds from the independence of $\mathbf{X}_0^y$ and $\boldsymbol{\varepsilon}$, and the last inequality is due to (H.1) and (H.2). Then we aim to bound the tail probability. For each $\xi > 0$, we have

$$\begin{aligned}
 \mathbb{P}\{\|\mathbf{X}_t^y\|_2 \geq \xi\} &= \mathbb{P}\left\{ \frac{\|\mathbf{X}_t^y\|_2^2}{2\mu_t^2 V_{SG}^2 + 8\sigma_t^2} \geq \frac{\xi^2}{2\mu_t^2 V_{SG}^2 + 8\sigma_t^2} \right\} \\
 &= \mathbb{P}\left\{ \exp \left(\frac{\|\mathbf{X}_t^y\|_2^2}{2\mu_t^2 V_{SG}^2 + 8\sigma_t^2} \right) \geq \exp \left(\frac{\xi^2}{2\mu_t^2 V_{SG}^2 + 8\sigma_t^2} \right) \right\} \\
 &\leq \exp \left(- \frac{\xi^2}{2\mu_t^2 V_{SG}^2 + 8\sigma_t^2} \right) \mathbb{E} \left[\exp \left(\frac{\|\mathbf{X}_t^y\|_2^2}{2\mu_t^2 V_{SG}^2 + 8\sigma_t^2} \right) \right] \\
 &\leq 2^d \kappa_y C_{SG} \exp \left(- \frac{\xi^2}{2\mu_t^2 V_{SG}^2 + 8\sigma_t^2} \right),
 \end{aligned}$$

where the first inequality invokes Markov's inequality, and the last inequality is due to (H.3). This completes the proof. $\square$

H.2 Convergence of Langevin dynamics. Consider the Langevin dynamics with an invariant density ν :

$$(H.4) \quad d\mathbf{Z}_t = \nabla \log \nu(\mathbf{Z}_t) dt + \sqrt{2} d\mathbf{B}_t, \quad \mathbf{Z}_0 \sim \nu_0,$$

where $(\mathbf{B}_t)_{t \geq 0}$ is a d -dimensional Brownian motion. Denote by ν_t the marginal density of $\mathbf{Z}_t$.

Convergence in Wasserstein distance.

Lemma H.5. *Suppose the invariant distribution ν is m -strongly log-concave. Then for each $t > 0$,*

$$\mathbb{W}_2^2(\nu_t, \nu) \leq \exp(-2mt) \mathbb{W}_2^2(\nu_0, \nu).$$

Proof of Lemma H.5. We first construct a Wasserstein coupling. Let γ_0 be the optimal coupling of (ν_0, ν) , which means,

$$(H.5) \quad \mathbb{E}_{(\mathbf{Z}_0, \mathbf{Z}_0^*) \sim \gamma_0} [\|\mathbf{Z}_0 - \mathbf{Z}_0^*\|_2^2] = \int \|\mathbf{z}_0 - \mathbf{z}_0^*\|_2^2 d\gamma_0(\mathbf{z}_0, \mathbf{z}_0^*) = \mathbb{W}_2^2(\nu_0, \nu).$$

We define two processes $\mathbf{Z}_t$ and $\mathbf{Z}_t^*$ by evolving the Langevin dynamics (H.4) with the same Brownian motion, that is,

$$(H.6) \quad d\mathbf{Z}_t = \nabla \log \nu(\mathbf{Z}_t) dt + \sqrt{2} d\mathbf{B}_t, \quad \text{and} \quad d\mathbf{Z}_t^* = \nabla \log \nu(\mathbf{Z}_t^*) dt + \sqrt{2} d\mathbf{B}_t.$$

Denote by γ_t the joint distributoin of $(\mathbf{Z}_t, \mathbf{Z}_t^*)$. It is apparent that γ_t is a coupling of (ν_t, ν) , as ν is the invariant distribution of the Langevin dynamics (H.4) and $\mathbf{Z}_t^* \sim \nu$ for each $t > 0$. Then

$$\begin{aligned}
 d\|\mathbf{Z}_t - \mathbf{Z}_t^*\|_2^2 &= 2(\mathbf{Z}_t - \mathbf{Z}_t^*, d\mathbf{Z}_t - d\mathbf{Z}_t^*) \\
 &= 2(\mathbf{Z}_t - \mathbf{Z}_t^*, \nabla \log \nu(\mathbf{Z}_t) - \nabla \log \nu(\mathbf{Z}_t^*)) dt \\
 &\leq -2m\|\mathbf{Z}_t - \mathbf{Z}_t^*\|_2^2 dt,
 \end{aligned}
 \tag{H.7}$$

where the second equality involves (H.6) and the fact that two Langevin dynamics share the same Brownian motion, and the inequality follows from the strongly log-concavity of ν . Applying Gronwall's inequality (Evans, 2010, Section B.2) to (H.7) yields

$$\|\mathbf{Z}_t - \mathbf{Z}_t^*\|_2^2 \leq \exp(-2mt)\|\mathbf{Z}_0 - \mathbf{Z}_0^*\|_2^2.$$

Taking expectation with respect to both sides of the inequality implies

$$\begin{aligned}
 W_2^2(\nu_t, \nu) &\leq \mathbb{E}_{(\mathbf{Z}_t, \mathbf{Z}_t^*) \sim \gamma_t} [\|\mathbf{Z}_t - \mathbf{Z}_t^*\|_2^2] \\
 &\leq \exp(-2mt) \mathbb{E}_{(\mathbf{Z}_0, \mathbf{Z}_0^*) \sim \gamma_0} [\|\mathbf{Z}_0 - \mathbf{Z}_0^*\|_2^2] \\
 &= \exp(-2mt) W_2^2(\nu_0, \nu),
 \end{aligned}$$

where the first inequality is owing to the definition of the Wasserstein distance, and the equality follows from (H.5). This completes the proof. $\square$

Convergence in χ^2 -divergence. Under log-Sobolev inequality of the invariant density ν , the convergence of the Langevin dynamics is stated as the following lemma, the proof of which can be found in Vempala and Wibisono (2019, Theorem 3) or Chewi et al. (2024).

Lemma H.6. *Suppose that the invariant density $\nu \in C^2(\mathbb{R}^d)$ satisfies the log-Sobolev inequality with constant $C_{\text{LSI}}(\nu)$, then for each initial density ν_0 , it holds that*

$$\chi^2(\nu_t \| \nu) \leq \exp\left(-\frac{2t}{C_{\text{LSI}}(\nu)}\right) \chi^2(\nu_0 \| \nu).$$

H.3 Log-concavity. Let μ be a probability distribution absolutely continuous with respect to the Lebesgue measure. Denote by ρ the density of μ , that is, $d\mu(\mathbf{x}) = \rho(\mathbf{x}) d\mathbf{x}$. Suppose that $\rho \in C^2(\mathbb{R}^d)$. In this work, we will consider some regimes of the distribution μ :

- (i) The distribution μ is called strongly log-concave, if there exists a constant $\alpha > 0$ such that $-\nabla^2 \log \rho(\mathbf{x}) \succeq \alpha \mathbf{I}_d$ for each $\mathbf{x} \in \mathbb{R}^d$.
- (ii) The distribution μ is called semi-log-concave, if there exists a constant $\alpha > 0$ such that $-\nabla^2 \log \rho(\mathbf{x}) \succeq -\alpha \mathbf{I}_d$ for each $\mathbf{x} \in \mathbb{R}^d$.
- (iii) The distribution μ on $\mathbb{R}^d$ satisfies the log-Sobolev inequality, if there exists a constant $C_{\text{LSI}}(\mu) > 0$ such that

$$(\text{LSI}) \quad \text{Ent}_\mu(f) \leq 2C_{\text{LSI}}(\mu) \mathbb{E}_\mu[\|\nabla f\|_2^2], \quad \text{for each } f \in C_0^\infty(\mathbb{R}^d),$$

where $\text{Ent}_\mu(f) = \mathbb{E}_\mu[f \log f] - \mathbb{E}_\mu[f] \log \mathbb{E}_\mu[f]$, and $\mathbb{E}_\mu[\cdot]$ denotes the expectation with respect to μ .

- (iv) The distribution μ on $\mathbb{R}^d$ has sub-Gaussian tails, if there exist constants $V_{\text{SG}} > 0$ and $C_{\text{SG}} > 0$, such that

$$\int \exp\left(\frac{\|\mathbf{x}\|_2^2}{V_{\text{SG}}^2}\right) d\mu(\mathbf{x}) \leq C_{\text{SG}}.$$

The conditions (i) to (iv) have close connections:

strongly log-concavity $\subset$ log-Sobolev inequality $\subset$ sub-Gaussian tails.

Specifically, the Bakry–Émery theorem ([Bakry and Émery, 1985](#)) shows that α -strong log-concavity implies that a log-Sobolev inequality with $C_{\text{LSI}}(\mu) \leq \alpha^{-1}$. A log-Sobolev inequality implies sub-Gaussian tails ([Ledoux, 1999](#)). See [Villani \(2003, Theorem 9.9\)](#) and [Bakry et al. \(2014, Section 5.4\)](#) for further details.